



Equipment

- 0103C [GC+ +PC]
- 0106D [Box Key Reset holder]
- 0108A [Add+USB]
- 0202D [AX5 X/Xa OCT Safety 2x6A]
- 0405A [Basic Parts TSc - CE]
- 0407A [Section-A-W600H800D400+MP]
- 0407D [Section-B-W800H800D400+MP]
- 0605M [LightBarrier SICK*]
- 1102C [Main Cabi supply M+N switched]
- 1107A [Transform 36V4x100VA]
- 1110A [EP PowerBox 4x4A]
- 1204A [Foot switch, endtruck, 2x2 FrontRear]
- 1501B [Cabinet cooling, Door, 300W]
- 1702A [Single OmniFlow - Supply A]
- 1705A [Gas-Safety-Valves Supply - A]
- 2504A [Monitoring Compressed Air]
- 2505C [Global Rail Grounding M12]
- 0501B [L1-SL 200*]
- 0701A [L1-Punch, HF]
- 0202G [L2-AX5 Y OCT Safety 3A]
- 0501C [L2-SL 200 HD*]
- 0801A [L2-Torch ALFA]
- 0810A [L2-Torch addressing]
- 1001A [L2-Laser Pointer]
- 0202G [L3-AX5 Y OCT Safety 3A]
- 0501C [L3-SL 200 HD*]
- 0801A [L3-Torch ALFA]
- 0810A [L3-Torch addressing]

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Customer:
Maghreb Steel

project 286002000
machine MultiTherm
length / width 19m / 5m
power 10 kVA
Toolbox Toolbox_88
drawing No. 286002000_Maghreb Steel

editor Johann.Rais
building year 2024
regulation EN60204-1

Number of pages 198
Eplan-version 2.9.4
15117

machines number: 286002000

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Docu	24.05.2024	Johann.Rais					Customer
Index Revision	Date	Name	Date	Name	Date	name	Remarks
	Created on		reviewed		Approved		



shop floor: EU

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directory to Modul Overview location aware

module		description		group box	page
[=]	1B	table of contents		Module Overview	2...5
	4A	topology EtherCAT		EtherCAT cable;SPS-Kartenübersicht	30...37
	Z6A			cable overview	146...155
	Z6B			cable connection plan	156...198
[+]	B (gantry)				
[=]	3C	grounding bar	grounding bar	connection overview;connection points	20;21
[=]	3E	grounding elements	grounding Location	connection point	26
[=]	BI	DC power supply interface	EP power - distributor	24VDC, EtherCat;rack overview	60;61
[=]	FO	flow controller oxyfuel	gas supply OmniFlow	error, RS422;heating, cutting, fuel gas;rack overview	84...87
[=]	IB	interface box	Fieldbus module	power supply, EtherCat;rack overview	91;92
[=]	IBB	interface box	Fieldbus module	power supply, EtherCat;rack overview	93;94
[=]	SP	sensor pressure	compressed air	monitoring;rack overview	134;135
[+]	E (end truck left)				
[=]	EA	emergency stop additions	emergency stop	connection	74
[=]	EF	emergency stop foot protection		sensor, safety EP box	76
[=]	EL	Reißleine	Reißleine vorne	Safety circuit	78
[=]	ELB	Reißleine	Reißleine hinten	Safety circuit	79
[=]	MX	motor X	AM8542 OCT X	connection motor encoder	112
[+]	G (end truck right)				
[=]	EAB	emergency stop additions	emergency stop	connection	75
[=]	EFB	emergency stop foot protection		locator, safety EP box	77
[=]	MXa	motor Xa	AM8542 OCT Xa	connection motor encoder	113
[=]	OP	operation panel CP7902	Operation interface	connection overview;interface cabinet;operating elements;operating elements Lifter;rack overview	120...127
[+]	HB (cabinet main)				
[=]	1A	cover sheet	cover sheet	Messer	1

directory to Modul Overview location aware

module	description		group box	page
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+ HB (cabinet main)

=	1C	diagram conventions	Cable Symbols;Cover page;General Description;Local Identifiers;Overview Device;Overview Terminal strip;Structure and Overview Locations	6...12
=	3B	grounding cabinet	cabinet;mounting plate +HB drive controller	grounding points in cabinet;structure cabinet
=	3BB	grounding cabinet	cabinet;mounting plate +HB control system	grounding points in cabinet;structure cabinet
=	3D	grounding machine	grounding machine	overview
=	AA	AC power supply cabinet	AC operating voltage	main switch, PE/N;overview terminal strip
=	AB	AC distribution	AC operating voltage	operating voltage;overview terminal strip
=	AL	AC levels	transformer	rack overview;supply
=	AS	AC service socket	service socket	feed-in
=	BA	DC power supply	operating voltage	connection overview;electronical fuse channel 1 / 2;electronical fuse channel 3 / 4;power supply;rack overview
=	BU	DC USV	control voltage buffered	connection overview;feed-in;rack overview
=	CXXa	controller X,Xa	AX5206 X,Xa	power supply 400 VAC;rack overview
=	CY2Y3	controller Y2,Y3	AX5203 Y2,Y3	entrances;power supply 400 VAC;rack overview
=	EP	emergency stop process	emergency stop process	Safety circuit interface;rack overview
=	ER	emergency stop relay	contact extension	Safety relais;rack overview
=	FO	flow controller oxyfuel		rack overview
=	IC	interface control	star hub	power supply, EtherCat;rack overview
=	IT	interface terminal	bus coupler;coupler, 8 x DI, 8 x DO;power supply terminal	EtherCAT, power supply;EtherCAT, refreshment
=	OC	operation control system	cabinet PC	connection overview;power supply, (rechargeable) battery, Bus;rack overview
=	UT	unit temperature	climatisation	supply

+ HG ()

=	FS	flow controller safety	gas supply OmniFlow	supply	89;90
---	----	------------------------	---------------------	--------	-------

+ HOK (operation panel key switch)

=	OK	operation panel key switch	key switch box	elements	119
---	----	----------------------------	----------------	----------	-----

+ L1 (Lifter)

=	3E1	grounding elements	grounding Location	connection point	27
---	-----	--------------------	--------------------	------------------	----

directory to Modul Overview location aware

module	description		group box	page
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+ L1 (Lifter)				
=	DL1	detection limiting	limit Y+;limiting Y+	cabinet wiring;connection overview70;71
=	LI1	lifter	SL 200	AC/DC power supply;rack overview105...107
=	SH1	sensor height	PAN	connection on lifter;rack overview128;129
=	UM1	unit marking	HF punch marker	connection141

+ L2 (Lifter)				
=	3E2	grounding elements	grounding Location	connection point28
=	LI2	lifter	SL 200 HD	AC/DC power supply;rack overview108;109
=	MY2	motor Y2	AM8531 OCT Y2	connection motor encoder114
=	SH2	sensor height	ARKI	power supply, operation;rack overview130;131
=	TO2	torch	type ALFA	gas supply;power supply collision137;138
=	UP2	unit pointer	laser diode	connection142
=	VA2	valves adresssing	adresssing	cutting gas, fuel gas gas flow, heating gas144

+ L3 (Lifter)				
=	3E3	grounding elements	grounding Location	connection point29
=	DL3	detection limiting	limit Y-	cabinet wiring;connection overview72;73
=	LI3	lifter	SL 200 HD	AC/DC power supply;rack overview110;111
=	MY3	motor Y3	AM8531 OCT Y3	connection motor encoder115
=	SH3	sensor height	ARKI	power supply, operation;rack overview132;133
=	TO3	torch	type ALFA	gas supply;power supply collision139;140
=	VA3	valves adresssing	adresssing	cutting gas, fuel gas gas flow, heating gas145

+ P (periphery)				
=	SPB	sensor pressure	differential pressure monitor	feedback exhaust136

+ PZ (customer)				
=	1S	supply connection plan	electrical connection plan	connection plan;connection plan exhaust Donaldson;connection plan machine13...15

module	description	group box	page
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+ PZ (customer)				
=	AA	AC power supply cabinet	rack overview	44
+ T (table)				
=	3CB	grounding bar	grounding bar	connection overview;connection points
=	3DB	grounding machine	grounding machine	excerpt machine, track

General explanations on Electrical Documentation

This electrical documentation contains the necessary technical information required to understand the basic idea of the global electrical schematics with the New Electrical Modular Concept.

This includes: **Cover page**

General Description

Structure and Overview locations

Local identifiers

Overview Terminal strip

Overview Device

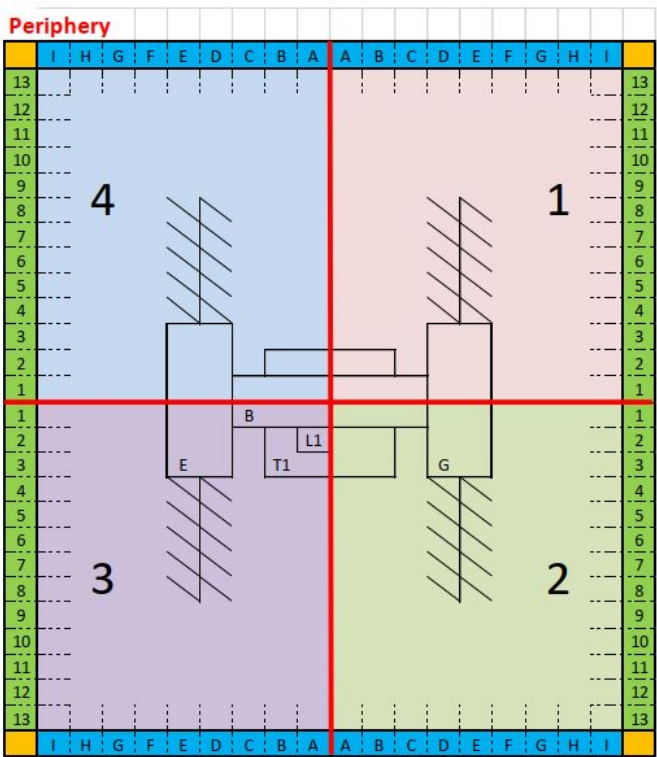
Structure and Overview Locations

Structure consists of total 3 levels.

"+" Location "=" Module "-" Device Tag

+HB=BA-F1

Overview grid with X and Y coordinates:



Eg: 1D3G
1 Sector
D X-Coordinate
3 Y-Coordinate
G Endtruck Right

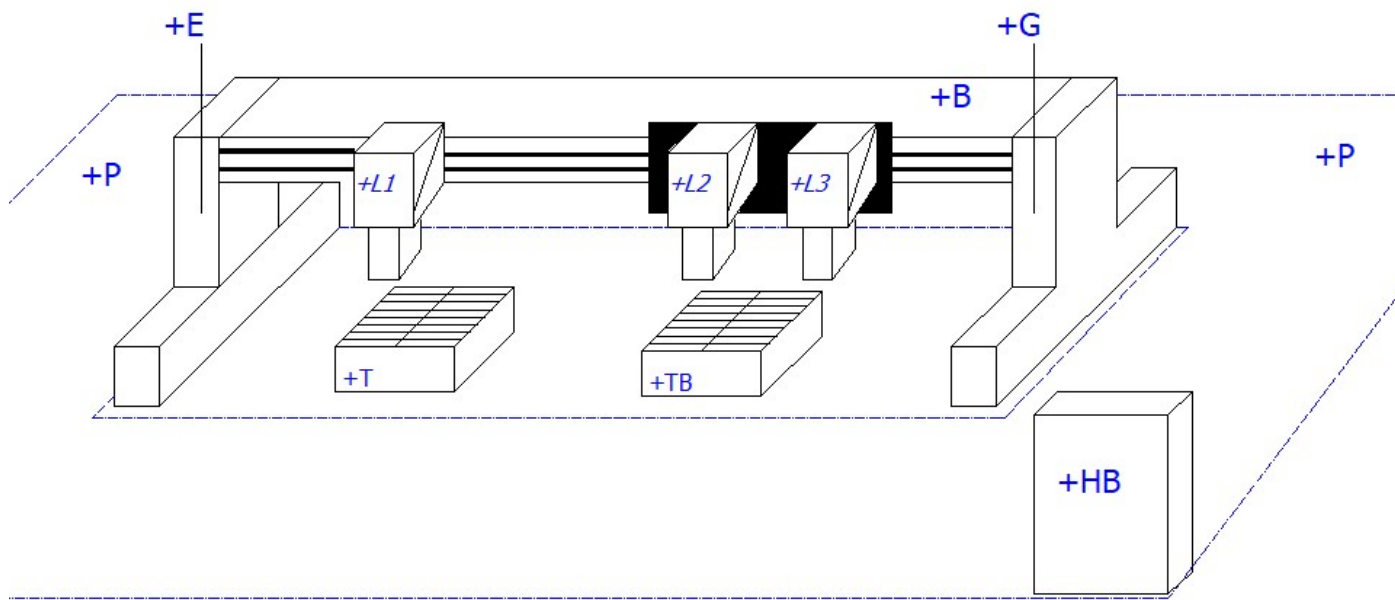
Overview Locations:

"+" - Location

- B Beam
- L Lifter
- E Endtruck Left
- G Endtruck Right
- P Periphery
- T Table
- C Cover / Housing

"+" - Cabinet

- HB Main Cabinet
- HP Cabinet Plasma Interface
- HL Cabinet Laser Interface
- HD Cabinet Drilling
- HC Cabinet Controller
- HO Cabinet Global Pedestal
- HQ Cabinet Laser Head
- HT Cutting Table Interface



Local Identifiers

Location box

Terminal

Connection definition

Cable length calculated
by ABIGAIL (Invisible)

Possible cables found
by ABIGAIL (Invisible)

Cable

Cable Shield

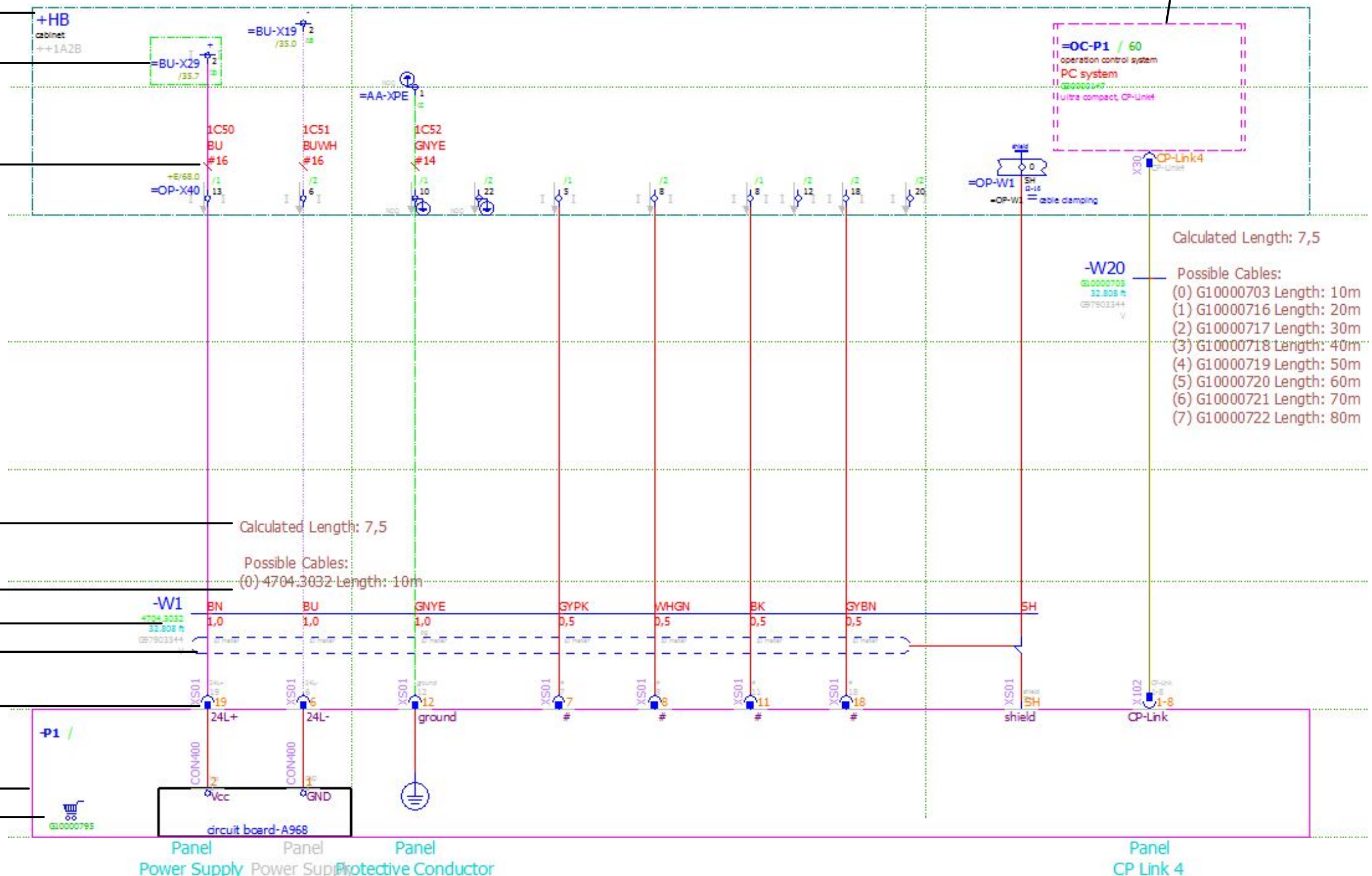
PLC Connection Point

PLC Device Box (Internal)

Part Definition Cart

Function Path text

PLC Device Box (External)



Overview Terminal strip

-X3 = L3 Anschluss

X3

G10000470

Klemme

Brücke

Seite

1	a	Einspeisung Phase	/19.7
	b	Höhenverstellung Trafoschutz	=AL/20.1
	c	Regler Versorgung AC	=CXB+G/34.1
	d		
2	a		
	b		
	c		
	d		

How to populate potential distribution terminals

Terminals are numbered ascending from left to right
Connection points from top to down with a,b,c,d...

G10000485
G10000431

Klemme

Brücke

Etage

Seite

1	,	1	-Absaugung aktiv	/129.8
2	,	2	-Absaugung aktiv	/129.9
3	,	1	-Absaugung aktiv	/129.9
4	,	2	-Absaugung aktiv	/129.9

How to populate feed-through terminals

Represent One to One connection of a Signal
Terminal has a number from left to right per level

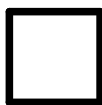
address:	G10001010		
	controller		
	-K1	AX5206, 2 channel, AC 2x 6A, FW >=2.10	
SteckerBMK			
GND			
Kanal	<u>grounding</u>		
M6	ground	controller grounding	/28.9
SteckerBMK			
Safety			
Kanal	<u>PCI</u>		
PCI	Bus	controller Safety Card	/28.5
SteckerBMK			
X01			
Kanal	<u>supply grid</u>		
1	L1	controller power supply AC	/28.1
2	L2	=	/28.2
3	L3	=	/28.2
PE	GND	controller ground	/28.2
SteckerBMK			
X02			
Kanal	<u>DC Link</u>		
1	DC+		
2	DC-		
SteckerBMK			
X03			
Kanal	<u>supply DC</u>		
1	Up+	controller power supply DC	/28.3
2	Us+	controller supply DC	/28.4
3	GND	controller power supply DC	/28.4
SteckerBMK			
X04			
Kanal	<u>Bus</u>		
1-4	Bus-IN	controller EtherCAT	/28.8
SteckerBMK			
X05			
1-4	Bus-OUT	controller EtherCAT	/28.8

A dashed box does not indicate a device location. You can have multiple module devices in one location.

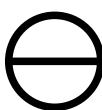
	Without Overview	Inside a module	Outside a module
Devices		K4 Relais 1 x G10000198 Sicherheitsrelais mit zwangsgeführten 1 x 77070641 Überspannungsbegrenzer	
With Overview		-K1 / 102 Relais Safety 77004457 Safety, 2x NO, 3x NO, abfallverzögert	=ER-K1 / 102 Sicherheitskreis Relais Relais Safety 77004457 Safety, 2x NO, 3x NO, abfallverzögert
PLC-Boxes		-K2 / 126 Busklemme G10000492 Digital, Input, 2 Kanal, 24 V DC, 3 m	=PS1-K2 / 126 Plasma Stromquelle Busklemme G10000492 Digital, Input, 2 Kanal, 24 V DC, 3 m

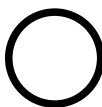
Cable Symbols

See the Z6A section of the electrical schematics to find the cable symbols associated with each cable.

 = Noisy

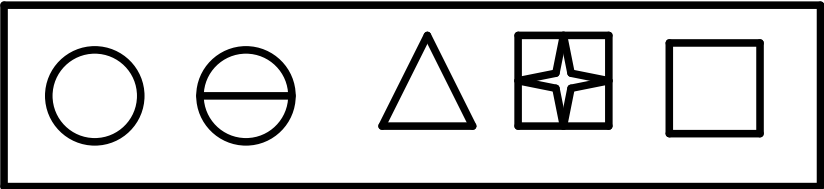
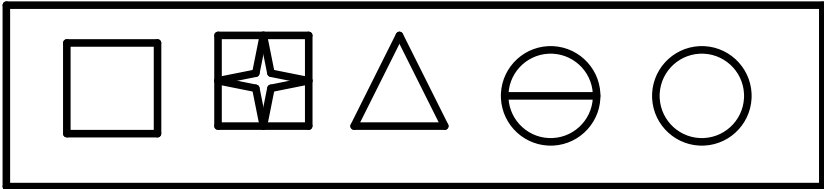
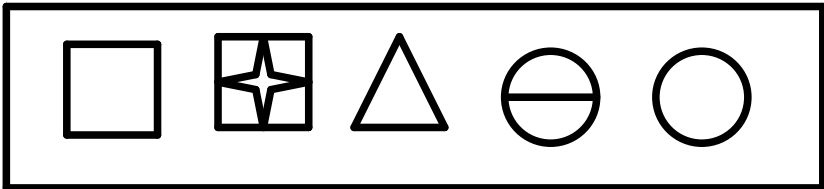
 = Medium Impact

 = Low Impact

 = Sensitive Signal

 = Isolating

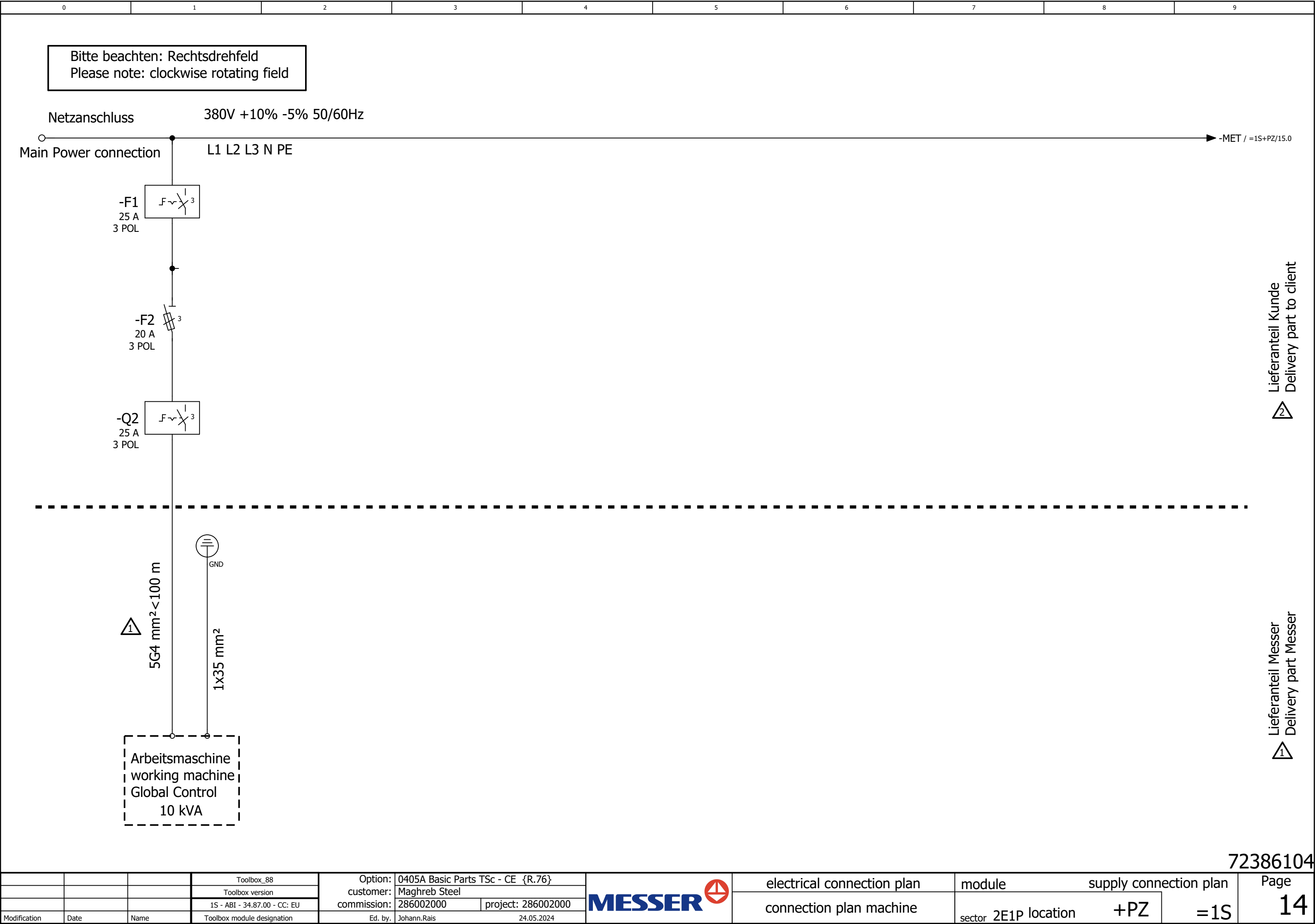
Examples of populating adjacent power tracks with cables

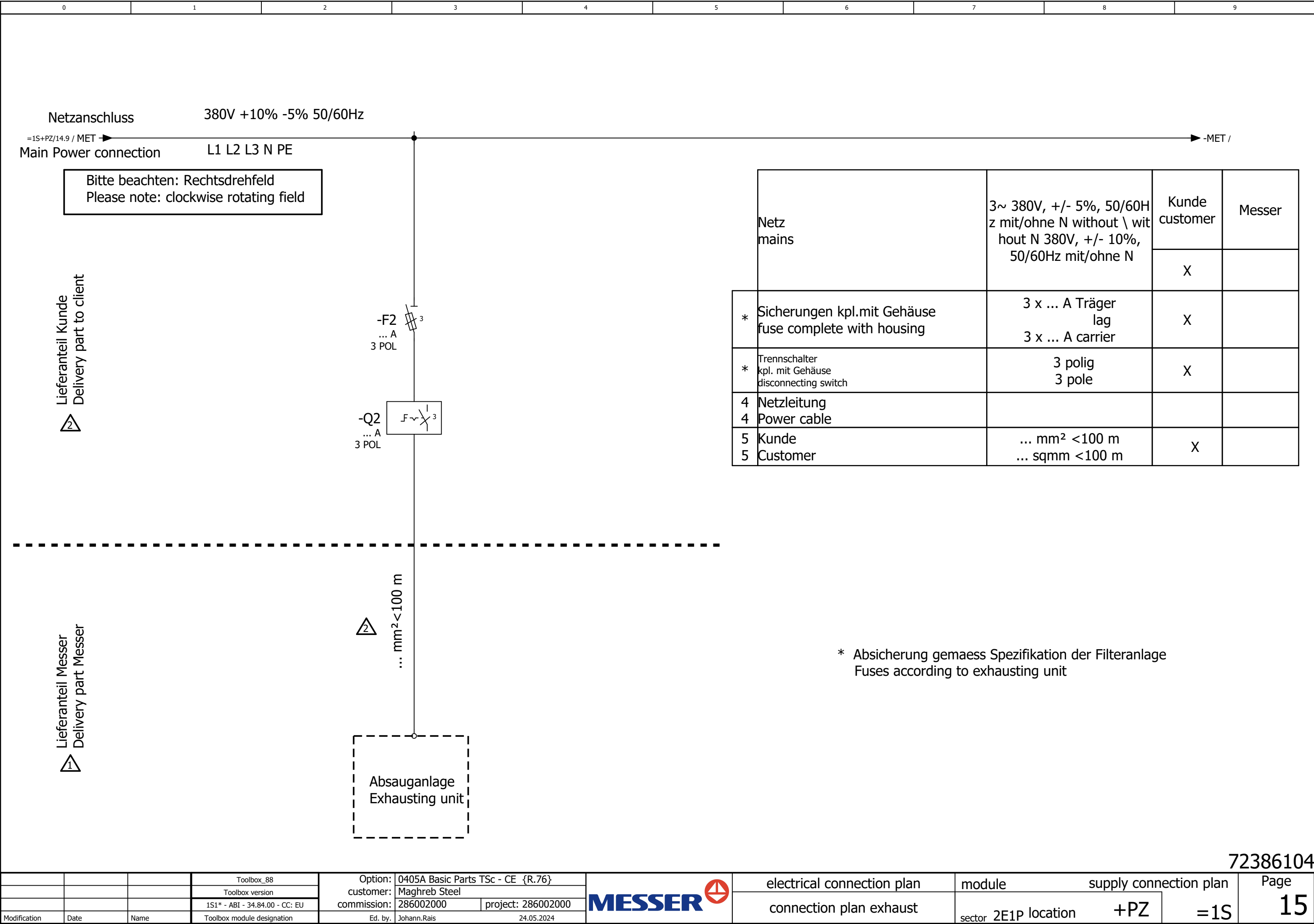


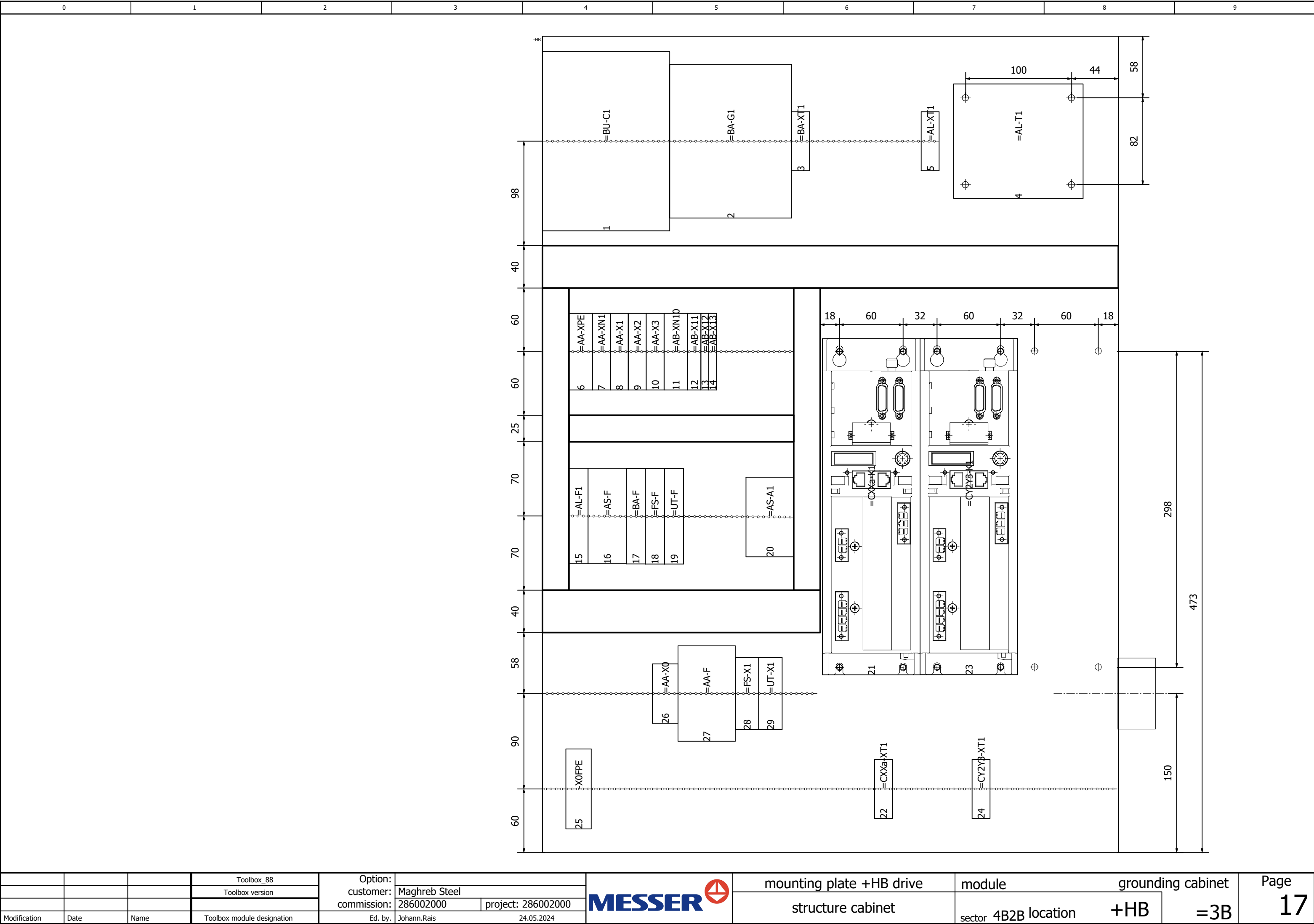
0		1		2		3		4		5		6		7		8		9									
Die Absicherungs- und Schalterbaugruppen sind entsprechend den örtlichen Bestimmungen der Energieversorgungsunternehmen auszulegen und stellen im Anschlussplan eine Empfehlung dar.										The main switchgear and circuit protection have to be arranged according to the local power supply authority regulations. The connection diagram represents a recommendation																	
Spannung: Maximale zulässige Abweichung +10-10% von der Sollspannung und Spannungsspitzen/Einbrüche von max. 5 ms Dauer und +/- 30V.										Line voltage: max. admissible fluctuations +6 -10% and spikes/cuts of 5 ms duration and 30V + or -																	
NC		Arbeitstemperaturbereich		+ 5 bis +45 Grad C		Lagerungstemperaturbereich		-10 bis +50 Grad C		Rel. Luftfeuchtigkeit		bis 90% (nichtkondensierend)		NC		Working temperature range		+ 5 to +45 degree centigrade		Storage temperature range		-10 to +50 degree centigrade		Relative humidity		max. 90% (condensation-free)	

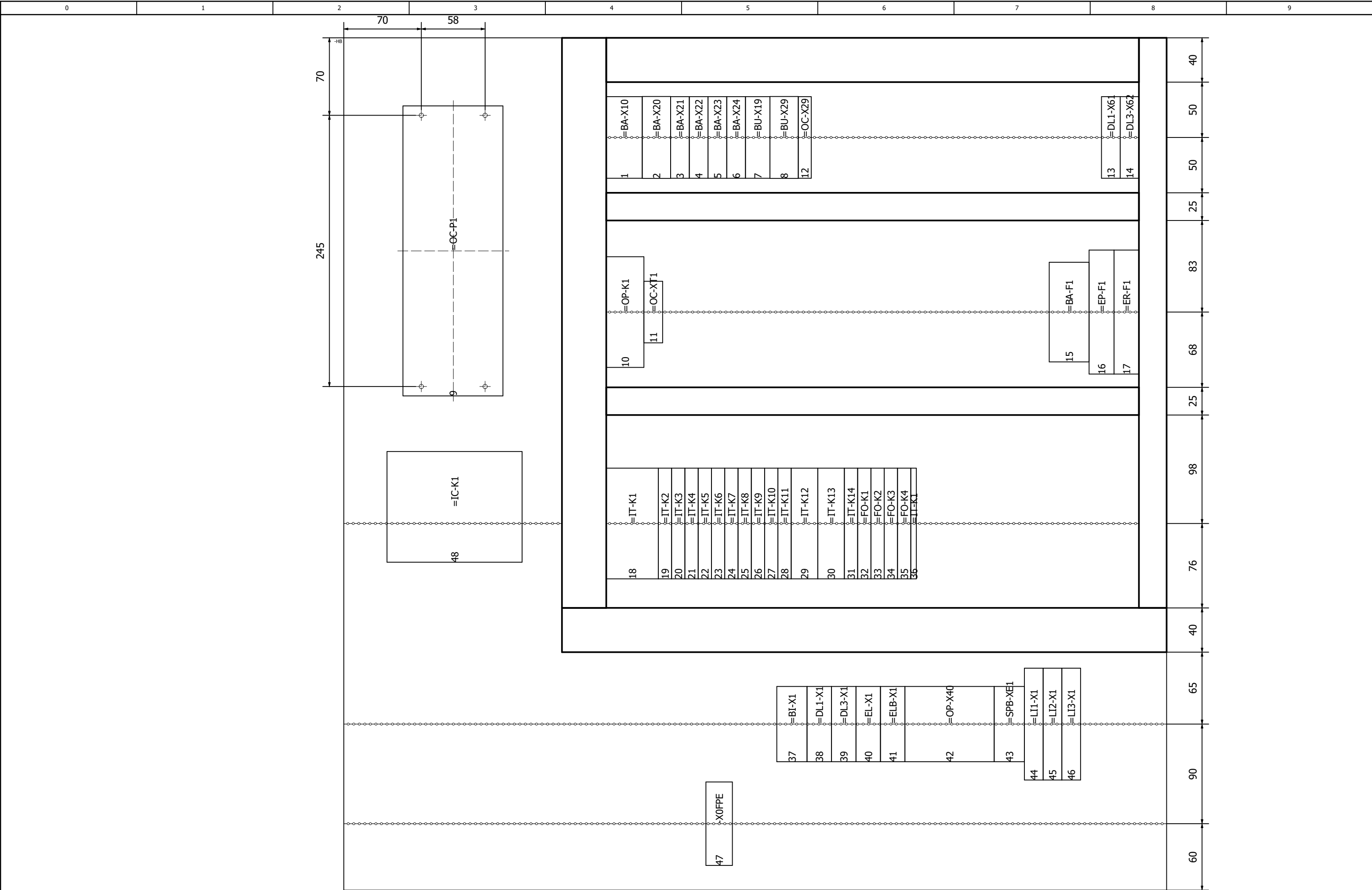
72386104

			Toolbox_88	Option: customer: commission:	0405A Basic Parts TSc - CE {R.76}			electrical connection plan	module	supply connection plan		Page 13
			Toolbox version		Maghreb Steel			connection plan				
			1S - ABI - 34.87.00 - CC: EU		286002000	project: 286002000			sector 2E1P location		+PZ	
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais 24.05.2024							

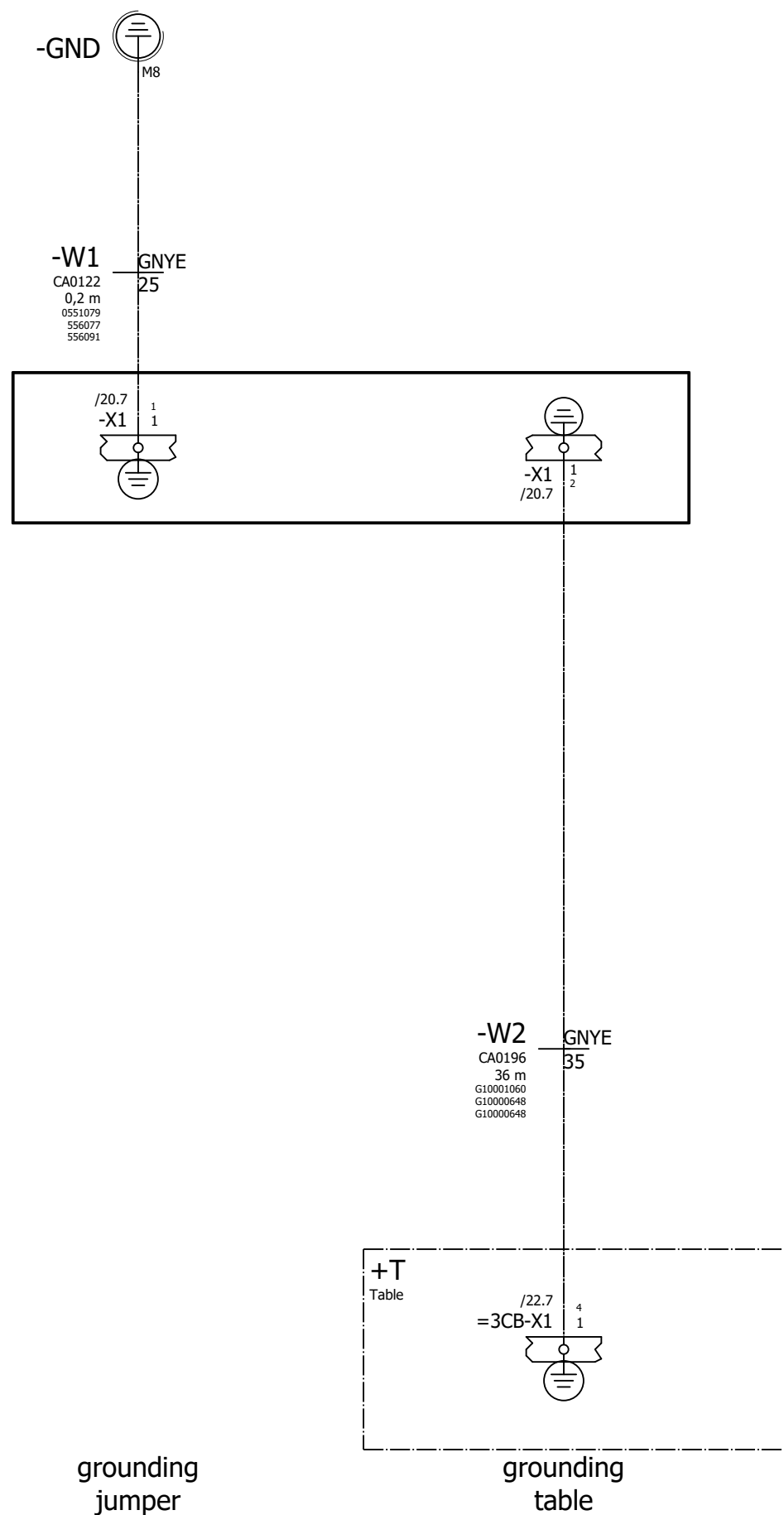




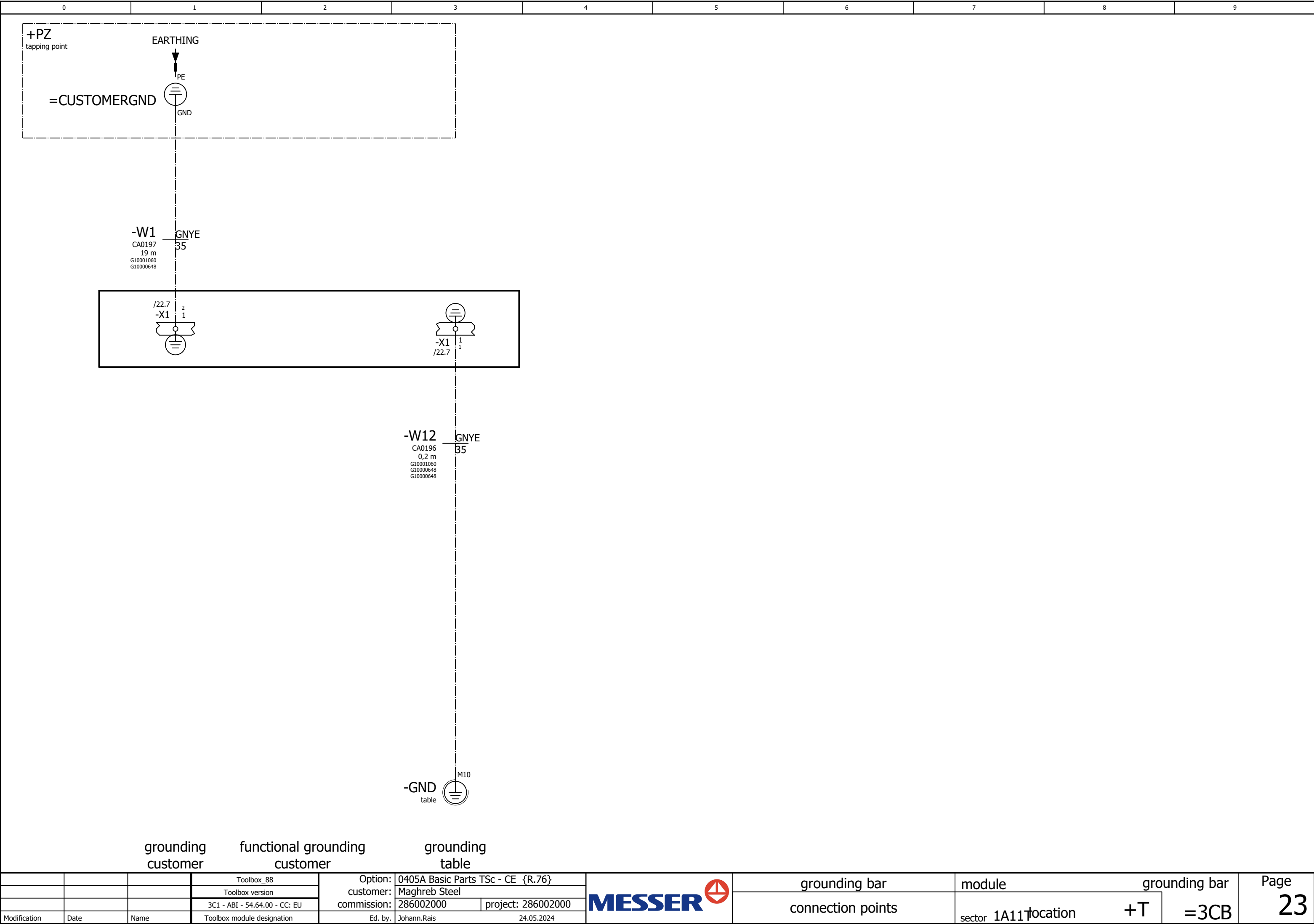




0	1	2	3	4	5	6	7	8	9		
X1 -X1 = potential equalization machine jumper 1 - grounding jumper / /21.1G00000775 2 - grounding table / /21.3 3 - cabinet / =3B+HB/16.1 4 - cabinet / =3BB+HB/18.2 5 - Mounting panel OmniFlow / =3E/26.1 6 - grounding carriage plate / =3E1+L1/27.1 7 - grounding carriage plate / =3E2+L2/28.1 8 - grounding carriage plate / =3E3+L3/29.1 9 - panel functional ground / =OP+G/124.8 10 - / 11 - / 12 - / 											
			Toolbox_88	Option:	0405A Basic Parts TSc - CE {R.76}		grounding bar		module	grounding bar	Page
			Toolbox version	customer:	Maghreb Steel		connection overview				20
			3C4 - ABI - No Rev - CC: EU	commission:	286002000	project: 286002000			sector	1A1B location	
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	24.05.2024			+B	=3C	



0	1	2	3	4	5	6	7	8	9	
X1 -X1 = potential equalization machine jumper • 1 - grounding table / /23.3G00000775 • 2 - grounding customer / /23.1 • 3 - track 2 / =3DB/25.9 • 4 - grounding table / =3C+B/21.3 • 5 - track 2 / =3DB/25.9 • 6 - / • 7 - / • 8 - / • 9 - / • 10 - / • 11 - / • 12 - / 										
			Toolbox_88	Option:	0405A Basic Parts TSc - CE {R.76}		MESSER	grounding bar		
			Toolbox version	customer:	Maghreb Steel				module	
			3C1 - ABI - No Rev - CC: EU	commission:	286002000	project: 286002000		connection overview	grounding bar	
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	24.05.2024		sector 1A11	location +T =3CB	



			Toolbox_88	Option:	0405A Basic Parts TSc - CE {R.76}		MESSER	grounding bar	module	grounding bar	Page	
			Toolbox version	customer:	Maghreb Steel			connection points	sector 1A11	location +T		=3CB
			3C1 - ABI - 54.64.00 - CC: EU	commission:	286002000	project: 286002000						
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	24.05.2024					23	

0	1	2	3	4	5	6	7	8	9
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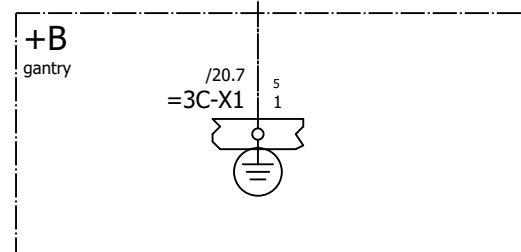
Me

-W10

CA0079
4 m
77070130
556018
556027

FPE

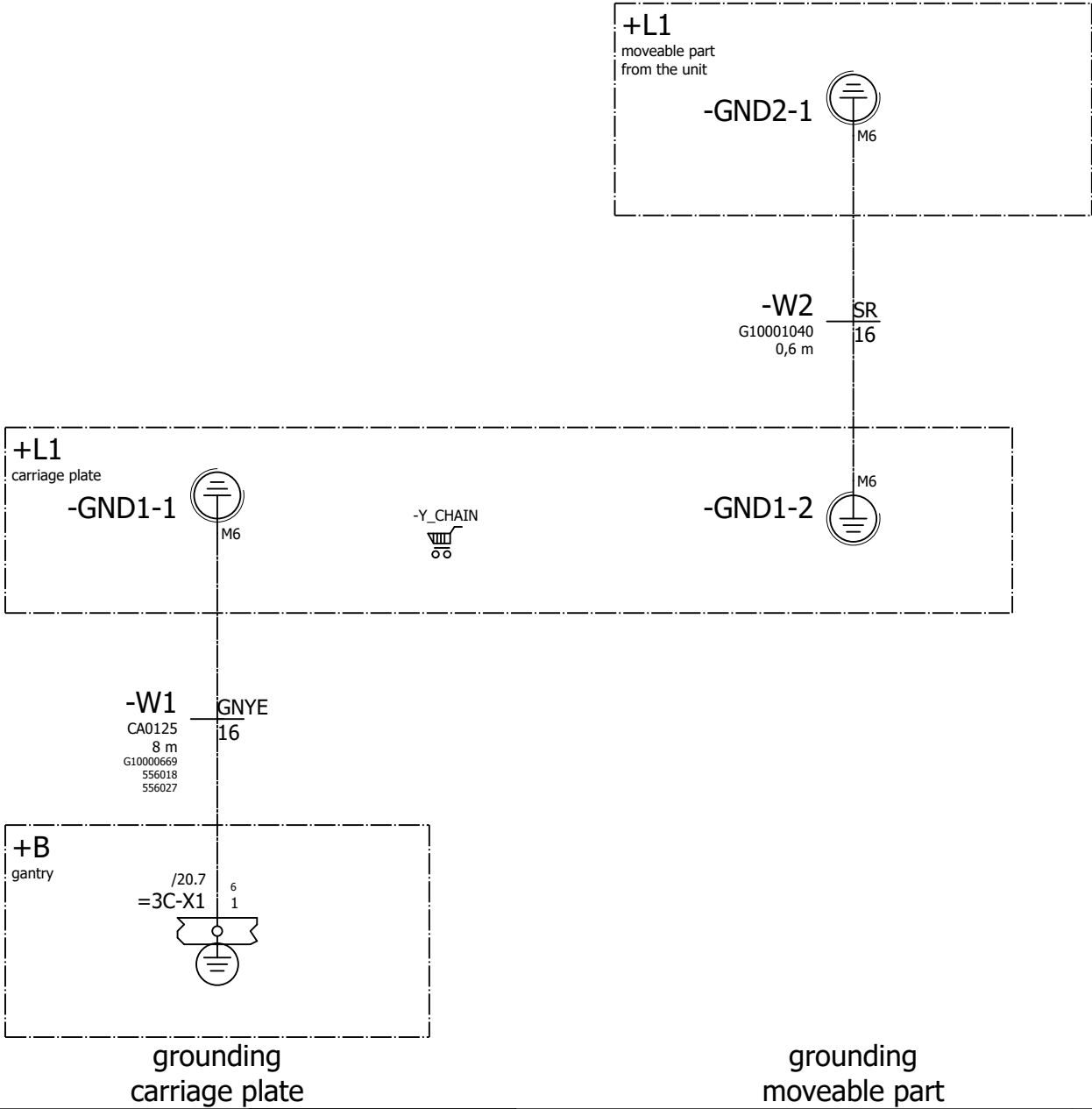
16

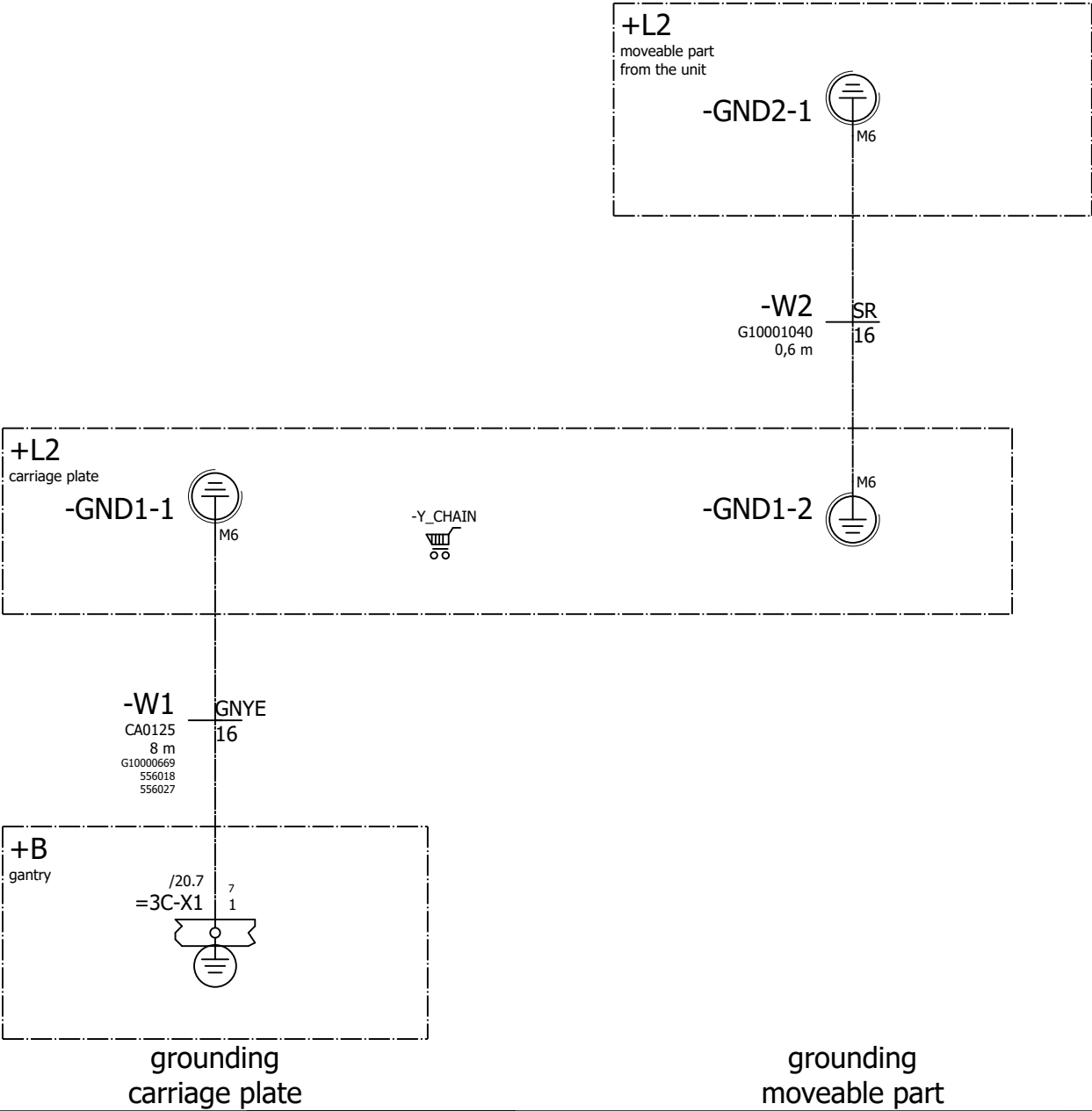


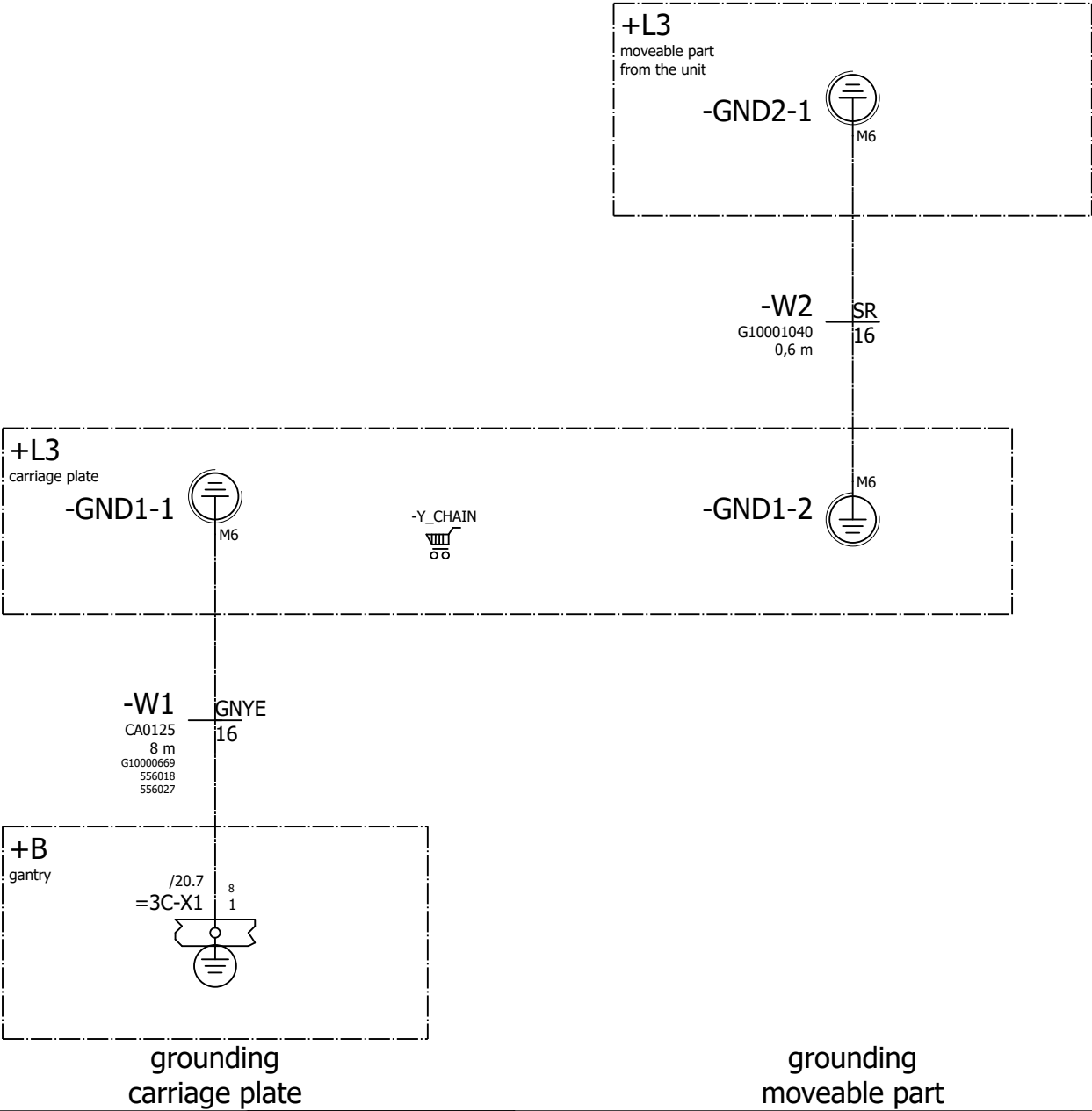
Mounting panel
OmniFlow

-Y_CHAIN

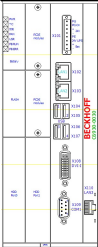
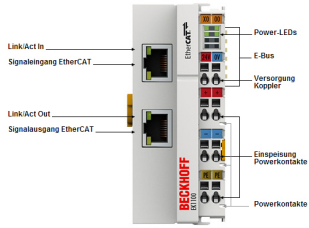
			Toolbox_88	Option:	1702A Single OmniFlow - Supply A			grounding Location	module	grounding elements	Page 26
			Toolbox version	customer:	Maghreb Steel			connection point sector 1B2B location +B =3E			
			3E1 - ABI - 33.51.00 - CC: EU	commission:	286002000	project: 286002000					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024					







EtherCAT-Topology

+HB		Schrank Haupt cabinet main																																
	=OC-P1	PC system	cabinet, DVI, USV, SSD, Win10 LTSC	G30000162	X110 LAN 1																													
		=OC+HB/118.0																																
Safety address: Hex<c8> / Dez<200>																																		
+HB		Schrank Haupt cabinet main																																
	=CXXa-K1	controller	AX5206, 2 channel, AC 2x 6A, FW >=2.10	G10001010	X04 Bus-IN X05 Bus-OUT																													
		ON <table border="1" data-bbox="599 823 848 898"> <tr><td></td><td></td><td></td><td>X</td><td></td><td></td><td>X</td><td>X</td><td></td><td></td></tr> <tr><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr> <tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr> </table>				X			X	X			X	X	X	X	X	X	X	X	X	X	1	2	3	4	5	6	7	8	9	10	=CXXa+HB/66.0	
			X			X	X																											
X	X	X	X	X	X	X	X	X	X																									
1	2	3	4	5	6	7	8	9	10																									
Safety address: Hex<c9> / Dez<201>																																		
+HB		Schrank Haupt cabinet main																																
	=CY2Y3-K1	controller	AX5203, 2 channel, AC 2x 3A, FW >=2.10	G10001008	X04 Bus-IN X05 Bus-OUT																													
		ON <table border="1" data-bbox="599 1218 848 1293"> <tr><td>X</td><td></td><td></td><td>X</td><td></td><td></td><td>X</td><td>X</td><td></td><td></td></tr> <tr><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr> <tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr> </table>	X			X			X	X			X	X	X	X	X	X	X	X	X	X	1	2	3	4	5	6	7	8	9	10	=CY2Y3+HB/68.0	
X			X			X	X																											
X	X	X	X	X	X	X	X	X	X																									
1	2	3	4	5	6	7	8	9	10																									
Safety address: Hex<d0> / Dez<202>																																		
+HB		Schrank Haupt cabinet main																																
	=IT-K1	bus terminal	bus coupler	G13000078	X1 IN X2 OUT																													
		=IT+HB/97.1																																

EtherCAT -Topologie

ECAT

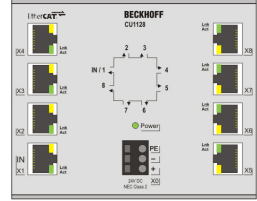
+HB=IT-K1:X2:BUS 2

=IC++4B2B+HB/96

+HB

Schrank Haupt
cabinet main

Introduction



=IC-K1

Verteiler aktiv
distributor active

EtherCAT Sternverteiler, 8x RJ45
EtherCAT star hub, 8x RJ45

=IC+HB/96.0

G10000188



X1
Bus

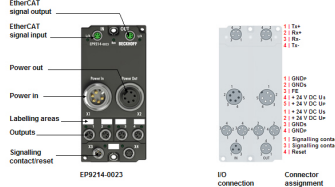


X8
Bus

+B

Brücke
gantry

EP9214 - Introduction



=BI-K1

Verteiler Power
distributor power

EtherCAT-Box, Powerverteilung, M8, 4 x 4A
EtherCAT-box, power distribution, M8, 4 x 4A

=BI+B/61.0

G10000103



L/A
Bus



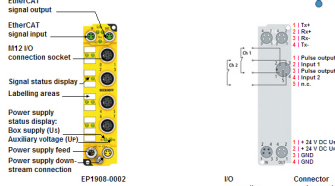
L/A
Bus

+B

Brücke
gantry

Safety address: Hex<8> / Dez<8>

Digital Input



=IBB-K1

Verteiler aktiv
distributor active

EtherCAT Safety Box, IN, 4 Kanal, M12
EtherCAT Safety box, IN, 4 channel, M12

=IBB+B/94.0

G10000100



L/A
Bus



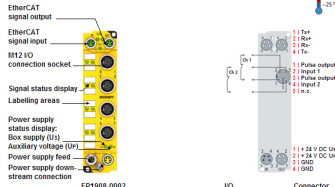
L/A
Bus

+B

Brücke
gantry

Safety address: Hex<9> / Dez<9>

Digital Input



=IB-K1

Verteiler aktiv
distributor active

EtherCAT Safety Box, IN, 4 Kanal, M12
EtherCAT Safety box, IN, 4 channel, M12

=IB+B/92.0

G10000100



L/A
Bus



L/A
Bus

Toolbox_88

Option:

Toolbox version

customer:

Maghreb Steel

automated

commission:

286002000

project:

286002000

Modification

Date

Name

evaluation has been generated

Ed. by:

Johann.Rais

24.05.2024

MESSER

EtherCAT cable

module

topology EtherCAT

sector

location

+A

=4A

Page

31

Johann.Rais

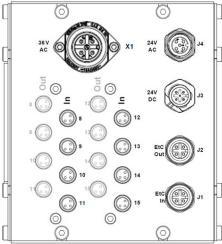
24.05.2024

EtherCAT -Topologie

ECAT_2

+L1

Lifter



=LI1-K1

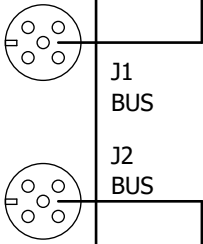
Höhenverstellung
lifter

=LI1+L1/107.1

SmartLifter 200, Plasma/ARKI/PAN, 2 x 4 DI, 2 x 4 DO

SmartLifter 200, plasma/ARKI/PAN, 2 x 4 DI, 2 x 4 DO

72163096



EtherCAT -Topologie

ECAT_3

+L2

Lifter

=LI2-K1

Höhenverstellung
lifter

=LI2+L2/109.1

SmartLifter 200, HD Plasma/ARKI/PAN, 2 x 4 DI, 2 x 4 DO

SmartLifter 200, HD plasma/ARKI/PAN, 2 x 4 DI, 2 x 4 DO

72163099

J1
BUS

J2
BUS

EtherCAT -Topologie

ECAT_4

+L3

Lifter

=LI3-K1

Höhenverstellung
lifter

=LI3+L3/111.1

SmartLifter 200, HD Plasma/ARKI/PAN, 2 x 4 DI, 2 x 4 DO

SmartLifter 200, HD plasma/ARKI/PAN, 2 x 4 DI, 2 x 4 DO

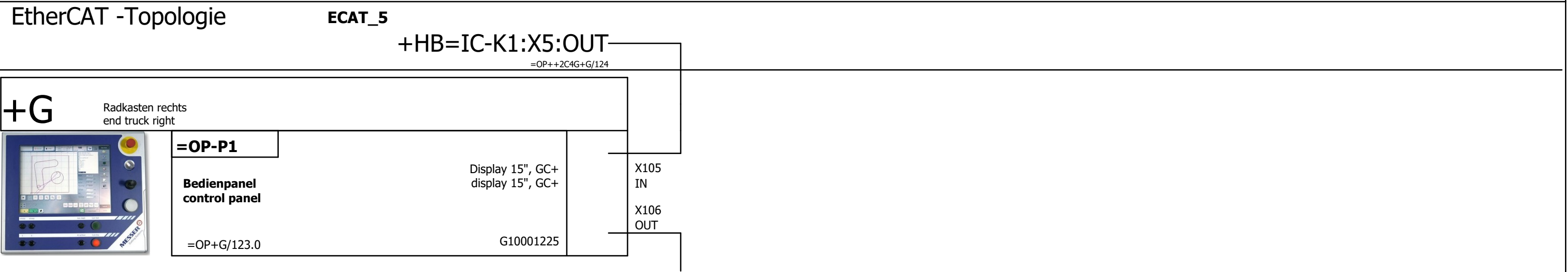
72163099

J1
BUS

J2
BUS

			Toolbox_88	Option: customer: commission:			MESSER		module	topology	EtherCAT	Page 34	
			Toolbox version		Maghreb Steel	project: 286002000		EtherCAT cable	sector	location	+A		=4A
			automated		286002000								
Modification	Date	Name	evaluation has been generated	Ed. by.	Johann.Rais	24.05.2024							

Johann.Rais
24.05.2024



+HB=IT-K1	+HB=IT-K2	+HB=IT-K3	+HB=IT-K4	+HB=IT-K5	+HB=IT-K6	+HB=IT-K7	+HB=IT-K8	+HB=IT-K9	+HB=IT-K10
G13000078 EK1100	G10001248 EL1904	G10001248 EL1904	G10001248 EL1904	G10001248 EL1904	G10000699 EL9184	G13000079 EL9410	G10000441 EL1859	G10001248 EL1904	G10001248 EL1904
	Twin Safe 	Twin Safe 	Twin Safe 	Twin Safe 				Twin Safe 	Twin Safe
=IT+HB/100.1	=IT+HB/102.1	=IT+HB/102.6	=IT+HB/103.1	=IT+HB/103.6	=IT+HB/104.1	=IT+HB/101.0	=IT+HB/104.6	=IT+HB/105.1	=IT+HB/105.6

Safety address: Dez. <11>

Safety address: Dez. <13>

Safety address: Dez. <4>

Safety address: Dez. <1>

Safety address: Dez. <37>

Safety address: Dez. <36>

+HB=IT-K11	+HB=IT-K12	+HB=IT-K13	+HB=IT-K14	+HB=FO-K1	+HB=FO-K2	+HB=FO-K3	+HB=FO-K4																																																																																																																		
G10001248 EL1904	G10001255 EL2904	G10001255 EL2904	G10001262 EL6900	G13000079 EL9410	G10001249 EL2008	G10000492 EL1002	G10001260 EL6021																																																																																																																		
Twin Safe ON <table><tr><td></td><td></td><td>X</td><td>X</td><td></td><td></td><td></td><td></td></tr><tr><td>X</td><td>X</td><td></td><td></td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr></table>			X	X					X	X			X	X	X	X	1	2	3	4	5	6	7	8	Twin Safe ON <table><tr><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr></table>	X	X	X											X	X	X	X	X	X	X	1	2	3	4	5	6	7	8	9	10	Twin Safe ON <table><tr><td></td><td></td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>X</td><td></td><td></td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr></table>			X	X							X			X	X	X	X	X	X	X	1	2	3	4	5	6	7	8	9	10	Twin Safe ON <table><tr><td></td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr></table>		X									X	X	X	X	X	X	X	X	X	X	1	2	3	4	5	6	7	8	9	10				
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24L-	24L-	24L-	24L-	24L-	24L-	24L-	24L-																																																																																																																		
				PE		PE																																																																																																																			
=IT+HB/106.1	=IT+HB/106.6	=IT+HB/107.1	=IT+HB/107.6	=FO+B/86.2	=FO+B/87.8	=FO+B/88.3	=FO+B/88.6																																																																																																																		

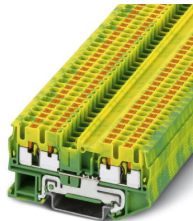
Safety address: Dez. <12> Safety address: Dez. <7> Safety address: Dez. <6> Safety address: Dez. <2>

0	1	2	3	4	5	6	7	8	9												
X1  jumper <table><tr><td>•</td><td>1</td><td>77070157</td></tr><tr><td colspan="3">a - operating voltage /43.8</td></tr><tr><td colspan="3">b - cabinet power supply /43.2</td></tr><tr><td colspan="3">c - service power socket =AS/52.3</td></tr></table>										•	1	77070157	a - operating voltage /43.8			b - cabinet power supply /43.2			c - service power socket =AS/52.3		
•	1	77070157																			
a - operating voltage /43.8																					
b - cabinet power supply /43.2																					
c - service power socket =AS/52.3																					
			Toolbox_88	Option:	1		module AC power supply cabinet		Page												
			Toolbox version	customer:	Maghreb Steel				38												
				commission:	286002000		project: 286002000														
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	24.05.2024	overview terminal strip	sector location +HB=AA													

0	1	2	3	4	5	6	7	8	9									
X3  jumper <table><tr><td>•</td><td>1</td><td>77070157</td></tr><tr><td></td><td>a - operating voltage</td><td>/43.8</td></tr><tr><td></td><td>b - cabinet power supply</td><td>/43.2</td></tr></table>										•	1	77070157		a - operating voltage	/43.8		b - cabinet power supply	/43.2
•	1	77070157																
	a - operating voltage	/43.8																
	b - cabinet power supply	/43.2																
			Toolbox_88	Option:	3	MESSER			module	AC power supply cabinet	Page							
			Toolbox version	customer:	Maghreb Steel						40							
				commission:	286002000			project: 286002000	overview terminal strip									
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	24.05.2024		sector	location	+HB	=AA							

0	1	2	3	4	5	6	7	8	9												
XN1  jumper <table><tr><td>•</td><td>1</td><td>G10000679</td></tr><tr><td colspan="3">a - cabinet power supply /43.0</td></tr><tr><td colspan="3">b - operating voltage /43.7</td></tr><tr><td colspan="3">c - service power socket =AS/52.3</td></tr></table>										•	1	G10000679	a - cabinet power supply /43.0			b - operating voltage /43.7			c - service power socket =AS/52.3		
•	1	G10000679																			
a - cabinet power supply /43.0																					
b - operating voltage /43.7																					
c - service power socket =AS/52.3																					
			Toolbox_88	Option:	4		module		AC power supply cabinet		Page										
			Toolbox version	customer:	Maghreb Steel		overview terminal strip	sector		location	+HB	=AA	41								
				commission:	286002000									project: 286002000							
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	24.05.2024															

XPE



jumper

1	77070158
a - cabinet power supply /43.0	
b - service power socket =AS/52.5	
c - OmniFlow operating voltage / PE =FO+B/85.2	
2	77070158
a - star hub supply / PE =IC/96.2	
b - panel power supply =OP+G/124.2	

XXX

-K1

P

L1	380	cabinet power supply	+HB/44.2
L2	380	=	+HB/44.2
L3	380	=	+HB/44.2
N	0	=	+HB/44.0
PE	PE	=	+HB/44.0

S

L1	400	cabinet power supply	+HB/44.2
L2	400	=	+HB/44.2
L3	400	=	+HB/44.2
N	0	=	+HB/44.0
PE	PE	=	+HB/44.0

X11



jumper

1	77070157
a - operating voltage =AA/43.8	
b - transformer supply =AL/51.1	
c - controller power supply AC / L1 =CXXa/66.1	
d - controller power supply AC / L1 =CY2Y3/68.1	
2	77070157
a - valve heating oxygen 2 =FS+HG/89.2	

0	1	2	3	4	5	6	7	8	9

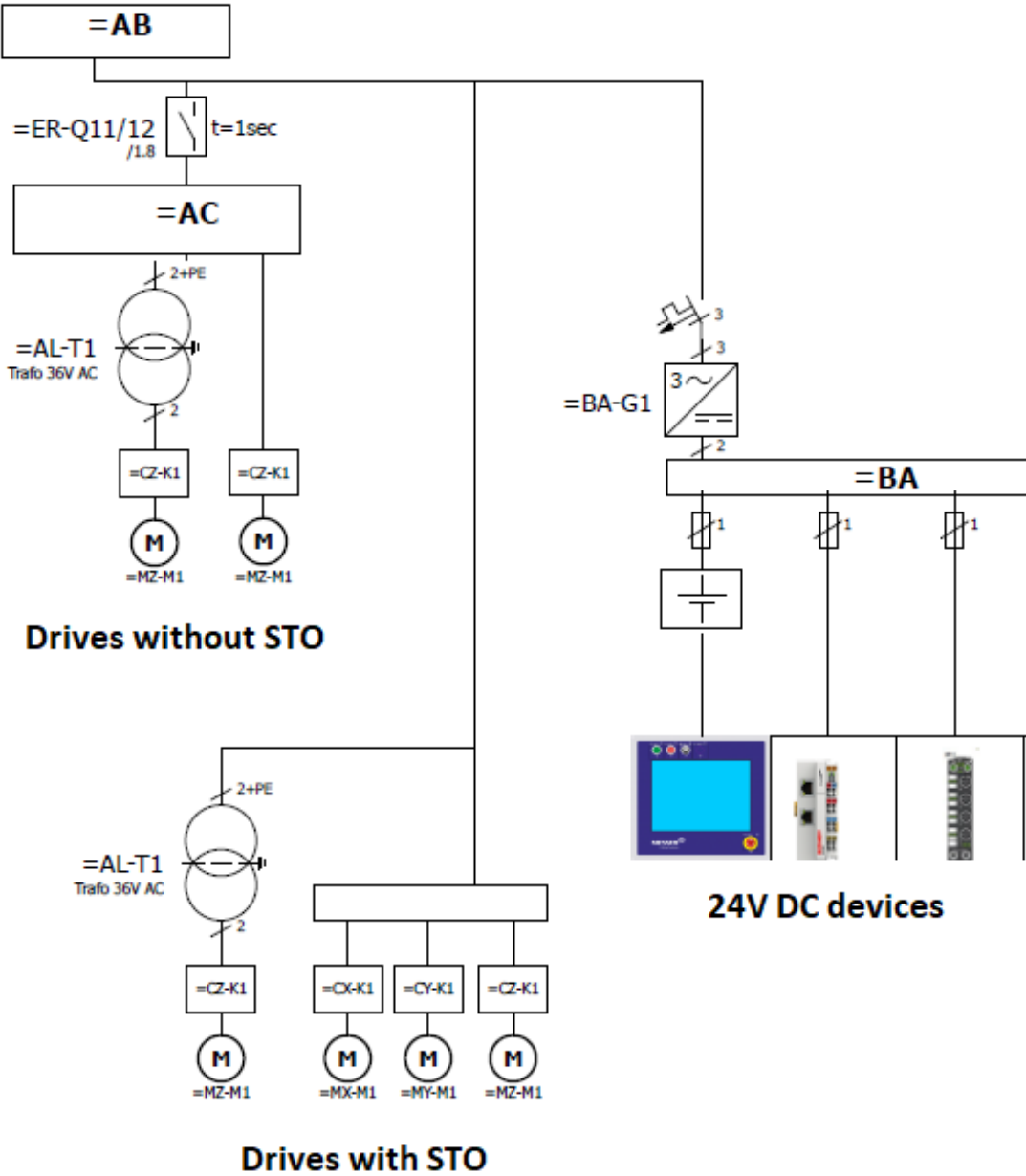
XN10



jumper

1	G10000679
a - operating voltage =AA/43.7	
b - air conditioner supply =UT/143.2	
c - transformer supply / 0 =AL/51.2	
d - operating voltage power supply 24VDC / 230N =BA/57.4	
2	G10000679
a - valve fuel gas 1 =FS+HG/89.3	

AC supply principle





77070862

transformer

-T1 1 phase, 4 x 100 VA, 230V/36V

X1

L	230	transformer supply	/52.1
N	0	=	/52.2
PE	ground	=	/52.2

X2.1

L	36	SmartLifter AC power supply	=LI1+L1/109.3
N	0	=	=LI1+L1/109.2

X2.2

L	36	SmartLifter AC power supply	=LI2+L2/111.3
N	0	=	=LI2+L2/111.2

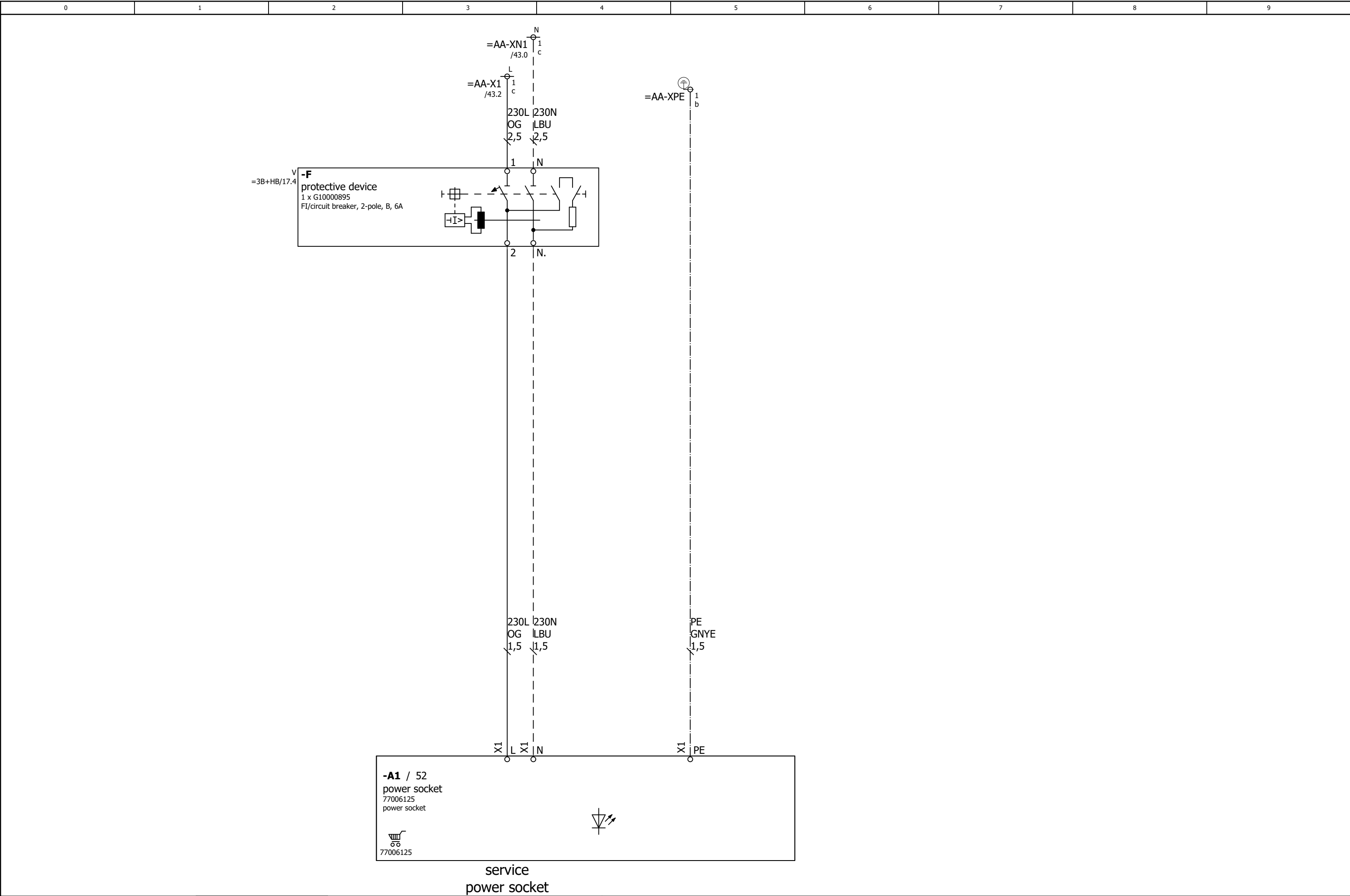
X2.3

L	36	SmartLifter AC power supply	=LI3+L3/113.3
N	0	=	=LI3+L3/113.2

X2.4

L	36
N	0







G10000827

protective device

-F1

electronic circuit breaker 10A

Contact		
13	NO	
14	NO	
IN+		
1	24L+	operating voltage power supply 24V DC
2	24L+	
IN-		
1	24L-	operating voltage power supply 24V DC
2	24L-	
OUT+		
1	1-10A	operating voltage 24L+ channel 1
2	1-10A	operating voltage 24L+ channel 2
3	1-10A	operating voltage 24L+ channel 3
4	1-10A	operating voltage 24L+ channel 4

X21

-X21 = operating voltage 24L+
channel 1



jumper

•	1	77070157
	a - operating voltage 24L+ channel 1 / 1-10A	/58.4
	b - potential distribution limiting Y+ =DL1+L1	/71.3
	c - EtherCAT coupler operating supply / 24L+	=IT/97.5

X22

-X22 = operating voltage 24L+
channel 2

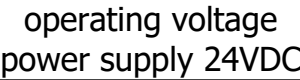


jumper

•	1	77070157
	a - operating voltage 24L+ channel 2 / 1-10A	/58.6
	b - locator limit Y- =DL3+L3/73.3	
	c - EtherCAT coupler powerbus / + =IT/97.6	

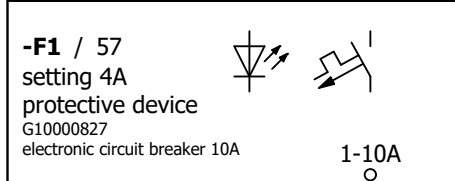
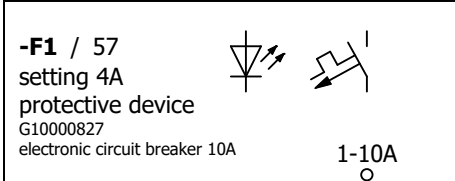
			Toolbox_88	Option:	1102C Main Cabi supply M+N switched			operating voltage		module		DC power supply		Page
			Toolbox version	customer:	Maghreb Steel			connection overview	sector	4B2B location	+HB	=BA	55	
			BA1 - ABI - No Rev - CC: EU	commission:	286002000	project: 286002000								
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024								

0	1	2	3	4	5	6	7	8	9							
X23 -X23 = operating voltage 24L+ channel 3  jumper • 1 77070157 a - operating voltage 24L+ channel 3 / 1-10A /59.4 b - light barrier receiver =EL+E/78.2 c - OmniFlow operating voltage / 24L+ =FO+B/85.0 d - EtherCAT operating supply / 24L+ =IT/98.5 X24 -X24 = operating voltage 24L+ channel 4  jumper • 1 77070157 a - operating voltage 24L+ channel 4 / 1-10A /59.6 b - light barrier receiver =ELB+E/79.2 c - star hub supply / 24L+ =IC/96.1 d - EtherCAT powerbus / 24L+ =IT/98.6																
			Toolbox_88		Option: customer: commission:	1102C Main Cabi supply M+N switched		operating voltage		module		DC power supply		Page 56		
			Toolbox version			Maghreb Steel										
			BA1 - ABI - No Rev - CC: EU			286002000	project: 286002000	connection overview		sector 4B2B location		+HB	=BA			
Modification	Date	Name	Toolbox module designation		Ed. by.	Johann.Rais										
				Ed. by.			MESSER									



			Toolbox_88	Option: customer: commission:	1102C Main Cabi supply M+N switched		MESSER	operating voltage	module	DC power supply	Page 57
			Toolbox version		Maghreb Steel			power supply	sector 4B2B location	+HB =BA	
			BA1 - ABI - 34.80.00 - CC: EU		286002000 project: 286002000						
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais						

0	1	2	3	4	5	6	7	8	9
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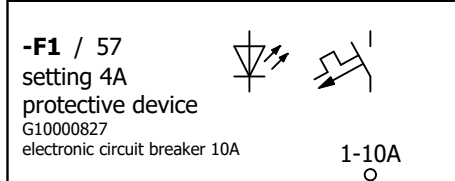
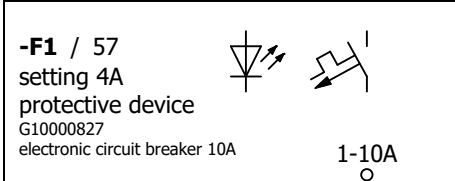


operating voltage 24L+
channel 1

operating voltage 24L+
channel 2

			Toolbox_88	Option:	1102C Main Cabi supply M+N switched			operating voltage	module	DC power supply	Page 58
			Toolbox version	customer:	Maghreb Steel			electronic fuse channel 1 / 2	sector 4B2B location	+HB =BA	
			BA1 - ABI - 34.80.00 - CC: EU	commission:	286002000	project: 286002000					
Modification	Date	Name	Toolbox module designation	Ed. bv.	Johann.Rais	24.05.2024					

0	1	2	3	4	5	6	7	8	9
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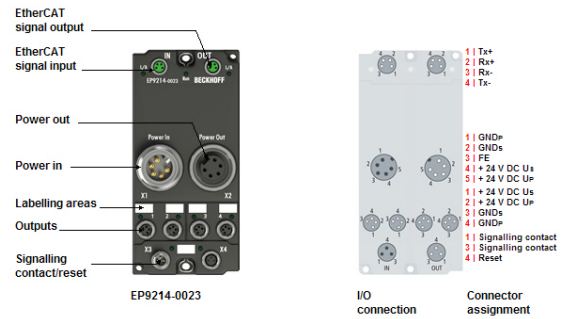


operating voltage 24L+
channel 3

operating voltage 24L+
channel 4

			Toolbox_88	Option:	1102C Main Cabi supply M+N switched			operating voltage	module	DC power supply	Page 59
			Toolbox version	customer:	Maghreb Steel			electronic fuse channel 3 / 4	sector 4B2B location +HB	=BA	
			BA1 - ABI - 34.80.00 - CC: EU	commission:	286002000	project: 286002000					
Modification	Date	Name	Toolbox module designation	Ed. bv.	Johann.Rais	24.05.2024					

EP9214 - Introduction



G10000103

distributor power

-K1 EtherCAT-box, power distribution, M8, 4 x 4A

1

OUT 1			
1	24L+s	EP module supply	=IBB/95.1
2	24L+p		= =IBB/95.2
3	24L-s		= =IBB/95.3
4	24L-p		= =IBB/95.3

2

OUT 2			
1	24L+s	SmartLifter DC operating voltage	=LI1+L1/109.5
2	24L+p	SmartLifter DC control voltage	=LI1+L1/109.6
3	24L-s	SmartLifter DC operating voltage	=LI1+L1/109.7
4	24L-p		= =LI1+L1/109.7

3

OUT 3			
1	24L+s	SmartLifter DC operating voltage	=LI2+L2/111.5
2	24L+p	SmartLifter DC control voltage	=LI2+L2/111.6
3	24L-s	SmartLifter DC operating voltage	=LI2+L2/111.7
4	24L-p		= =LI2+L2/111.7

4

OUT 4			
1	24L+s	SmartLifter DC operating voltage	=LI3+L3/113.5
2	24L+p	SmartLifter DC control voltage	=LI3+L3/113.6
3	24L-s	SmartLifter DC operating voltage	=LI3+L3/113.7
4	24L-p		= =LI3+L3/113.7

L/A

Bus			
IN	Bus	EtherCAT	/62.8
OUT	Bus	=	/62.8

X1

Power IN			
1	24L-s	distribution 24 VDC external	/62.1
2	24L-p		= /62.2
3	PE		= /62.3
4	24L+s		= /62.3
5	24L+p		= /62.4

X2

Power OUT			
1	24L-s		
2	24L-p		
3	PE		
4	24L+s		
5	24L+p		

X3

diagnosis IN			
1	NO		
3	COM		
4	reset		

X4

diagnosis OUT			
1	NO		
3	COM		
4	reset		



G10000082

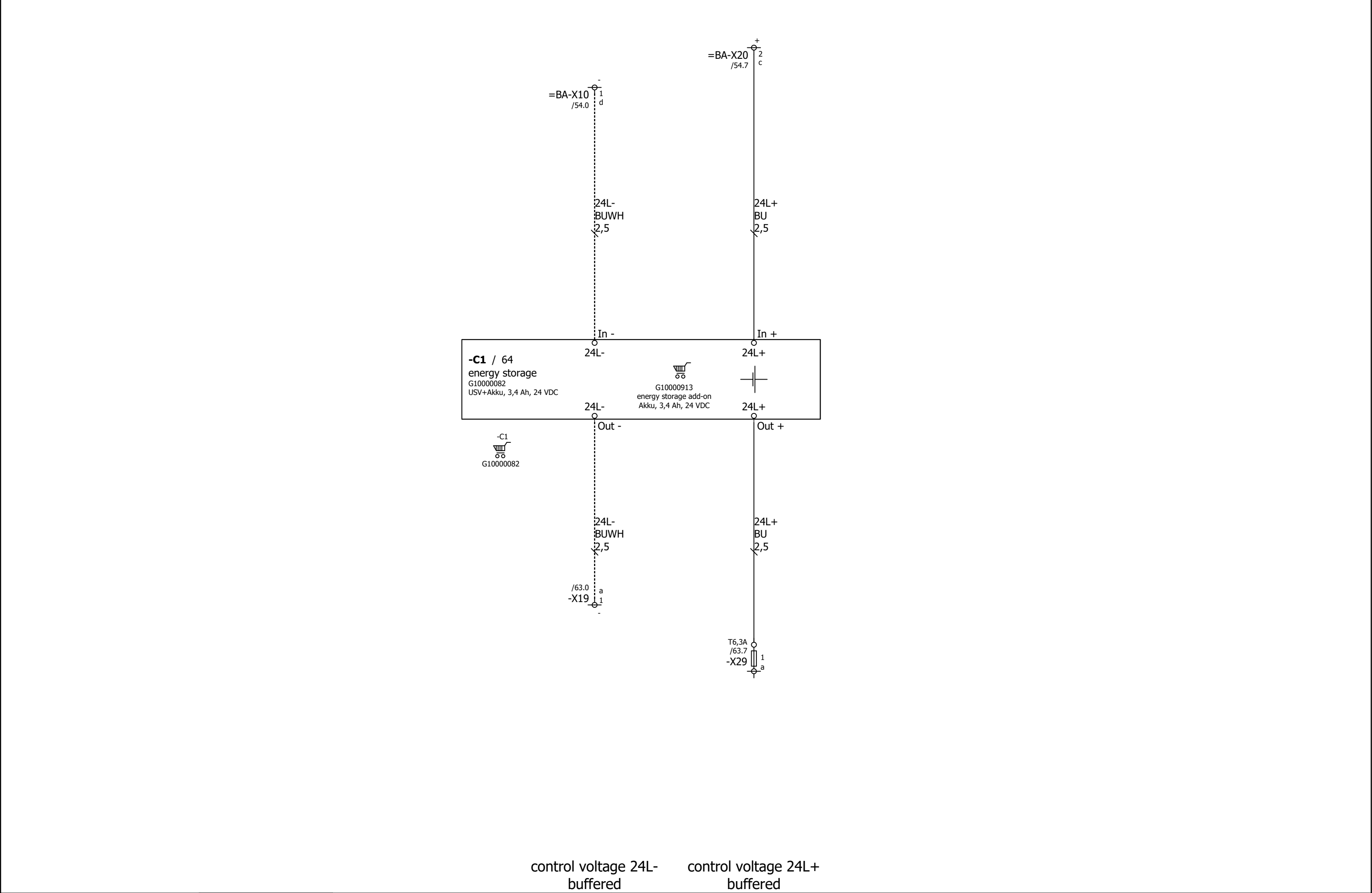
energy storage

-C1

USV+Akkus, 3,4 Ah, 24 VDC

In +	24L+	control voltage 24L+ buffered	/65.5
In -	24L-	control voltage 24Lbuffered	/65.4
Out +	24L+	control voltage 24L+ buffered	/65.5
Out +	24L+ 0,2A		
Out -	24L-	control voltage 24Lbuffered	/65.4
Alarm			
13	Alarm		
14	Alarm		
Bat. Charge			
33	Bat. Charge		
34	Bat. Charge		
Bat. Mode			
23	Bat. Mode		
24	Bat. Mode	PC turn off	=OC/120.5
Remote			
R1	Remote		
R2	Remote in	USV turn off	=OC/120.6

0	1	2	3	4	5	6	7	8	9			
X19 -X19 = jumper 1 77070157 a - control voltage 24Lbuffered / 24L- /64.4 b - controller power supply DC / GND =CXXa/66.4 c - controller power supply DC / GND =CY2Y3/68.4 d - PC operating voltage / 24L- =OC/118.2 2 77070157 a - panel power supply =OP+G/124.1 b - panel power supply =OP+G/124.3 X29 -X29 = control voltage 24L+ buffered jumper 1 G10000602;G10000459;65541022 a - control voltage 24L+ buffered / 24L+ /64.5 b - panel power supply =OP+G/124.3 c - panel power supply =OP+G/124.1 2 G10000602;G10000506 a - PC operating voltage / 24L+ =OC/118.2 b - controller supply DC / Us+ =CXXa/66.4 c - controller supply DC / Us+ =CY2Y3/68.4												
			Toolbox_88	Option:	0405A Basic Parts TSc - CE {R.76}		MESSER	control voltage buffered	module	DC USV	Page	
			Toolbox version	customer:	Maghreb Steel			connection overview	sector 4B2B location +HB	=BU	63	
			BU2 - ABI - No Rev - CC: EU	commission:	286002000	project: 286002000						
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais	24.05.2024						





G10001010

controller

-K1

AX5206, 2 channel, AC 2x 6A,
FW >=2.10

GND

grounding		
M6	ground	controller grounding /67.9

Safety

PCI		
PCI	Bus	controller Safety Card /67.5

X01

supply grid		
1	L1	controller power supply AC /67.1
2	L2	= /67.2
3	L3	= /67.2
PE	GND	controller ground /67.2

X02

DC Link		
1	DC+	
2	DC-	

X03

supply DC		
1	Up+	controller power supply DC /67.3
2	Us+	controller supply DC /67.4
3	GND	controller power supply DC /67.4

X04

Bus		
1-4	Bus-IN	controller EtherCAT /67.8

X05

1-4	Bus-OUT	controller EtherCAT /67.8
-----	---------	---------------------------

X06

digital I/O's	
0	input
0V	24L-
1	input
2	input
3	input
4	input
5	input
6	input
7	in/out
24	24L+

X11

Hiperface	
1	SIN+
2	GND-9V
3	COS+
4	n.c.
5	Data+
6	Us-9V
7	n.c.
8	n.c.
9	REFSIN
10	n.c.
11	REF COS
12	n.c.
13	Data-
14	n.c.
15	n.c.
SH	shield

X12

Resolver	
1	Temp.
2	AGND
3	COS-
4	SIN-
5	REF-
6	n.c.
7	n.c.
8	n.c.
9	Temp.GND
10	COS+

X12

11	SIN+
12	REF+
13	n.c.
14	n.c.
15	n.c.
SH	shield

X13

Motor	
PE	GND motor coil =MX+E/114.2
U	Motor = =MX+E/114.1
V	Motor = =MX+E/114.2
W	Motor = =MX+E/114.2

X13/X14

SH	shield connection shield =MX+E/114.6
----	--------------------------------------

X14

Temp./brake	
B+	brake + connection # =MX+E/114.3
B-	brake GND = =MX+E/114.4
PE	shield
T+	Temp.+ connection encoder =MX+E/114.5
T-	Temp.- = =MX+E/114.5

X21

Hiperface	
1	SIN+
2	GND-9V
3	COS+
4	n.c.
5	Data+
6	Us-9V
7	n.c.
8	n.c.
9	REFSIN
10	n.c.
11	REF COS
12	n.c.
13	Data-
14	n.c.
15	n.c.
SH	shield

X22

Resolver	
1	Temp.
2	AGND
3	COS-
4	SIN-
5	REF-
6	n.c.
7	n.c.
8	n.c.
9	Temp.GND
10	COS+
11	SIN+
12	REF+
13	n.c.
14	n.c.
15	n.c.
SH	shield

X23

Motor	
PE	GND motor coil =MXa+G/115.2
U	Motor = =MXa+G/115.1
V	Motor = =MXa+G/115.2
W	Motor = =MXa+G/115.2

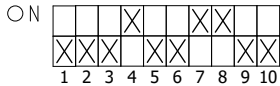
X23/X24

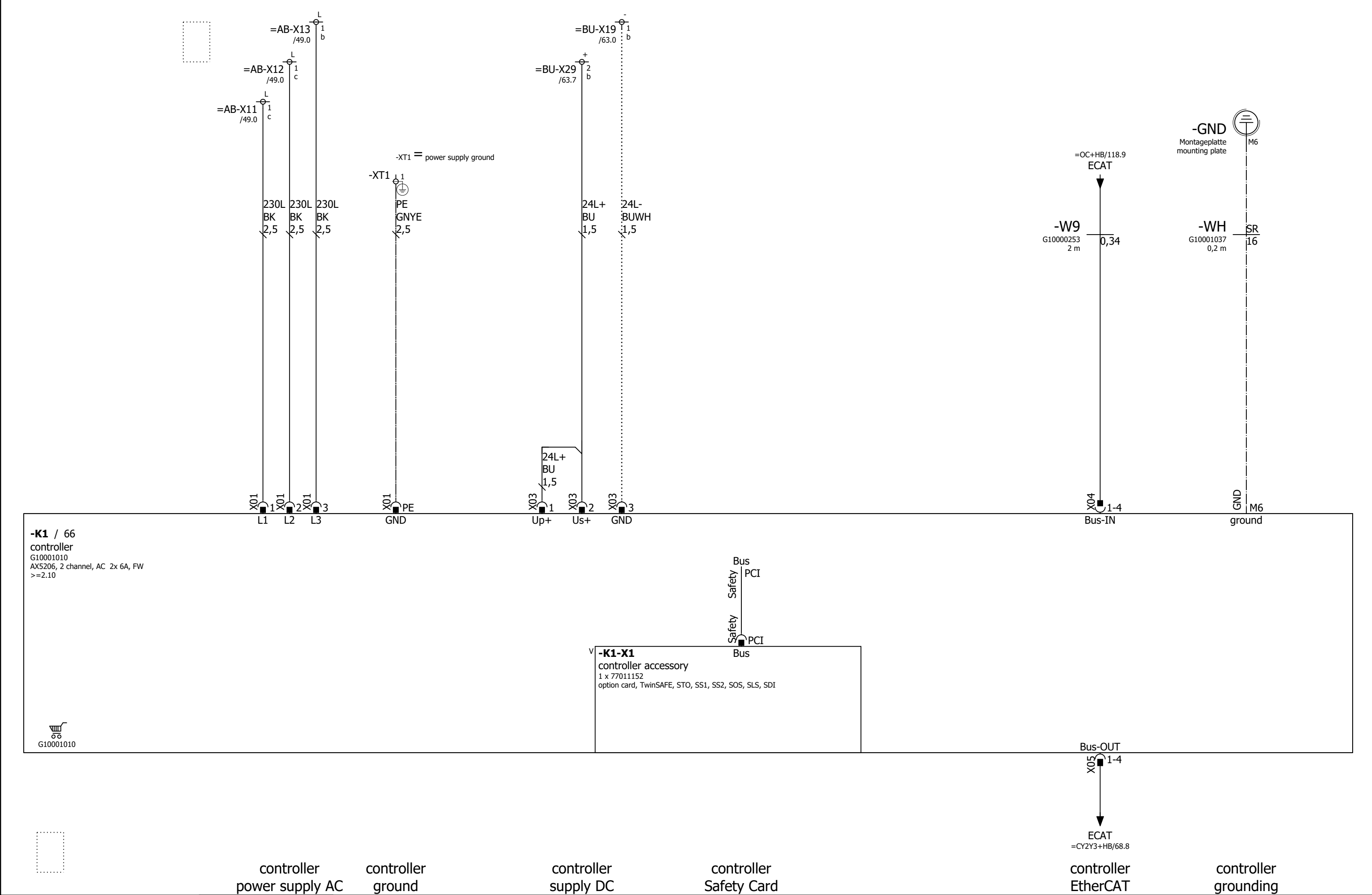
SH	shield connection shield =MXa+G/115.6
----	---------------------------------------

X24

Temp./brake	
B+	brake + connection # =MXa+G/115.3
B-	brake GND = =MXa+G/115.4
PE	shield
T+	Temp.+ connection encoder =MXa+G/115.5
T-	Temp.- = =MXa+G/115.5

Safety adress: HEX -<c8> / Dez: <200>







G10001008
controller

-K1 AX5203, 2 channel, AC 2x 3A,
FW >=2.10

GND		
	grounding	
M6	ground	controller grounding /69.9

Safety		
	PCI	
PCI	Bus	controller Safety Card /69.5

X01		
	supply grid	
1	L1	controller power supply AC /69.1
2	L2	= /69.2
3	L3	= /69.2
PE	GND	controller ground /69.2

X02		
	DC Link	
1	DC+	
2	DC-	

X03		
	supply DC	
1	Up+	controller power supply DC /69.3
2	Us+	controller supply DC /69.4
3	GND	controller power supply DC /69.4

X04		
	Bus	
1-4	Bus-IN	controller EtherCAT /69.8

X05		
1-4	Bus-OUT	controller EtherCAT /69.8

X06		
	digital I/O's	
0	input	
0V	24L-	
1	input	working area beam Y + /70.2
2	input	working area beam Y - /70.3
3	input	
4	input	
5	input	
6	input	
7	in/out	
24	24L+	

X11		
	Hiperface	
1	SIN+	
2	GND-9V	
3	COS+	
4	n.c.	
5	Data+	
6	Us-9V	
7	n.c.	
8	n.c.	
9	REFSIN	
10	n.c.	
11	REF COS	
12	n.c.	
13	Data-	
14	n.c.	
15	n.c.	
SH	shield	

X12		
	Resolver	
1	Temp.	
2	AGND	
3	COS-	
4	SIN-	
5	REF-	
6	n.c.	
7	n.c.	
8	n.c.	
9	Temp.GND	
10	COS+	

X12		
11	SIN+	
12	REF+	
13	n.c.	
14	n.c.	
15	n.c.	
SH	shield	

X13		
	Motor	
PE	GND	motor coil =MY2+L2/116.2
U	Motor	= =MY2+L2/116.1
V	Motor	= =MY2+L2/116.2
W	Motor	= =MY2+L2/116.2

X13/X14		
SH	shield	connection shield =MY2+L2/116.6

X14		
	Temp./brake	
B+	brake +	connection # =MY2+L2/116.3
B-	brake GND	= =MY2+L2/116.4
PE	shield	
T+	Temp.+	connection encoder =MY2+L2/116.5
T-	Temp.-	= =MY2+L2/116.5

X21		
	Hiperface	
1	SIN+	
2	GND-9V	
3	COS+	
4	n.c.	
5	Data+	
6	Us-9V	
7	n.c.	
8	n.c.	
9	REFSIN	
10	n.c.	
11	REF COS	
12	n.c.	
13	Data-	
14	n.c.	
15	n.c.	
SH	shield	

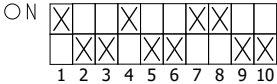
X22		
	Resolver	
1	Temp.	
2	AGND	
3	COS-	
4	SIN-	
5	REF-	
6	n.c.	
7	n.c.	
8	n.c.	
9	Temp.GND	
10	COS+	
11	SIN+	
12	REF+	
13	n.c.	
14	n.c.	
15	n.c.	
SH	shield	

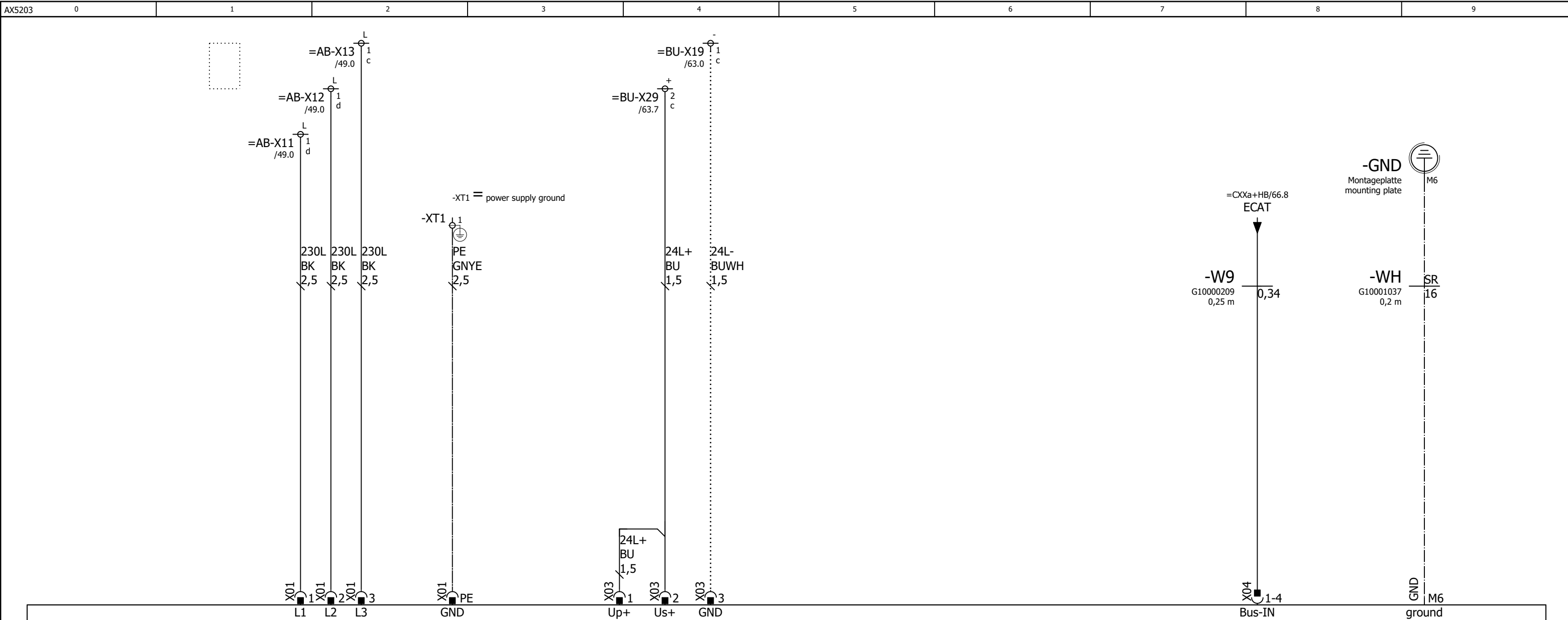
X23		
	Motor	
PE	GND	motor coil =MY3+L3/117.2
U	Motor	= =MY3+L3/117.1
V	Motor	= =MY3+L3/117.2
W	Motor	= =MY3+L3/117.2

X23/X24		
SH	shield	connection shield =MY3+L3/117.6

X24		
	Temp./brake	
B+	brake +	connection # =MY3+L3/117.3
B-	brake GND	= =MY3+L3/117.4
PE	shield	
T+	Temp.+	connection encoder =MY3+L3/117.5
T-	Temp.-	= =MY3+L3/117.5

Safety adress: HEX -<c9> / Dez: <201>

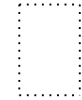




-K1 / 68
controller
G10001008
AX5203, 2 channel, AC 2x 3A, FW
>=2.10



G10001008



controller
power supply AC

controller
ground

controller
supply DC

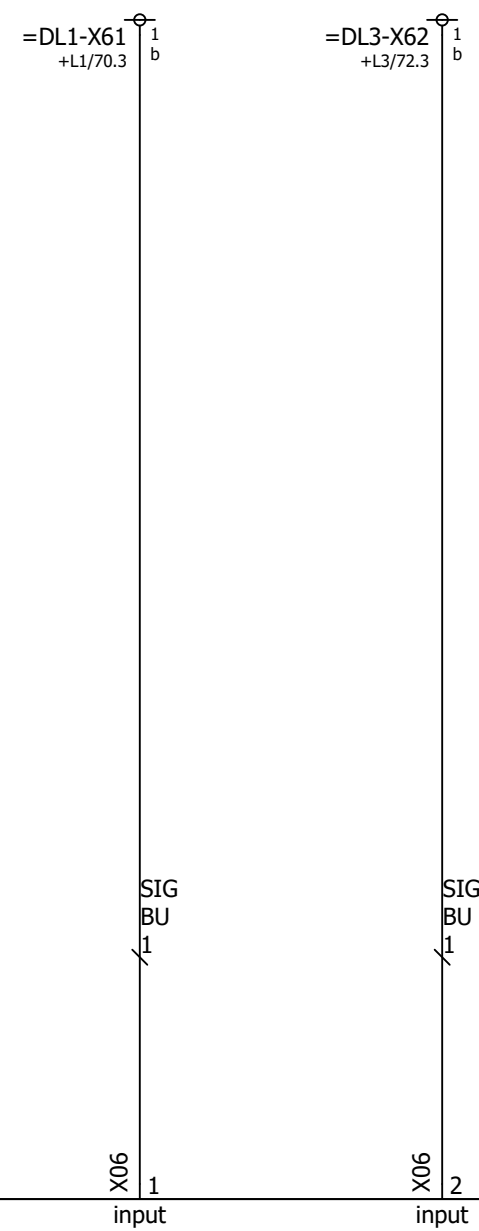
controller
Safety Card

controller
EtherCAT

controller
grounding

			Toolbox_88	Option: customer: commission:	0202G,0202G AX5 Y OCT Safety 3A			AX5203 Y2,Y3		module		controller Y2,Y3		Page 68
			Toolbox version		Maghreb Steel			power supply 400 VAC		sector 4B2B location		+HB =CY2Y3		
			CDA9 - ABI - 41.83.00 - CC: EU		286002000 project: 286002000									
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais 24.05.2024									

0	1	2	3	4	5	6	7	8	9
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=CY2Y3-K1 / 68
controller
G10001008
AX5203, 2 channel, AC 2x 3A, FW
>=2.10

working area beam Y +	working area beam Y -
--------------------------	--------------------------

			Toolbox_88	Option:	0202G L2-AX5 Y OCT Safety 3A {R.53}			AX5203 Y2,Y3	module	controller Y2,Y3	Page 69
			Toolbox version	customer:	Maghreb Steel			entrances	sector 4B2B location	+HB =CY2Y3	
			CX2* - ABI - 31.32.00 - CC: EU	commission:	286002000	project: 286002000					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024					

+HB

cabinet

X61

-X61 = potential distribution limit Y+



jumper

•	1	77070157
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a - potential distribution limiting Y+ /71.6

b - working area beam Y + / input =CY2Y3+HB/69.2

+HB

cabinet

X62

-X62 = potential distribution limit Y-

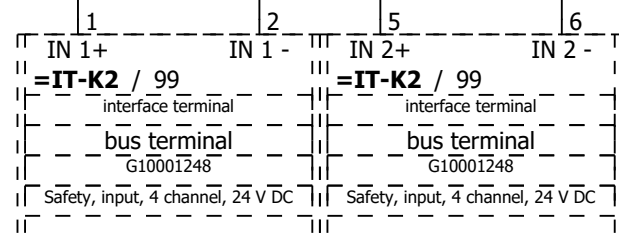
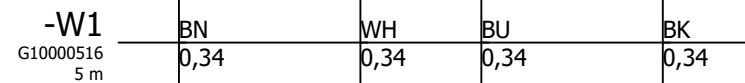
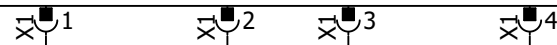
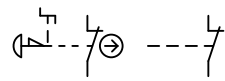
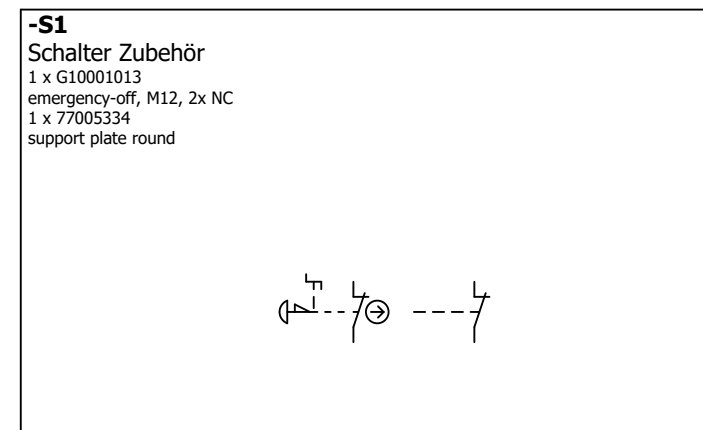


jumper

•	1	77070157
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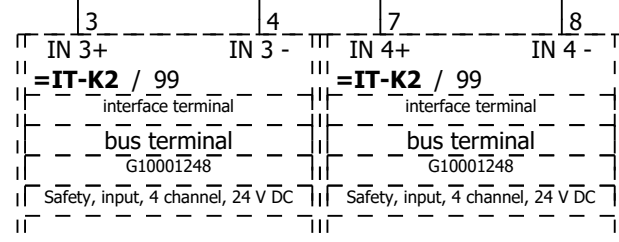
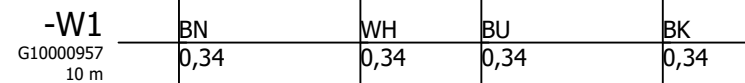
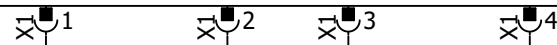
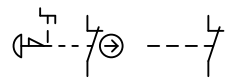
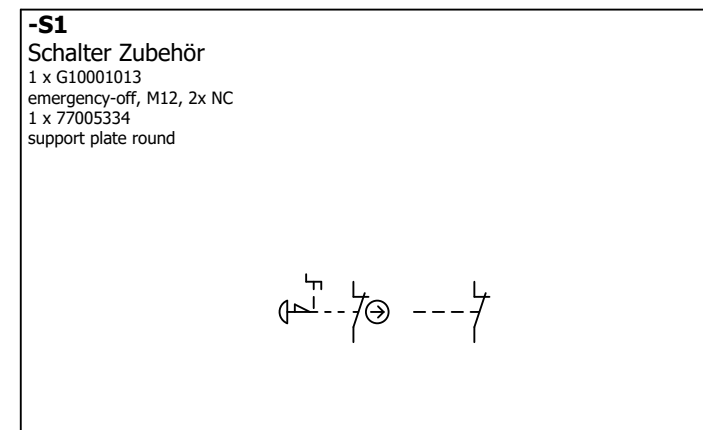
a - potential distribution limit Y- /73.6

b - working area beam Y - / input =CY2Y3+HB/69.3



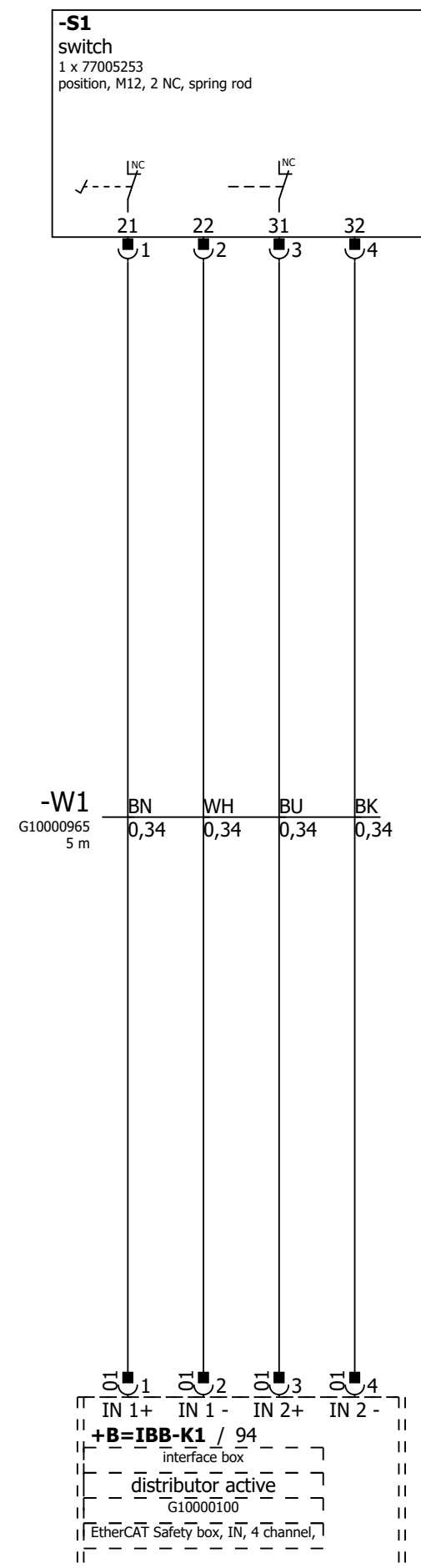
emergency stop

			Toolbox_88	Option:	0405A Basic Parts TSc - CE {R.76}			emergency stop	module	emergency stop additions	Page 74
			Toolbox version	customer:	Maghreb Steel			connection	sector 4D1E location	+E =EA	
			EA1 - ABI - 33.59.00 - CC: EU	commission:	286002000	project: 286002000					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024					

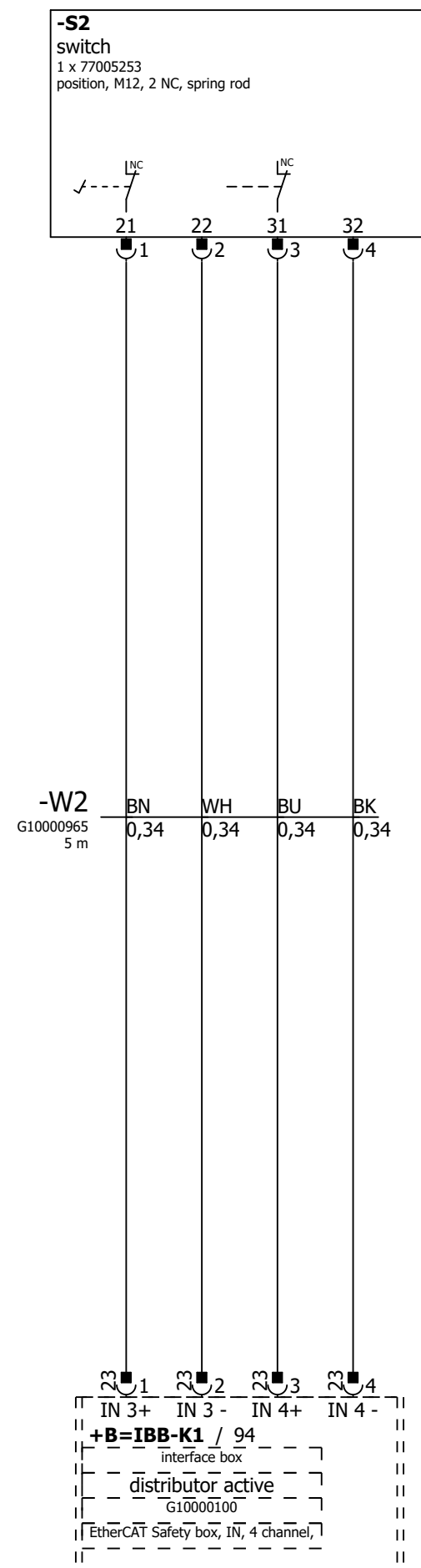


emergency stop

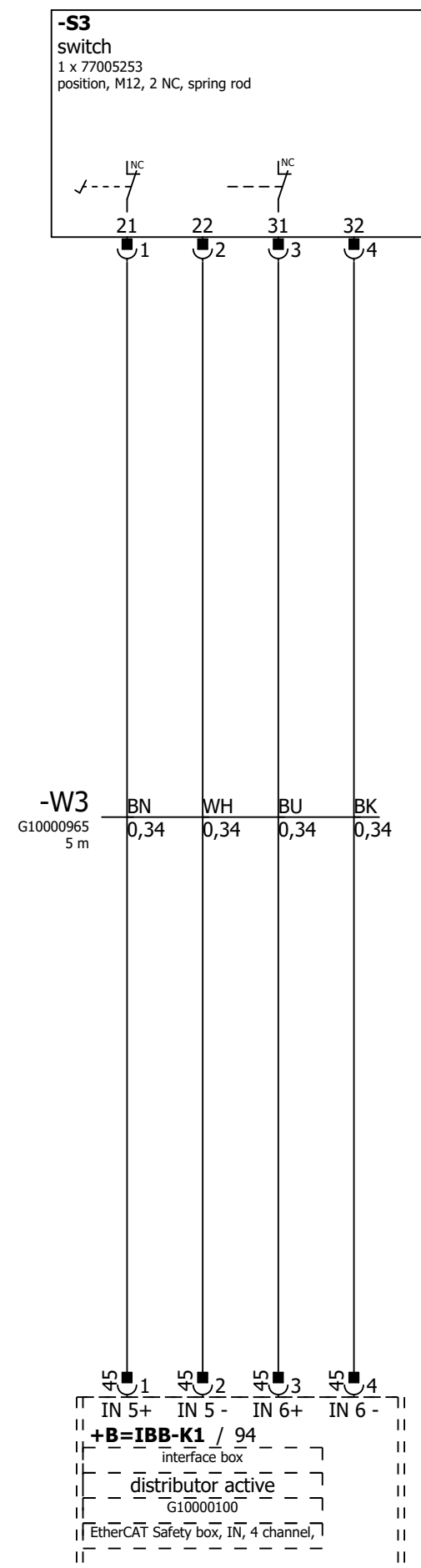
			Toolbox_88	Option:	0405A Basic Parts TSc - CE {R.76}			emergency stop	module	emergency stop additions	Page 75
			Toolbox version	customer:	Maghreb Steel			connection	sector 1D1G location	+G =EAB	
			EA1 - ABI - 33.59.00 - CC: EU	commission:	286002000	project: 286002000					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024					



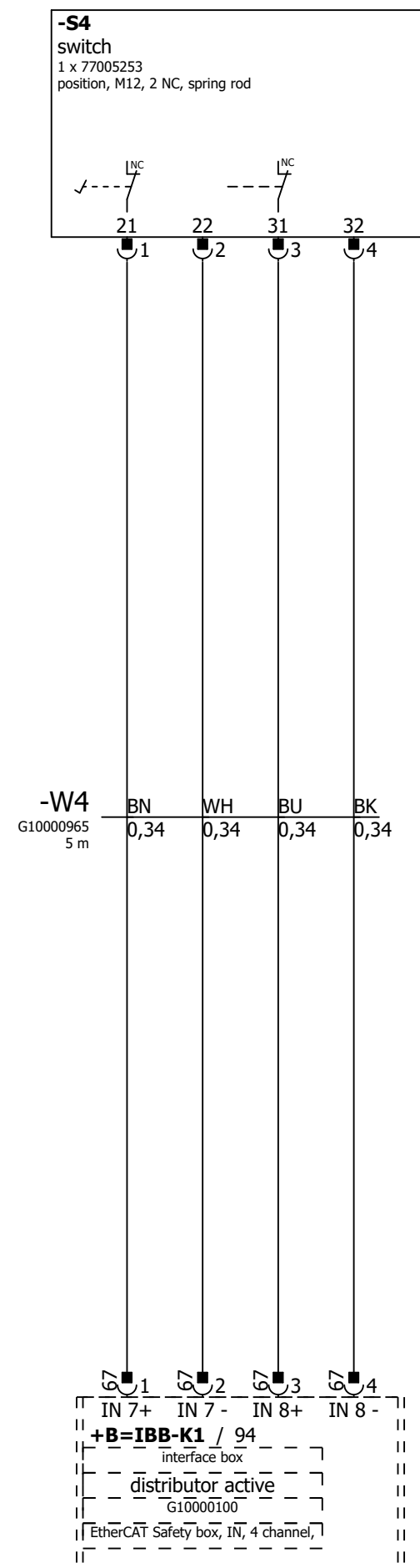
foot safety trigger
rear inside



foot safety trigger
rear outside



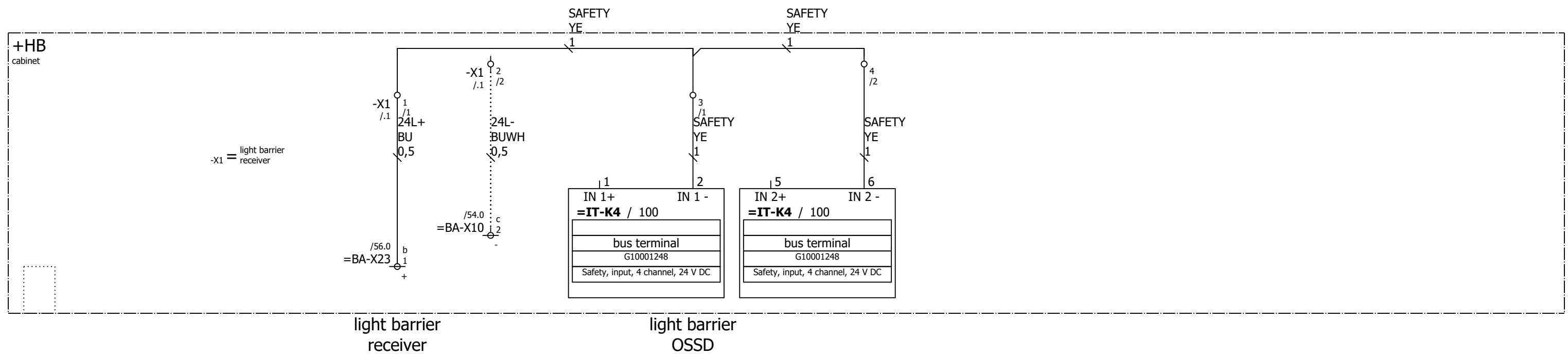
foot safety trigger
front outside



foot safety trigger
front inside

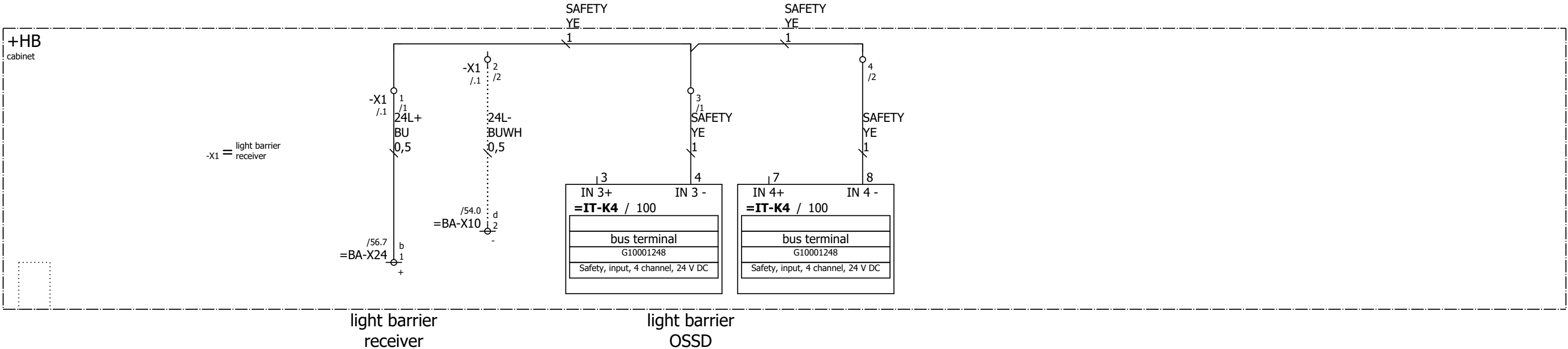
			Toolbox_88	Option: customer: commission:	1204A Foot switch, endtruck, 2x2 Maghreb Steel 286002000 project: 286002000				module emergency stop foot protection		Page 76
			Toolbox version								
			EF2 - ABI - 31.33.01 - CC: EU								
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024		sensor, safety EP box	sector 3C1E	location +E =EF	

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---



			Toolbox_88	Option:	0605M Ripcord {R.38}			Reißleine vorne	module	Reißleine	Page 78
			Toolbox version	customer:	Maghreb Steel			Safety circuit			
			EL19 - ABI - 31.88.01 - CC: EU	commission:	286002000	project: 286002000				sector 3C5E location +E	
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024					

0	1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---	---



			Toolbox_88	Option: customer: commission:	0605M Ripcord customer {R.38}			Reißleine hinten	module	Reißleine =ELB	Page 79
			Toolbox version		Maghreb Steel			Safety circuit	sector 4C5E location +E		
			EL19 - ABI - 31.88.01 - CC: EU		286002000 project: 286002000						
Modification	Date	Name	Toolbox module designation	Ed. bv.	Johann.Rais 24.05.2024						



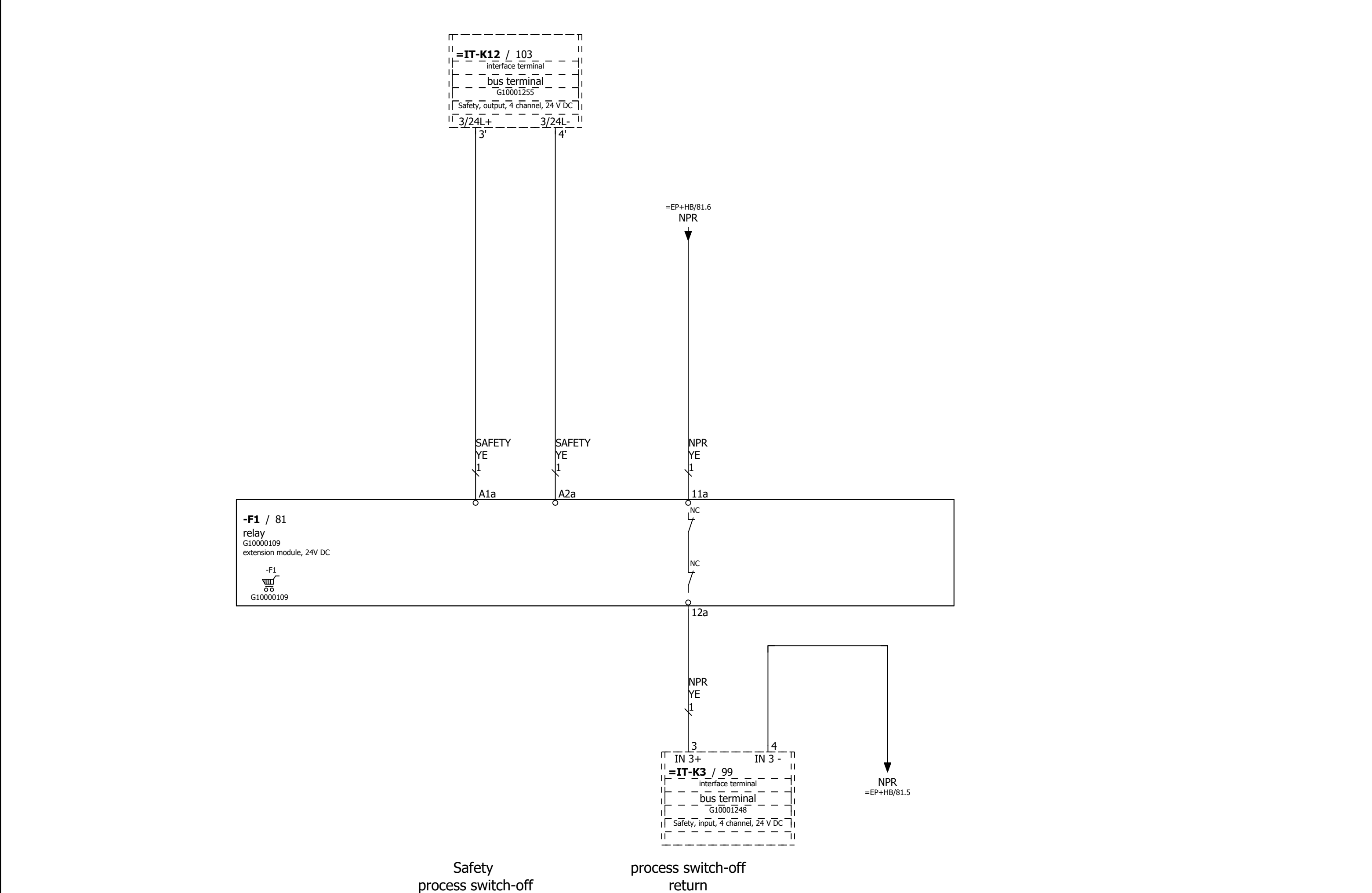
G10000109

relay

-F1 extension module, 24V DC

72a		
72b		
A1a	Safety process switch-off	/82.3
A1b		
A2a	Safety process switch-off	/82.4
A2b		

Logic		
11a	process switch-off return	/82.5
11b		
12a	process switch-off return	/82.5
12b		
23a	valve heating oxygen 2	=FS+HG/90.2
23b		
24a	valve heating oxygen 2	=FS+HG/90.2
24b		
33a		
33b		
34a		
34b		
43a		
43b		
44a		
44b		
53a		
53b		
54a		
54b		
63a		
63b		
64a		
64b		
71a		
71b		





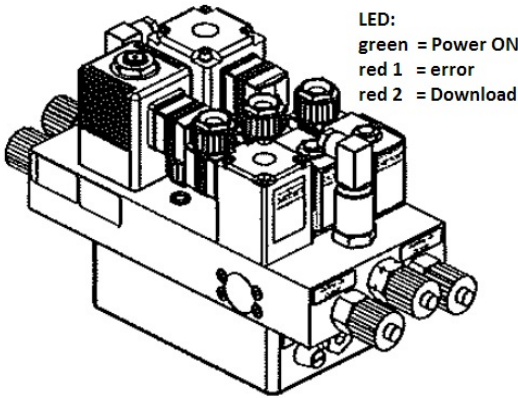
G10000109

relay

-F1 extension module, 24V DC

72a		
72b		
A1a	safety-related control voltage	/84.0
A1b		
A2a	safety-related control voltage	/84.1
A2b		

Logic		
11a	safety relay feedback	/84.2
11b		
12a	safety relay feedback	/84.3
12b		
23a		
23b		
24a		
24b		
33a		
33b		
34a		
34b		
43a		
43b		
44a		
44b		
53a		
53b		
54a		
54b		
63a		
63b		
64a		
64b		
71a		
71b		



77070548
gas regulator

-R1 OmniFlow, propane, 4,5A, IP54

A1

GAS OUT 1		
1	heating oxygen	ALFA torch heating gas O2 =TO2+L2/139.7
2	fuel gas	ALFA torch fuel gas =TO2+L2/139.8
3	cutting oxygen	=TO2+L2/139.9

A2

GAS OUT 2		
1	heating oxygen	ALFA torch heating gas O2 =TO3+L3/141.7
2	fuel gas	ALFA torch fuel gas =TO3+L3/141.8
3	cutting oxygen	=TO3+L3/141.9

A3

GAS OUT 3		
1	heating oxygen	
2	fuel gas	
3	cutting oxygen	

A4

GAS OUT 4		
1	heating oxygen	
2	fuel gas	
3	cutting oxygen	

A5

GAS OUT 5		
1	heating oxygen	
2	fuel gas	
3	cutting oxygen	

A6

GAS OUT 6		
1	heating oxygen	
2	fuel gas	
3	cutting oxygen	

E

GAS IN		
1	heating oxygen	OmniFlow heating gas /86.4
2	fuel gas	OmniFlow fuel gas /86.5
3	cutting oxygen	OmniFlow cutting gas /86.6

GND

GND		
M6	ground	grounding OmniFlow /86.9

X1

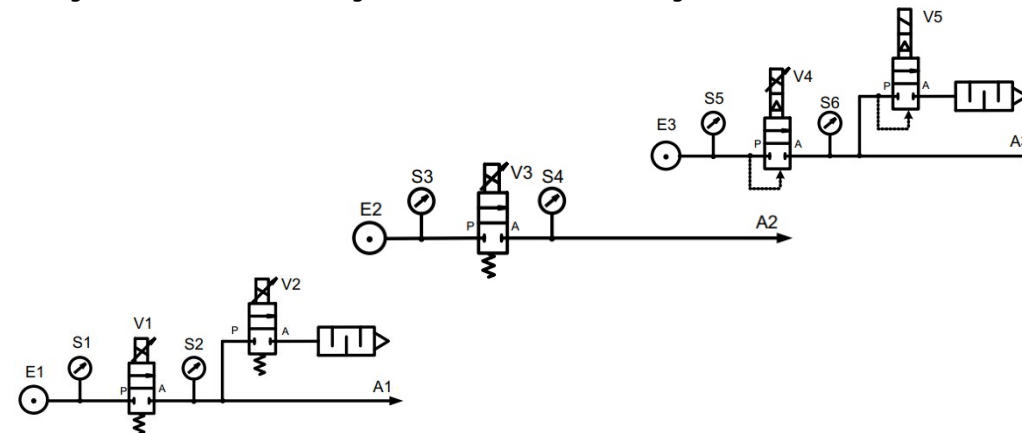
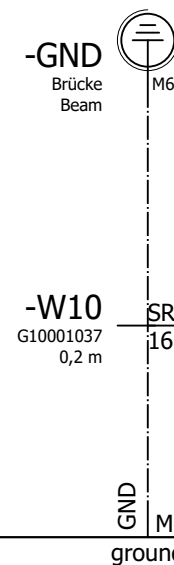
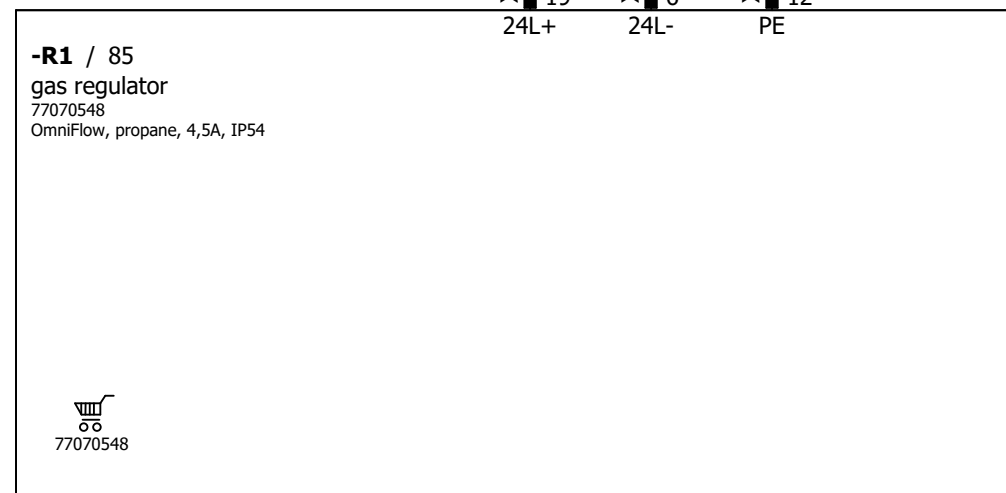
RS422		
1	RX+	RS422 T/R /88.4
2	RX-	= /88.4
3	GND	/88.6
4	Tx+	RS422 T/R /88.5
5	Tx-	= /88.5

X2

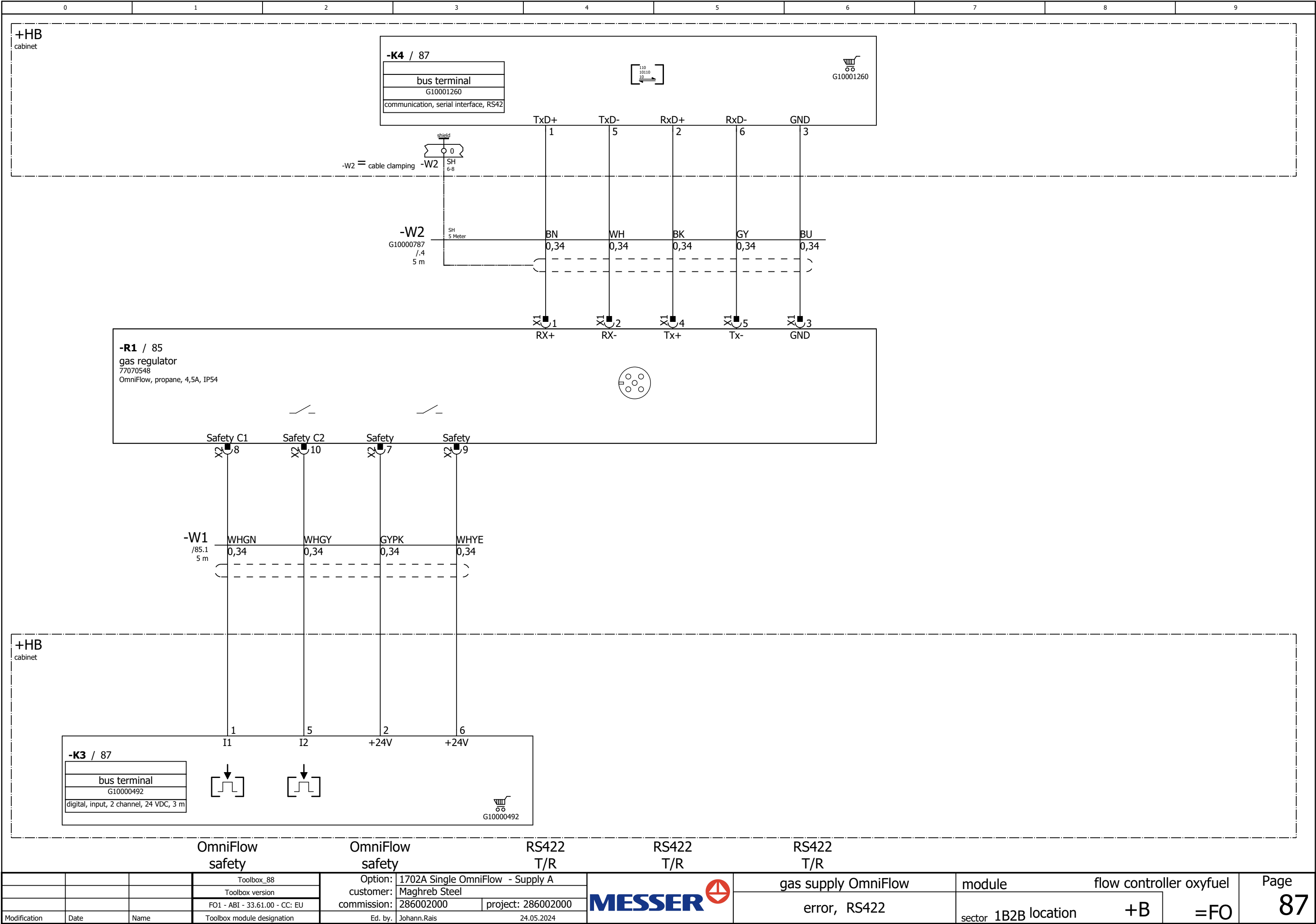
Digital IO		
1	Reserve	
2	DI Heiz -O2 niedrig	O2 low heating /87.1
3	DI Brenngas niedrig	fuel gas low /87.5
4	DI Reserve	spare /87.7
5	DI Reserve	OmniFlow X2 /87.8
6	24L-	OmniFlow operating voltage /86.2
7	Safety C1.com	OmniFlow safety /88.2
8	Safety C1	= /88.1
9	Safety C2.com	= /88.3
10	Safety C2	= /88.2
11	DI Brenngas hoch	fuel gas high /87.6
12	PE	OmniFlow operating voltage /86.2
13	n.c.	
14	n.c.	
15	DI Schneid -O2 niedrig	O2 low piercing /87.3
16	DI Schneid -O2 hoch	O2 high cutting /87.4
17	DI Heiz -O2 hoch	O2 high heating /87.2
18	n.c.	
19	24L+	/86.1

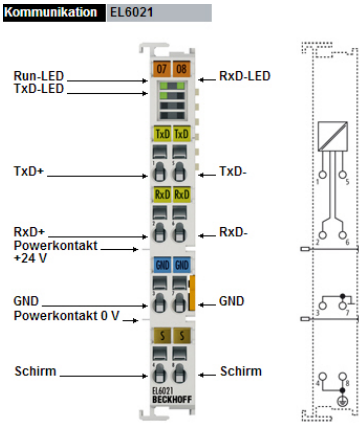
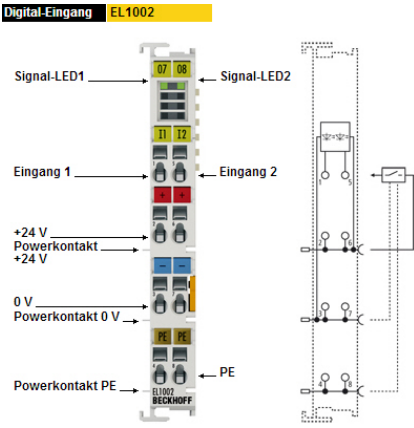
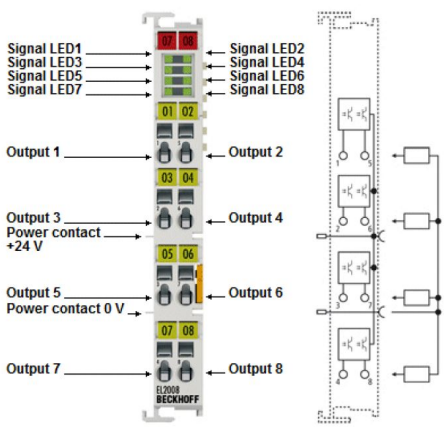
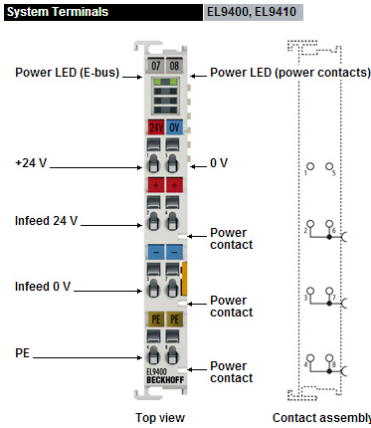
X3

CAN		
1	PE	
2	24L+	
3	24L-	
4	CAN H	
5	CAN L	

grounding
OmniFlow

MESSER 





G13000079

bus terminal

-K1

power supply terminal, E-bus refreshment

G10001249

bus terminal

-K2

digital, output, 8 channel, 24 V DC, 0,5 A

G10000492

bus terminal

-K3

digital, input, 2 channel, 24 VDC, 3 ms

G10001260

bus terminal

-K4

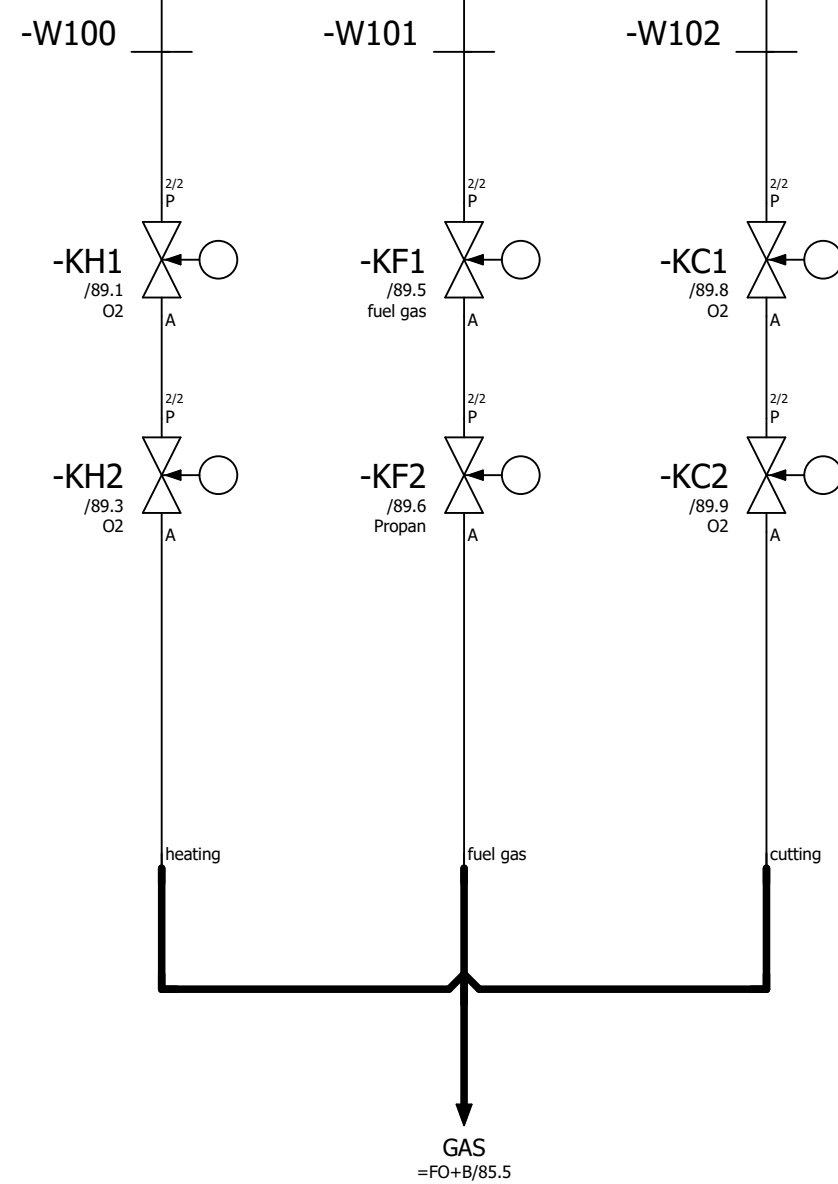
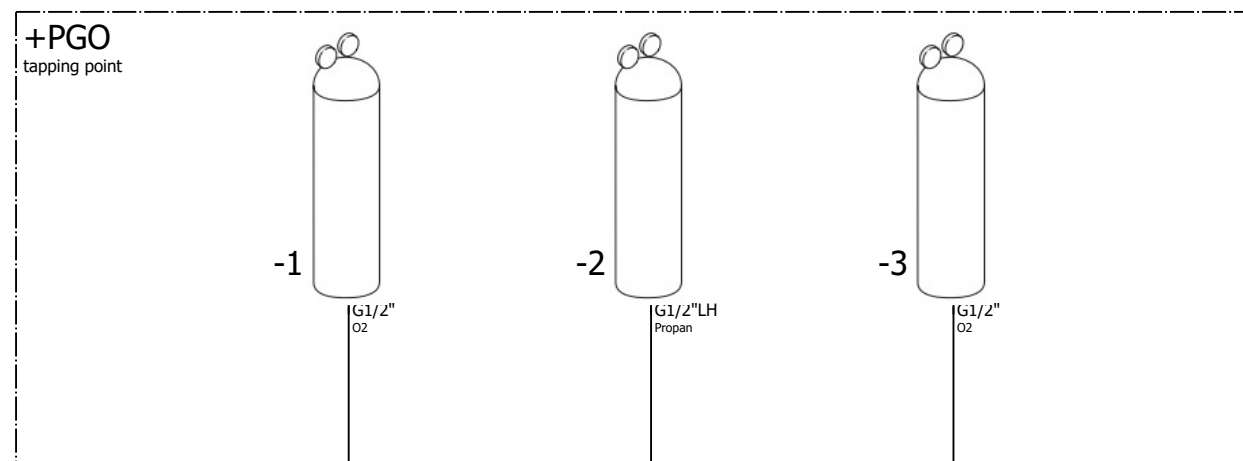
communication, serial interface, RS422 / RS485

power contacts		
2	24L+	+B/86.1
3	24L- OmniFlow operating voltage	+B/86.2
4	PE	= +B/86.2
6	24L+	+B/86.1
7	24L- OmniFlow operating voltage	+B/86.2
8	PE	= +B/86.2
supply		
1	24L+ OmniFlow operating voltage	+B/86.0
5	24L-	= +B/86.1

O1		
1	O1	O2 low heating +B/87.1
O2		
5	O2	O2 high heating +B/87.2
O3		
2	O3	O2 low piercing +B/87.3
O4		
6	O4	O2 high cutting +B/87.4
O5		
3	O5	fuel gas low +B/87.5
O6		
7	O6	fuel gas high +B/87.6
O7		
4	O7	spare +B/87.7
O8		
8	O8	OmniFlow X2 +B/87.8

I2		
6	+24V	OmniFlow safety +B/88.3
I1		
1	I1	OmniFlow safety +B/88.1
2	+24V	= +B/88.2
3	0V	
4	PE	
I2		
5	I2	OmniFlow safety +B/88.2
7	0V	
8	PE	

1	TxD+	RS422 T/R	+B/88.4
2	RxD+	=	+B/88.5
3	GND		+B/88.6
4	SH		
5	TxD-	RS422 T/R	+B/88.4
6	RxD-	=	+B/88.5
7	GND		
8	SH		



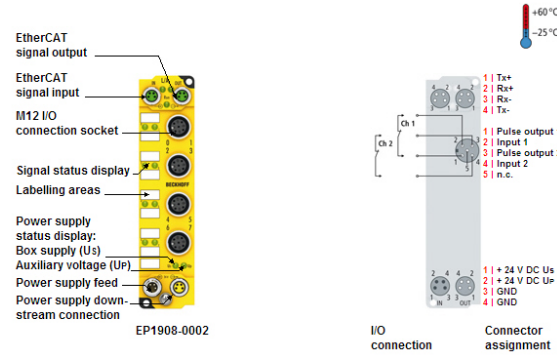
heating gas

fuel gas

cutting gas

			Toolbox_88	Option:	1705A Gas-Safety-Valves Supply - A			gas supply OmniFlow		module	flow controller safety		Page		
			Toolbox version	customer:	Maghreb Steel									90	
			FS1 - ABI - 33.76.00 - CC: EU	commission:	286002000	project: 286002000			supply						
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024					sector 2E9p	location +HG	=FS		

Digital Input EP1908-0002



G10000100
distributor active
-K1 EtherCAT Safety box, IN, 4 channel, M12

01

IN 01			
1	IN 1+	foot safety trigger rear inside	=EFB+G/78.0
2	IN 1 -	=	=EFB+G/78.1
IN 02			
3	IN 2+	foot safety trigger rear inside	=EFB+G/78.1
4	IN 2 -	=	=EFB+G/78.1

23

IN 03			
1	IN 3+	foot safety trigger rear outside	=EFB+G/78.2
2	IN 3 -	=	=EFB+G/78.2
IN 04			
3	IN 4+	foot safety trigger rear outside	=EFB+G/78.3
4	IN 4 -	=	=EFB+G/78.3

24V IN

Power IN			
1	24L+s		/93.1
2	24L+p	EP module supply	/93.2
3	24L-s	=	/93.3
4	24L-p	=	/93.3

24V OUT

Power OUT			
1	24L+s		
2	24L+p		
3	24L-s		
4	24L-p		

45

IN 05			
1	IN 5+	foot safety trigger front outside	=EFB+G/78.4
2	IN 5 -	=	=EFB+G/78.5
IN 06			
3	IN 6+	foot safety trigger front outside	=EFB+G/78.5
4	IN 6 -	=	=EFB+G/78.5

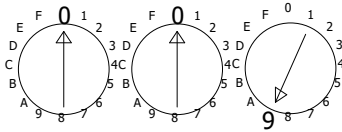
67

IN 07			
1	IN 7+	foot safety trigger front inside	=EFB+G/78.8
2	IN 7 -	=	=EFB+G/78.8
IN 08			
3	IN 8+	foot safety trigger front inside	=EFB+G/78.9
4	IN 8 -	=	=EFB+G/78.9

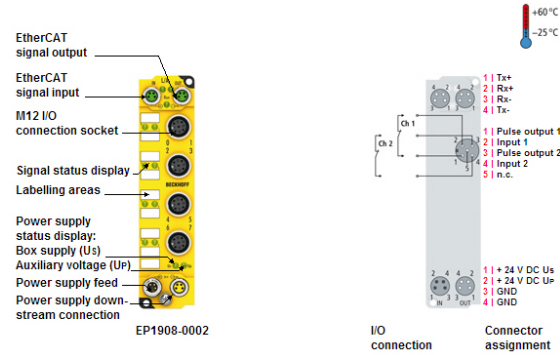
L/A

Bus			
IN	Bus	EtherCAT	/93.8
OUT	Bus	=	/93.8

Safety adress: HEX -<9> / Dez: <9>



Digital Input EP1908-0002



G10000100
distributor active
-K1 EtherCAT Safety box, IN, 4 channel, M12

01

IN 01	
1	IN 1+ foot safety trigger rear inside =EF+E/77.1
2	IN 1 - =EF+E/77.2
IN 02	
3	IN 2+ foot safety trigger rear inside =EF+E/77.2
4	IN 2 - =EF+E/77.2

23

IN 03	
1	IN 3+ foot safety trigger rear outside =EF+E/77.3
2	IN 3 - =EF+E/77.4
IN 04	
3	IN 4+ foot safety trigger rear outside =EF+E/77.4
4	IN 4 - =EF+E/77.4

24V IN

Power IN	
1	24L+s EP module supply /95.1
2	24L+p = /95.2
3	24L-s = /95.3
4	24L-p = /95.3

24V OUT

Power OUT	
1	24L+s =IB/93.1
2	24L+p EP module supply =IB/93.2
3	24L-s = =IB/93.3
4	24L-p = =IB/93.3

45

IN 05	
1	IN 5+ foot safety trigger front outside =EF+E/77.6
2	IN 5 - =EF+E/77.6
IN 06	
3	IN 6+ foot safety trigger front outside =EF+E/77.6
4	IN 6 - =EF+E/77.6

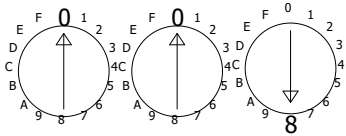
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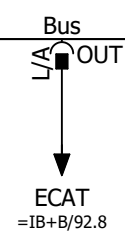
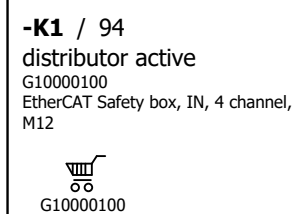
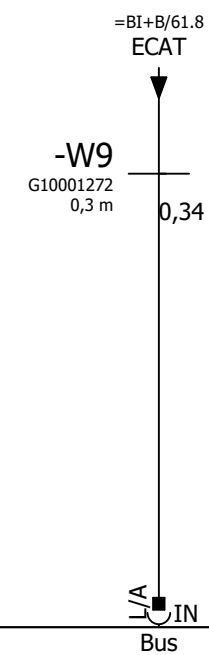
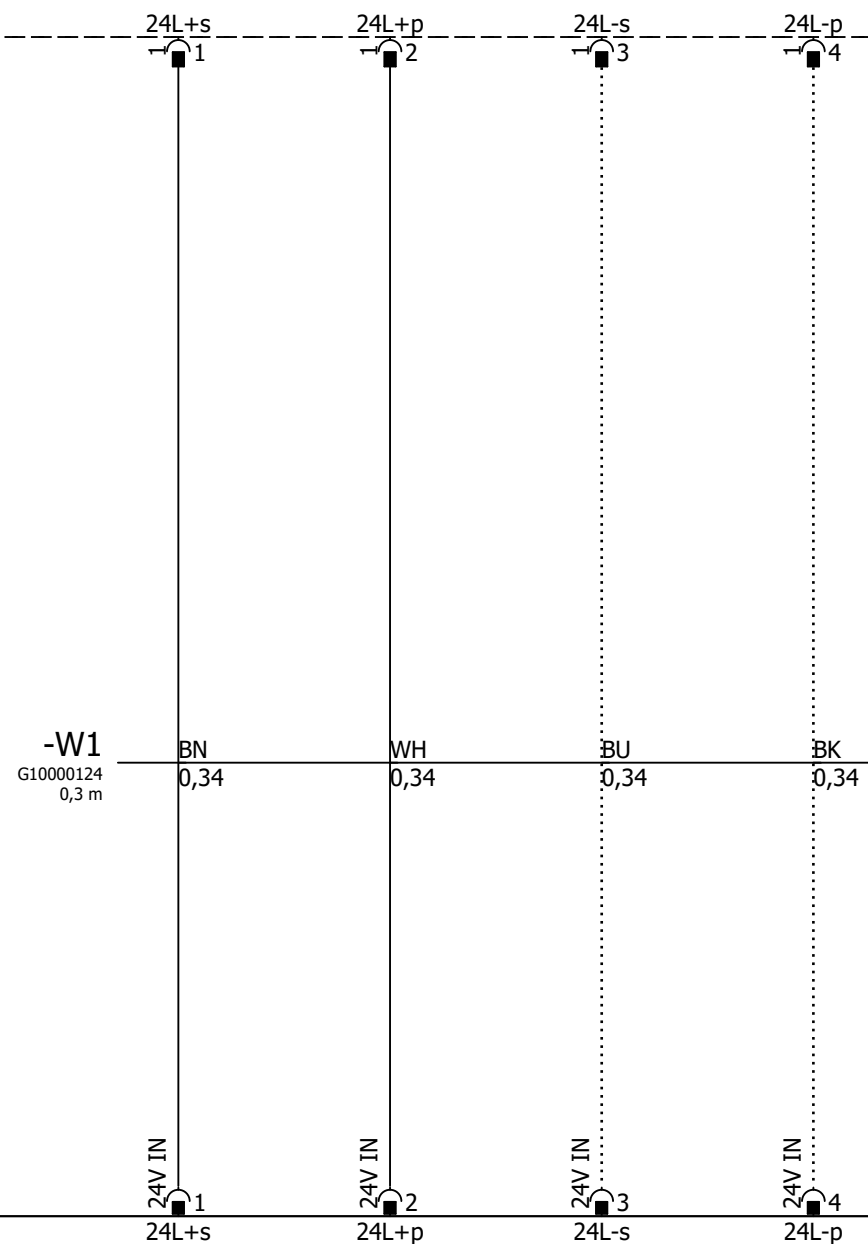
IN 07	
1	IN 7+ foot safety trigger front inside =EF+E/77.8
2	IN 7 - =EF+E/77.8
IN 08	
3	IN 8+ foot safety trigger front inside =EF+E/77.8
4	IN 8 - =EF+E/77.8

L/A

Bus	
IN	Bus EtherCAT /95.8
OUT	Bus = /95.8

Safety adress: HEX -<8> / Dez: <8>



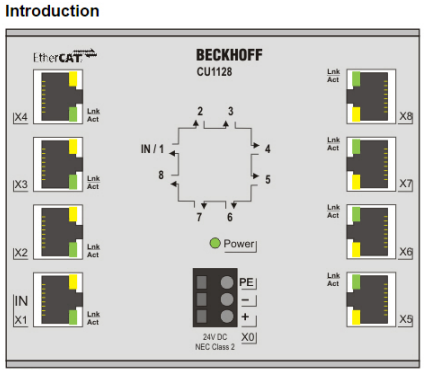


EP module
supply

EtherCAT

			Toolbox_88	Option:	1204A Foot switch, endtruck, 2x2			Fieldbus module		module		interface box		Page
			Toolbox version	customer:	Maghreb Steel			power supply, EtherCat		sector 4D1B location		+B		94
			IB1 - ABI - 34.80.00 - CC: EU	commission:	286002000	project: 286002000								
Modification	Date	Name	Toolbox module designation	Ed. bv.	Johann.Rais	24.05.2024								

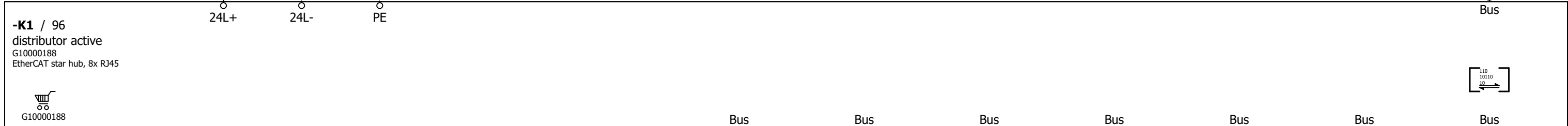
CU1128 | EtherCAT hub



CU1128

G10000188
distributor active
-K1 EtherCAT star hub, 8x RJ45

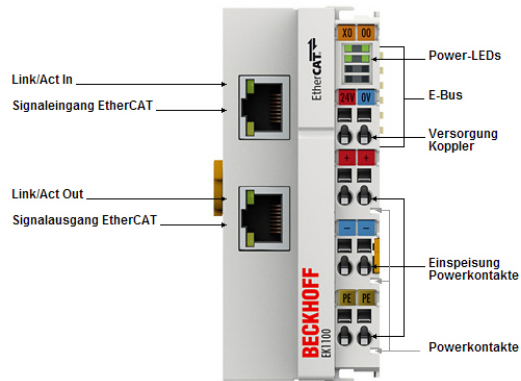
X0			
supply			
1	24L+	star hub supply	/99.1
2	24L-	=	/99.2
3	PE	=	/99.2
X1			
Bus			
IN	Bus	EtherCAT	/99.9
X2			
OUT	Bus	EtherCAT distribution	/99.4
X3			
OUT	Bus	EtherCAT distribution	/99.5
X4			
OUT	Bus	EtherCAT distribution	/99.6
X5			
OUT	Bus	EtherCAT distribution	/99.7
X6			
OUT	Bus	EtherCAT distribution	/99.7
X7			
OUT	Bus	EtherCAT distribution	/99.8
X8			
OUT	Bus	EtherCAT	/99.9



EtherCAT

MESSER 

EtherCAT EK1100



G13000078

bus terminal

-K1

bus coupler

supply

1 24L+ EtherCAT coupler operating supply /97.5

power contacts

2 + EtherCAT coupler powerbus /97.6

3 - = /97.7

4 PE

supply

5 24L- EtherCAT coupler operating supply /97.6

power contacts

6 +

7 -

8 PE

X1

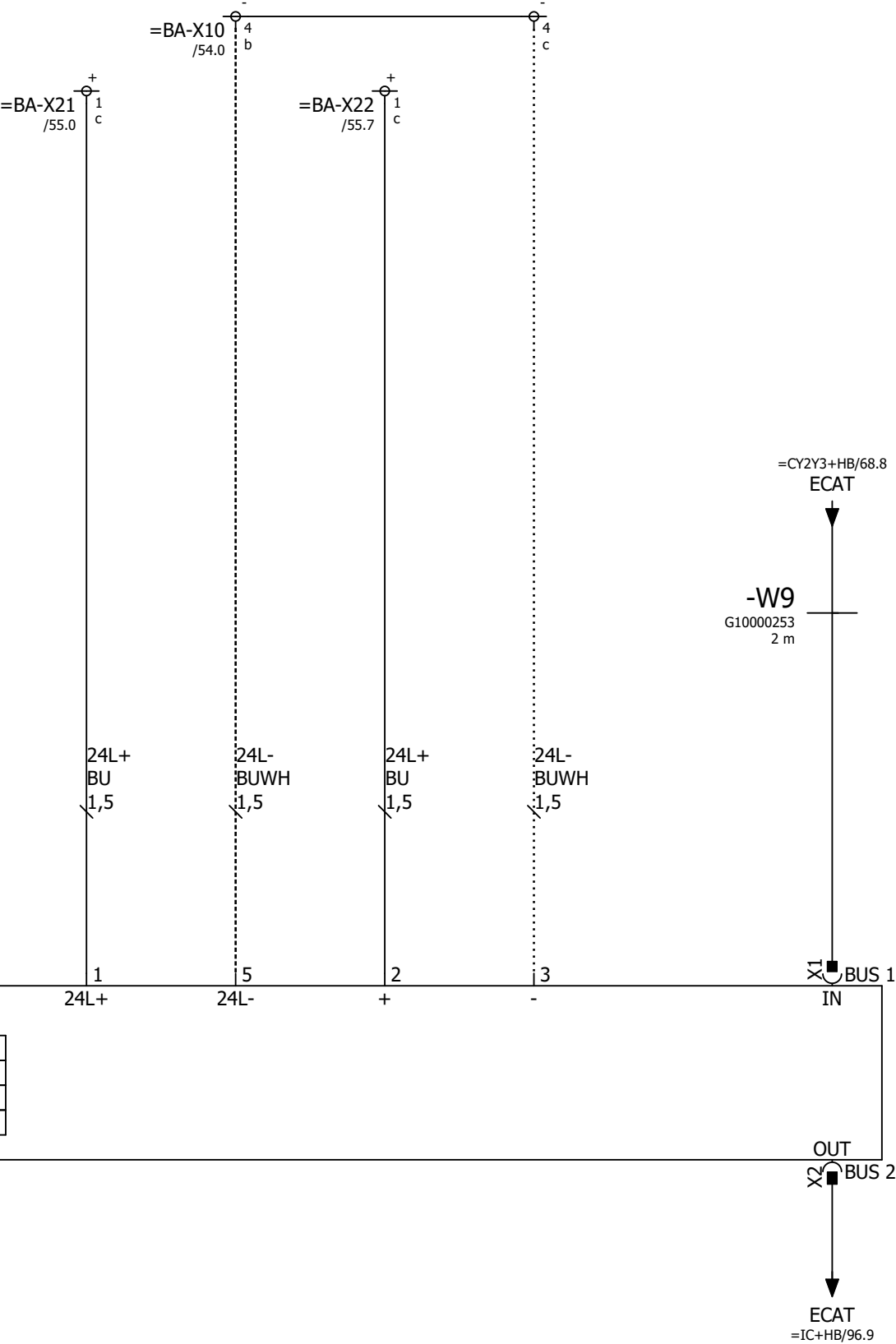
Bus 1

BUS 1 IN EtherCAT coupler Bus /97.8

X2

Bus 2

BUS 2 OUT EtherCAT coupler Bus /97.8



-K1 / 97

bus terminal
G13000078
bus coupler

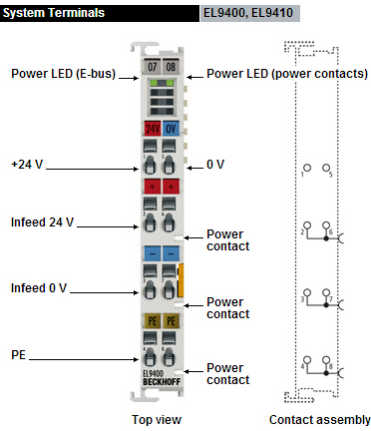
EtherCAT coupler



EtherCAT coupler
operating supply

EtherCAT coupler
powerbus

EtherCAT coupler
Bus

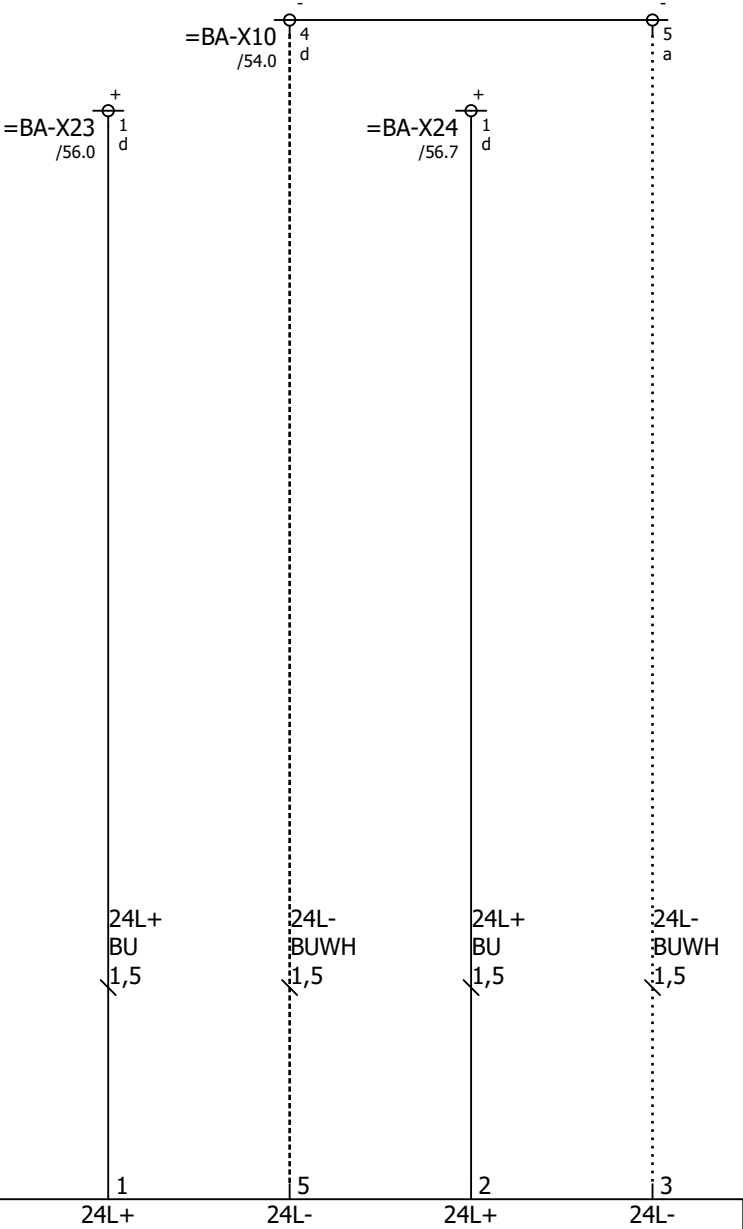


G13000079

bus terminal

-K7power supply terminal, E-bus refreshment

supply		
1	24L+EtherCAT operating supply	/98.5
power contacts		
2	24L+EtherCAT powerbus	/98.6
3	24L- =	/98.7
4	PE	
supply		
5	24L-EtherCAT operating supply	/98.6
power contacts		
6	24L+	
7	24L-	
8	PE	



-K7 / 98
bus terminal
G13000079
power supply terminal, E-bus refresh

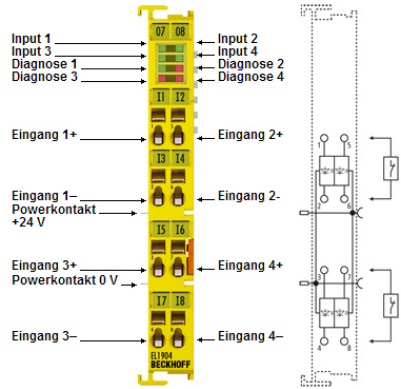


power supply terminal
EtherCAT

EtherCAT
operating supply

EtherCAT
powerbus

Digital-Eingang EL1904



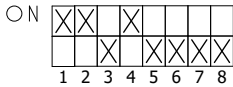
G10001248

bus terminal

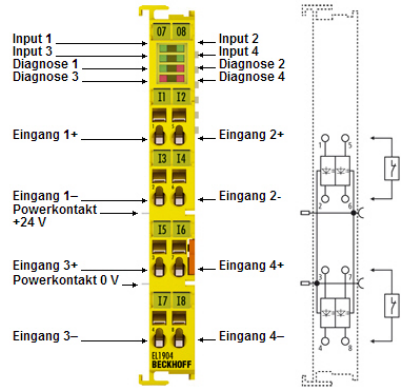
-K2

Safety, input, 4 channel, 24 V
DC

IN 01		
1	IN 1+	emergency stop =EA+E/74.0
2	IN 1 -	= =EA+E/74.1
IN 02		
5	IN 2+	emergency stop =EA+E/74.1
6	IN 2 -	= =EA+E/74.2
IN 03		
3	IN 3+	emergency stop =EAB+G/75.0
4	IN 3 -	= =EAB+G/75.1
IN 04		
7	IN 4+	emergency stop =EAB+G/75.1
8	IN 4 -	= =EAB+G/75.2



Digital-Eingang EL1904



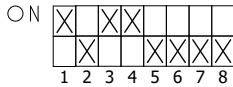
G10001248

bus terminal

-K3

Safety, input, 4 channel, 24 V
DC

IN 01		
1	IN 1+	
2	IN 1 -	
IN 02		
5	IN 2+	
6	IN 2 -	
IN 03		
3	IN 3+	process switch-off return =EP/81.5
4	IN 3 -	=EP/81.5
IN 04		
7	IN 4+	
8	IN 4 -	

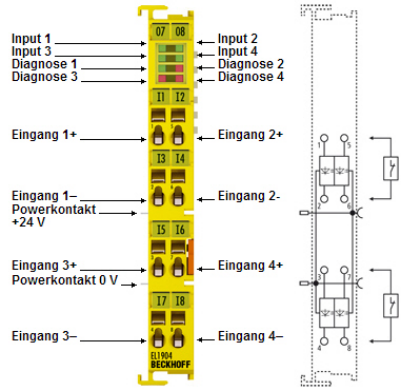


G10001248



G10001248

Digital-Eingang EL1904



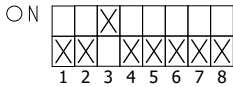
G10001248

bus terminal

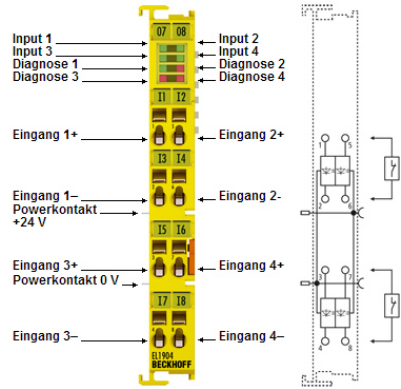
-K4

Safety, input, 4 channel, 24 V
DC

IN 01		
1	IN 1+	light barrier OSSD =EL+E/78.3
2	IN 1 -	= =EL+E/78.4
IN 02		
5	IN 2+	light barrier OSSD =EL+E/78.4
6	IN 2 -	= =EL+E/78.5
IN 03		
3	IN 3+	light barrier OSSD =ELB+E/79.3
4	IN 3 -	= =ELB+E/79.4
IN 04		
7	IN 4+	light barrier OSSD =ELB+E/79.4
8	IN 4 -	= =ELB+E/79.5



Digital-Eingang EL1904



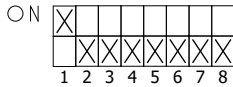
G10001248

bus terminal

-K5

Safety, input, 4 channel, 24 V
DC

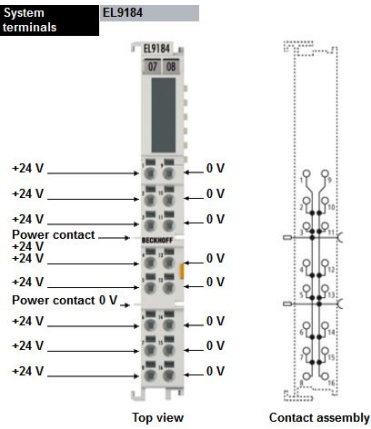
IN 01		
1	IN 1+	safety relay feedback =ER/83.3
2	IN 1 -	= =ER/83.2
IN 02		
5	IN 2+	
6	IN 2 -	
IN 03		
3	IN 3+	
4	IN 3 -	
IN 04		
7	IN 4+	
8	IN 4 -	



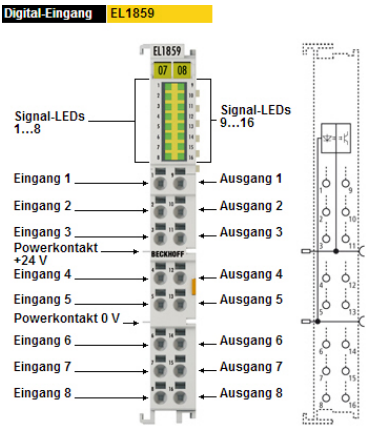
G10001248



G10001248



	supply-13
13	-
	supply-14
14	-
	supply-15
15	-
	supply-16
16	-



	O5
13	O5
	O6
14	O6
	O7
15	O7
	O8
16	O8

G10000699

bus terminal

-K6

power distributors, 2x8 power contacts

G10000441

bus terminal

-K8

digital, E/A, 2 x 8 channel, 24 VDC

	supply-1
1	+
	supply-2
2	+
	supply-3
3	+
	supply-4
4	+
	supply-5
5	+
	supply-6
6	+
	supply-7
7	+
	supply-8
8	+
	supply-9
9	- signal pointer =OK+HOK/119.9
	supply-10
10	- operator action request =OK+HOK/119.7
	supply-11
11	-
	supply-12
12	-

	I1
1	I1 potential distribution limiting Y+ =DL1+L1/71.4
	I2
2	I2 locator limit Y- =DL3+L3/73.4
	I3
3	I3
	I4
4	I4
	I5
5	I5
	I6
6	I6
	I7
7	I7
	I8
8	I8
	O1
9	O1
	O2
10	O2
	O3
11	O3 operator action request =OK+HOK/119.7
	O4
12	O4 signal pointer =OK+HOK/119.9

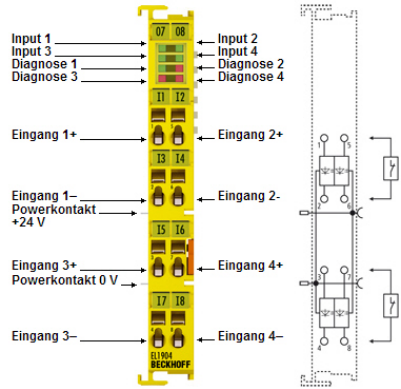


G10000699



G10000441

Digital-Eingang EL1904

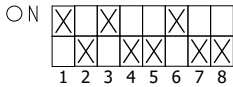


G10001248

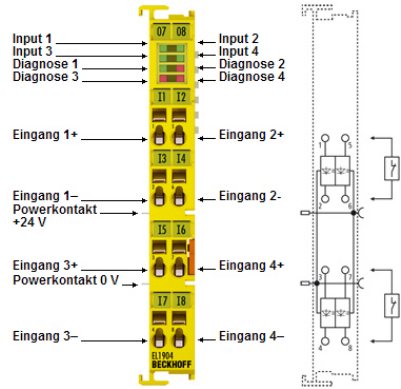
bus terminal

-K9 Safety, input, 4 channel, 24 V DC

IN 01		
1	IN 1+	light barrier fade out =OK+HOK/119.4
2	IN 1 -	= =OK+HOK/119.2
IN 02		
5	IN 2+	light barrier fade out =OK+HOK/119.3
6	IN 2 -	= =OK+HOK/119.3
IN 03		
3	IN 3+	
4	IN 3 -	
IN 04		
7	IN 4+	
8	IN 4 -	



Digital-Eingang EL1904

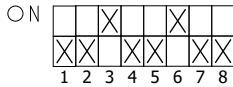


G10001248

bus terminal

-K10 Safety, input, 4 channel, 24 V DC

IN 01		
1	IN 1+	machine reset I =OP+G/125.3
2	IN 1 -	= =OP+G/125.4
IN 02		
5	IN 2+	panel Safety channel =OP+G/125.0
6	IN 2 -	= =OP+G/125.1
IN 03		
3	IN 3+	panel Safety channel =OP+G/125.1
4	IN 3 -	= =OP+G/125.2
IN 04		
7	IN 4+	monitoring safety circuit activate =OK+HOK/119.6
8	IN 4 -	= =OK+HOK/119.7

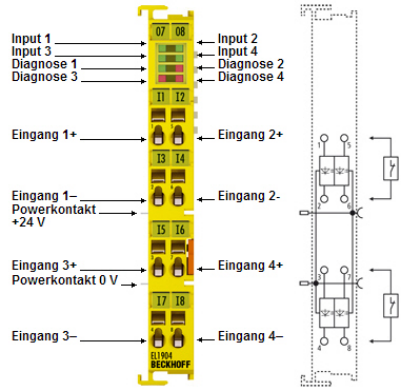


G10001248

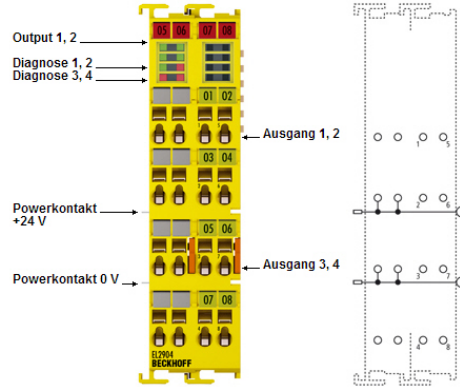


G10001248

Digital-Eingang EL1904



Digital-Ausgang EL2904



G10001248

bus terminal

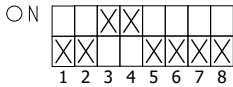
-K11 Safety, input, 4 channel, 24 V DC

G10001255

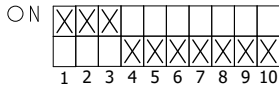
bus terminal

-K12 Safety, output, 4 channel, 24 V DC

IN 01		
1	IN 1+	monitoring exhaust =SPB+P/136.3
2	IN 1 -	= =SPB+P/136.3
IN 02		
5	IN 2+	monitoring exhaust =SPB+P/136.5
6	IN 2 -	= =SPB+P/136.6
IN 03		
3	IN 3+	
4	IN 3 -	
IN 04		
7	IN 4+	
8	IN 4 -	



Out 03		
4'	3/24L-	Safety process switch-off =EP/81.4
Out 01		
1'	1/24L+	
2'	1/24L-	
Out 02		
5'	2/24L+	
6'	2/24L-	
Out 03		
3'	3/24L+	Safety process switch-off =EP/81.3
Out 04		
7'	4/24L+	
8'	4/24L-	
Power		
2	24L+	
3	24L-	
6	24L+	
7	24L-	

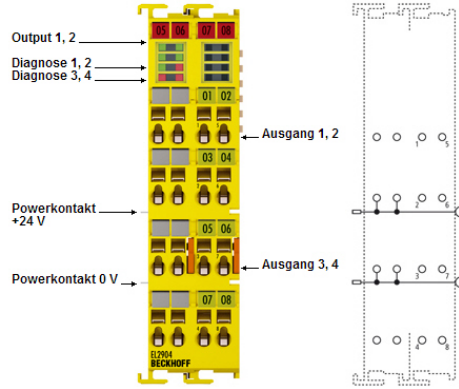


G10001248



G10001255

Digital-Ausgang EL2904



G10001255

bus terminal

-K13 Safety, output, 4 channel, 24 V DC

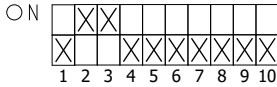
Out 01		
1'	1/24L+ safety-related control voltage	=ER/83.0
2'	1/24L-	=ER/83.1

Out 02		
5'	2/24L+	
6'	2/24L-	

Out 03		
3'	3/24L+	
4'	3/24L-	

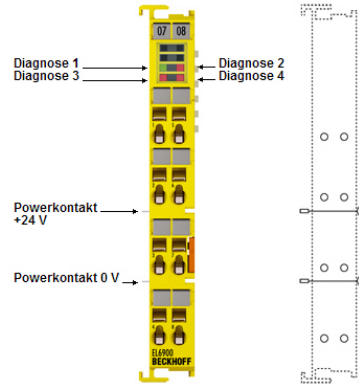
Out 04		
7'	4/24L+	
8'	4/24L-	

Power		
2	24L+	
3	24L-	
6	24L+	
7	24L-	



G10001255

Kommunikation EL6900

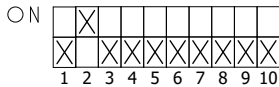


G10001262

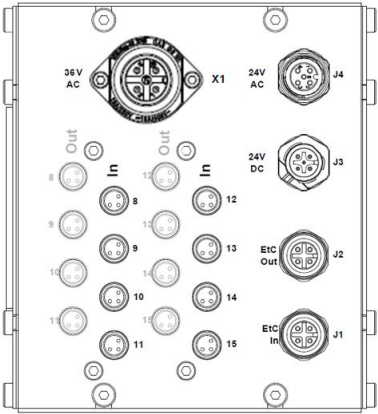
bus terminal

-K14 Safety, TwinSAFE-logic-terminal

1	n.c.
2	n.c.
3	n.c.
4	n.c.
5	n.c.
6	n.c.
7	n.c.
8	n.c.



G10001262



72163096

lifter

-K1

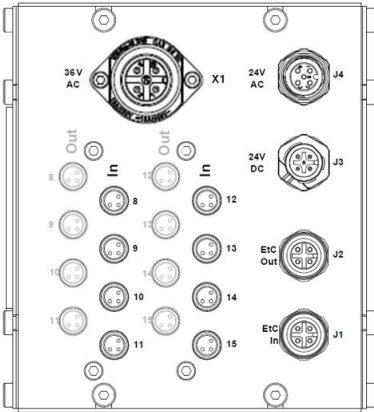
SmartLifter 200, plasma/ARKI/PAN,
2 x 4 DI, 2 x 4 DO

J10_PAN	
PAN	
SH	=SH1/131.3
PwrDrive	
3	36Ls /109.3
J1	
Bus	
IN	BUS
EtherCAT	
Bus	IN SmartLifter EtherCAT /109.9
J2	
Bus	
OUT	BUS
EtherCAT	
Bus	OUT SmartLifter EtherCAT /109.9
J3	
PwrDC	
1	Us SmartLifter DC operating voltage /109.5
2	Up SmartLifter DC control voltage /109.6
3	GND SmartLifter DC operating voltage /109.7
4	n.c. = /109.7
5	n.c. /109.8
SH	
J4	
PwrALFA	
1	24L

J4	
2	nc
3	24N
4	nc
5	nc
SH	
J5	
PwrDrive	
1	36Lp SmartLifter AC power supply /109.3
2	36N = /109.2
4	PE = /109.1
SH	/109.3
J7_I8	
I8	
1	24L+
3	24L-
4	I8
J7_I9	
I9	
1	24L+
3	24L-
4	I9
J7_I10	
I10	
1	24L+
3	24L-
4	I10
J7_I11	
I11	
1	24L+
3	24L-
4	I11
J7_I12	
I12	
1	24L+
3	24L-
4	I12
J7_I13	
I13	
1	24L+

J7_I13	
3	24L-
4	I13
J7_I14	
I14	
1	24L+
3	24L-
4	I14
J7_I15	
I15	
1	24L+
3	24L-
4	I15
J7_O8	
O8	
1	24L+ =UM1/143.2
3	24L- plate marking =UM1/143.2
4	O8 = =UM1/143.2
J7_O9	
O9	
1	24L+
3	24L-
4	O9
J7_O10	
O10	
1	24L+
3	24L-
4	O10
J7_O11	
O11	
1	24L+
3	24L-
4	O11
J7_O12	
O12	
1	24L+
3	24L-
4	O12

J7_O13	
O13	
1	24L+
3	24L-
4	O13
J7_O14	
O14	
1	24L+
3	24L-
4	O14
J7_O15	
O15	
1	24L+
3	24L-
4	O15
J10_PAN	
PAN	
1	+/-10V PAN sensors desired value =SH1/131.5
2	GND Signal = =SH1/131.4
7	24L- PAN sensors supply =SH1/131.2
9	+/-1V PAN sensors desired value =SH1/131.5
11	collision PAN sensors collision =SH1/131.6
15	24L+ PAN sensors supply =SH1/131.2
J11_ALFA	
ALFA	
1	
2	
3	
4	
5	
6	
7	
9	
10	
11	
12	
13	
14	
15	



72163096

lifter

-K1

SmartLifter 200, plasma/ARKI/PAN,
2 x 4 DI, 2 x 4 DO

J11_ALFA

SH

72163099

lifter

-K1

SmartLifter 200, HD
plasma/ARKI/PAN, 2 x 4 DI, 2 x 4 DO

J5			
	PwrDrive		
3	36Ls		/111.3
J1			
	Bus		
IN	BUS		
	EtherCAT		
Bus	IN	SmartLifter EtherCAT	/111.9
J2			
	Bus		
OUT	BUS		
	EtherCAT		
Bus	OUT	SmartLifter EtherCAT	/111.9
J3			
	PwrDC		
1	Us	SmartLifter DC operating voltage	/111.5
2	Up	SmartLifter DC control voltage	/111.6
3	GND	SmartLifter DC operating voltage	/111.7
4	n.c	=	/111.7
5	n.c		/111.8
SH			
J4			
	PwrALFA		
1	24L		
2	nc		
3	24N		

J4		
4	nc	
5	nc	
SH		
J5		
	PwrDrive	
1	36Lp SmartLifter AC power supply	/111.3
2	36N	= /111.2
4	PE	= /111.1
SH		/111.3
J7_I8		
	I8	
1	24L+	
3	24L-	
4	I8	
J7_I9		
	I9	
1	24L+	
3	24L-	
4	I9	
J7_I10		
	I10	
1	24L+	
3	24L-	
4	I10	
J7_I11		
	I11	
1	24L+	
3	24L-	
4	I11	
J7_I12		
	I12	
1	24L+	
3	24L-	
4	I12	
J7_I13		
	I13	
1	24L+	
3	24L-	
4	I13	

J7_I14			
	I14		
1	24L+		
3	24L-		
4	I14		
J7_I15			
	I15		
1	24L+		
3	24L-		
4	I15		
J7_O8			
	O8		
1	24L+	ALFA Torch ignition	=TO2/139.4
3	24L-	=	=TO2/139.4
4	O8	ALFA torch ignition	=TO2/139.4
J7_O9			
	O9		
1	24L+	ALFA Torch ignition valve	=TO2/139.2
3	24L-	=	=TO2/139.2
4	O9	=	=TO2/139.3
J7_O10			
	O10		
1	24L+		=UP2/144.2
3	24L-	laser diode	=UP2/144.2
4	O10	=	=UP2/144.2
J7_O11			
	O11		
1	24L+	addressing heating gas	=VA2/146.6
3	24L-		=VA2/146.6
4	O11	addressing heating gas	=VA2/146.6
J7_O12			
	O12		
1	24L+	addressing cutting gas	=VA2/146.2
3	24L-	=	=VA2/146.1
4	O12	=	=VA2/146.2
J7_O13			
	O13		
1	24L+	addressing fuel gas	=VA2/146.4
3	24L-	=	=VA2/146.3
4	O13	=	=VA2/146.4

J7_O14		
	O14	
1	24L+	
3	24L- addressing gas flow	=VA2/146.8
4	O14	= =VA2/146.8
J7_O15		
	O15	
1	24L+	
3	24L-	
4	O15	
J10_PAN		
	PAN	
1	+/-10V	
2	GND Signal	
7	24L-	
9	+/-1V	
11	collision	
15	24L+	
SH		
J11_ALFA		
	ALFA	
1	ARKI sensors desired value	=SH2/133.6
2		= =SH2/133.7
3		=SH2/133.6
4		=SH2/133.8
5		=SH2/133.8
6	ARKI sensors up	=SH2/133.3
7	ARKI sensors 24 VDC	=SH2/133.0
9		=SH2/133.9
10	ARKI sensors height set	=SH2/133.4
11	ARKI sensors collision	=SH2/133.5
12		=SH2/133.9
13	ARKI sensors automatic ON	=SH2/133.1
14	ARKI sensors lowering	=SH2/133.2
15	ARKI sensors 24 VDC	=SH2/133.0
SH		=SH2/133.4

72163099 lifter -K1SmartLifter 200, HD plasma/ARKI/PAN, 2 x 4 DI, 2 x 4 DO			
J5 PwrDrive 336Ls			
J1 Bus INBUS EtherCAT BusINSmartLifter EtherCAT			
J2 Bus OUTBUS EtherCAT BusOUTSmartLifter EtherCAT			
J3 PwrDC 1UsSmartLifter DC operating voltage 2UpSmartLifter DC control voltage 3GNDSmartLifter DC operating voltage 4n.c.= 5n.c. SH			
J4 PwrALFA 124L 2nc 324N			

/113.3

/113.9

/113.9

/113.8

J4 4nc 5nc SH			
J5 PwrDrive 136LpSmartLifter AC power supply 236N= 4PE= SH			
J7_I8 I8 124L+ 324L- 4I8			
J7_I9 I9 124L+ 324L- 4I9			
J7_I10 I10 124L+ 324L- 4I10			
J7_I11 I11 124L+ 324L- 4I11			
J7_I12 I12 124L+ 324L- 4I12			
J7_I13 I13 124L+ 324L- 4I13			

/113.3

/113.2

/113.1

/113.3

J7_I14 I14 124L+ 324L- 4I14			
J7_I15 I15 124L+ 324L- 4I15			
J7_O8 O8 124L+ALFA Torch ignition 324L-= 4O8ALFA torch ignition			
J7_O9 O9 124L+ALFA Torch ignition valve 324L-= 4O9=			
J7_O10 O10 124L+addressing heating gas 324L-= 4O10addressing heating gas			
J7_O11 O11 124L+ 324L+addressing gas flow 4O11=			
J7_O12 O12 124L+addressing cutting gas 324L-= 4O12=			
J7_O13 O13 124L+addressing fuel gas 324L-= 4O13=			

=TO3/141.4

=TO3/141.4

=TO3/141.4

=TO3/141.2

=TO3/141.2

=TO3/141.3

=VA3/147.6

=VA3/147.6

=VA3/147.6

=VA3/147.8

=VA3/147.8

=VA3/147.2

=VA3/147.1

=VA3/147.2

=VA3/147.4

=VA3/147.3

=VA3/147.4

J7_O14 O14 124L+ 324L- 4O14			
J7_O15 O15 124L+ 324L- 4O15			
J10_PAN PAN 1+/-10V 2GND Signal 724L- 9+/-1V 11collision 1524L+ SH			
J11_ALFA ALFA 1ARKI sensors desired value 2= 3 4 5 6ARKI sensors up 7ARKI sensors 24 VDC 9 10ARKI sensors height set 11ARKI sensors collision 12 13ARKI sensors automatic ON 14ARKI sensors lowering 15ARKI sensors 24 VDC SH			

=SH3/135.6

=SH3/135.7

=SH3/135.6

=SH3/135.8

=SH3/135.8

=SH3/135.3

=SH3/135.0

=SH3/135.9

=SH3/135.4

=SH3/135.5

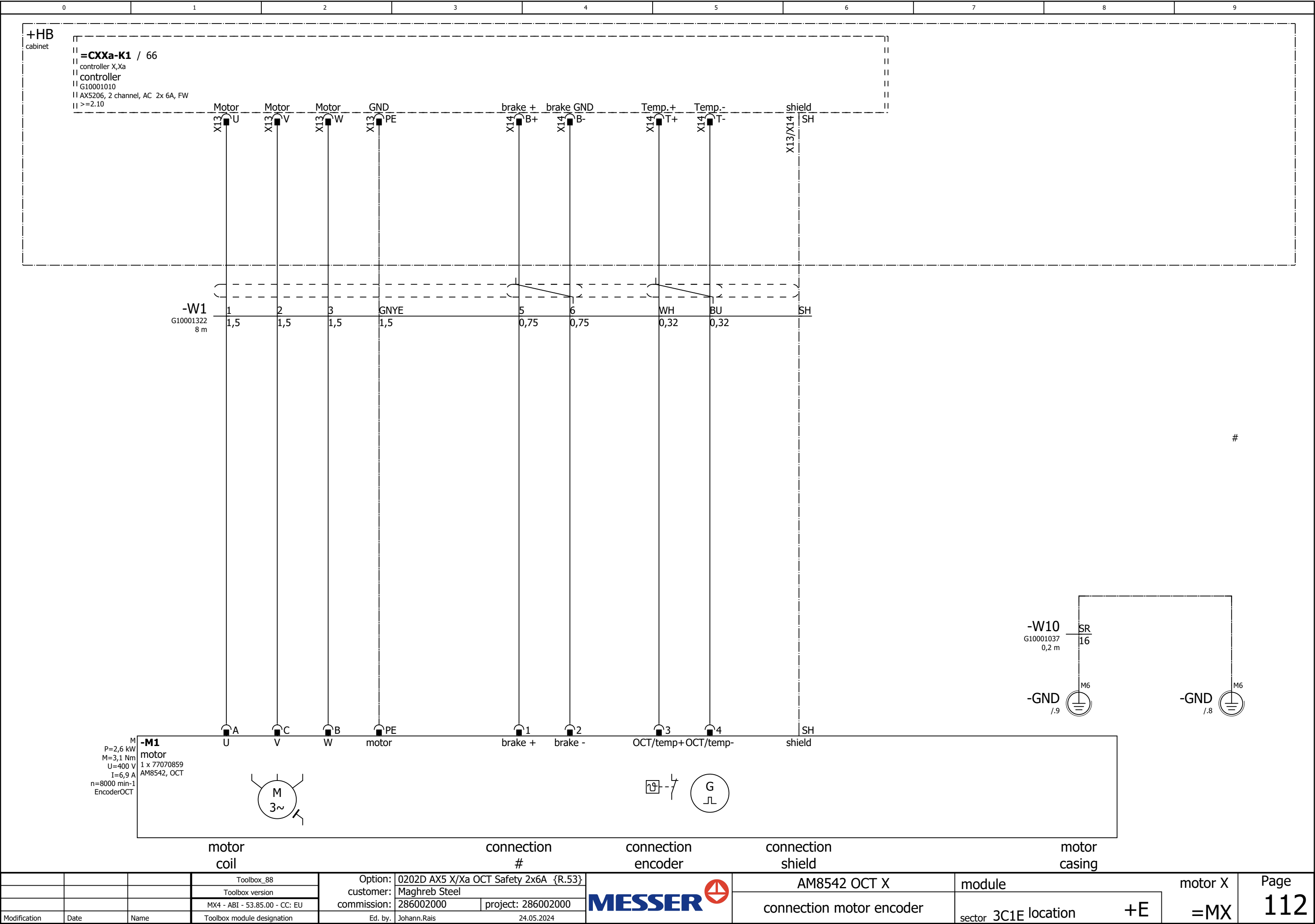
=SH3/135.9

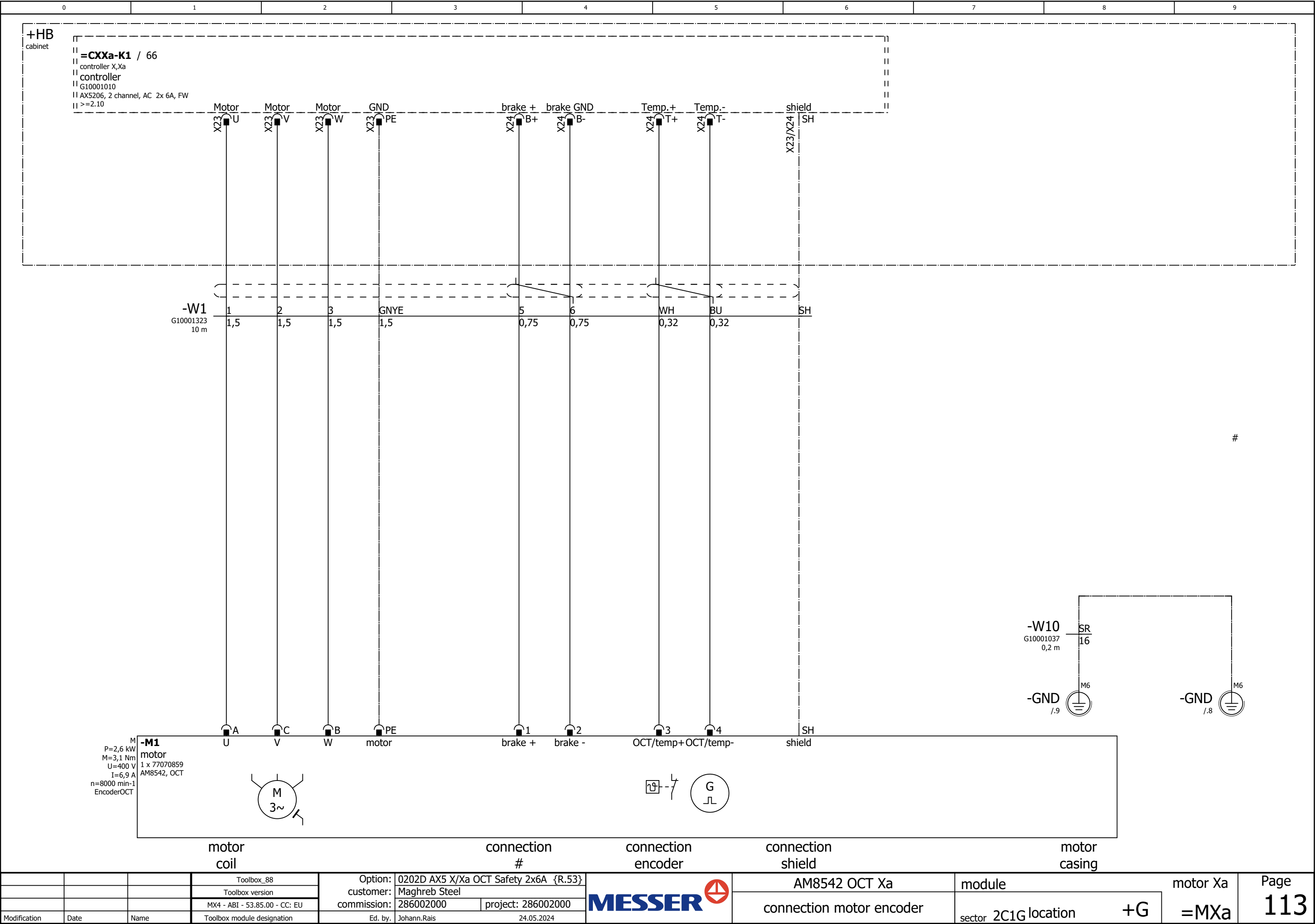
=SH3/135.1

=SH3/135.2

=SH3/135.0

=SH3/135.4







G10001225
control panel

-P1 display 15", GC+

S3		
	Joystick	
9	up	Joystick high
	Shield	
	Shield	
PE		
	grounding	
M6	grounding	panel functional ground
S3		
	Joystick	
5	left	Joystick left
7	down	Joystick down
8	right	Joystick right
10	Vcc	Joystick left
S4		
	Encoder	
1	5L+	control panel encoder
2	5L-	=
5	A	=
6	A-	=
7	B	=
8	B-	=
S5		
	Start	
		machine start
		=

S6		
	Stop	
		machine Stop
		=
S10		
	Lifter control	
24L+		panel Lifter All
DI10.1	basic-Position	=
DI10.2	work-Position	=
S11		
24L+		panel Lifter 1
DI11.1	basic-Position	=
DI11.2	work-Position	=
S12		
24L+		panel Lifter 2
DI12.1	basic-Position	=
DI12.2	work-Position	=
S13		
24L+		panel Lifter 3
DI13.1	basic-Position	=
DI13.2	work-Position	=
S14		
24L+		
DI14.1	basic-Position	
DI14.2	work-Position	
S15		
24L+		
DI15.1	basic-Position	
DI15.2	work-Position	
S16		
24L+		
DI16.1	basic-Position	
DI16.2	work-Position	
S20		
24L+		panel Lifter All
DI20.1	Lifter-up	=
DI20.2	Lifter-down	=
S21		
24L+		panel Lifter 1
DI21.1	Lifter-up	=

S21		
DI21.2	Lifter-down	panel Lifter 1
S22		
24L+		panel Lifter 2
DI22.1	Lifter-up	=
DI22.2	Lifter-down	=
S23		
24L+		panel Lifter 3
DI23.1	Lifter-up	=
DI23.2	Lifter-down	=
S24		
24L+		
DI24.1	Lifter-up	
DI24.2	Lifter-down	
S25		
24L+		
DI25.1	Lifter-up	
DI25.2	Lifter-down	
S26		
24L+		
DI26.1	Lifter-up	
DI26.2	Lifter-down	
USB		
	USB-OUT-S	
USB	side 1.1	
USB	side 1.2	
X101		
	DVI-E	
1-8	DVI	panel transmission image
X102		
	Interface	
1	PIN	panel Safety channel
2	PIN	=
3	PIN	=
4	PIN	=
5	PIN	machine reset I
6	PIN	/126.1
7	PIN	control panel #
8	PIN	/126.3
9	PIN	control panel #

X102		
10	PIN	control panel #
11	PIN	panel power supply
12	PIN	=
13	PIN	option key switch
14	PIN	/127.5
15	PIN	machine reset I
16	PIN	option key switch
17	PIN	control panel #
18	PIN	=
19	PIN	panel power supply
SH	shield	/126.4
X103		
	USB-E-IN	
USB	USB-E IN	panel transmission USB
X104		
	USB-OUT	
USB	USB-OUT	
X105		
	Bus	
BUS	IN	panel EtherCAT
X106		
BUS	OUT	panel EtherCAT
X107		
	K-Bus	
K-Bus		
i5L+		
	Potentialverteilung	
1		control panel encoder
i5L-		
2		control panel encoder
i24L+		
1		
5		machine start
6		machine Stop
i24L-		
1		
5		machine start

\$-FOS-F19-PLC



G10001225
control panel

-P1 display 15", GC+

i24L-
6 machine Stop /127.8

+HB
Schaltschrank
cabinet

X40

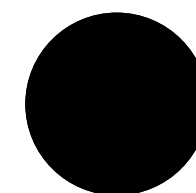
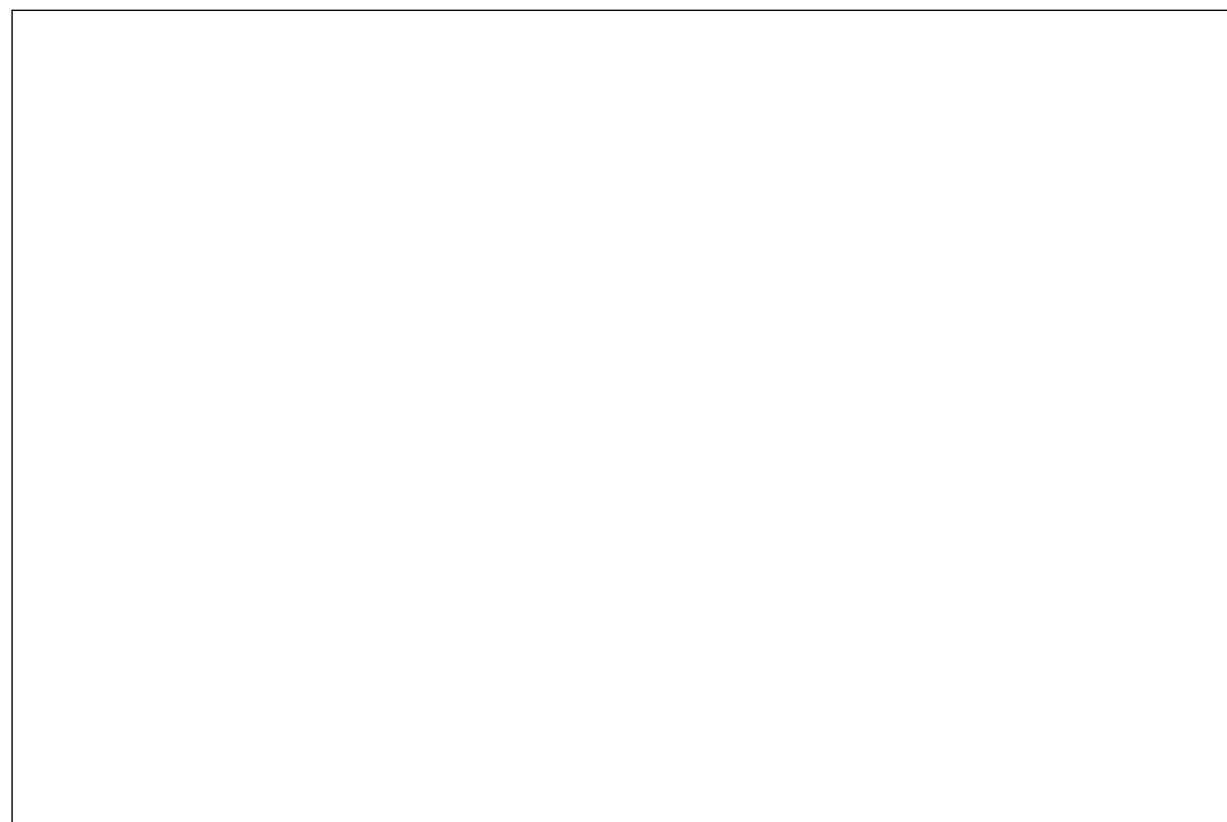
G10000287



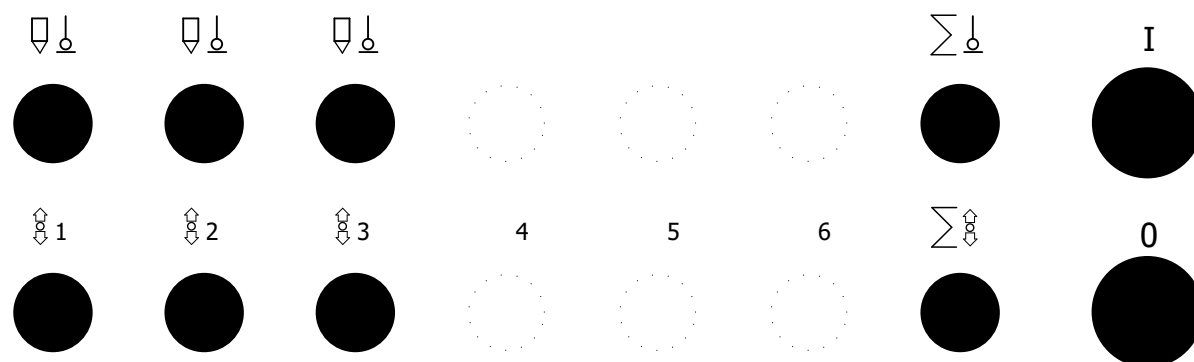
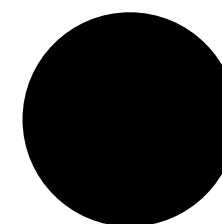
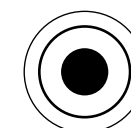
-X40 = Panel Anschluss
panel connection

Terminal	jumper	Etage		Page
1	.	1	-panel Safety channel	/125.0
2	.	2	-panel Safety channel	/125.1
3	.	1	-panel Safety channel	/125.1
4	.	2	-panel Safety channel	/125.2
5	.	1	-machine reset I	/125.3
6	.	2	-panel power supply	/124.1
7	.	1	-control panel #	/126.5
8	.	2	-panel power supply	/124.3
9	.	1	-control panel #	/126.5
10	.	2	-control panel #	/126.6
11	.	1	-panel power supply	/124.3
12	.	2	-panel power supply	/124.4
21	.	1	-panel power supply	/124.2
22	.	2	-panel power supply	/124.2
13	.	1	-option key switch	/125.4
14	.	2	-option key switch	/125.5
15	.	1	-machine reset I	/125.4
16	.	2	-option key switch	/125.5
17	.	1	-control panel #	/126.6
18	.	2	-control panel #	/126.7
19	.	1	-panel power supply	/124.1
20	.	2	-panel power supply	/124.4

Control Panel



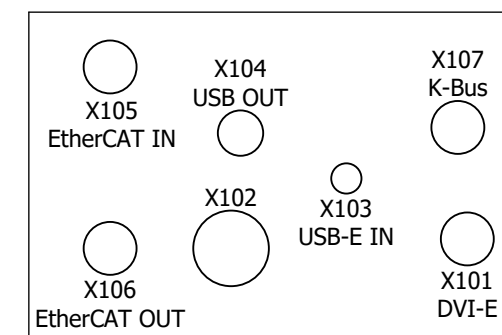
0 1



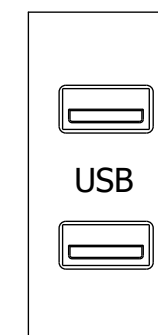
S1
S2

S3
S4

backplane



side wall

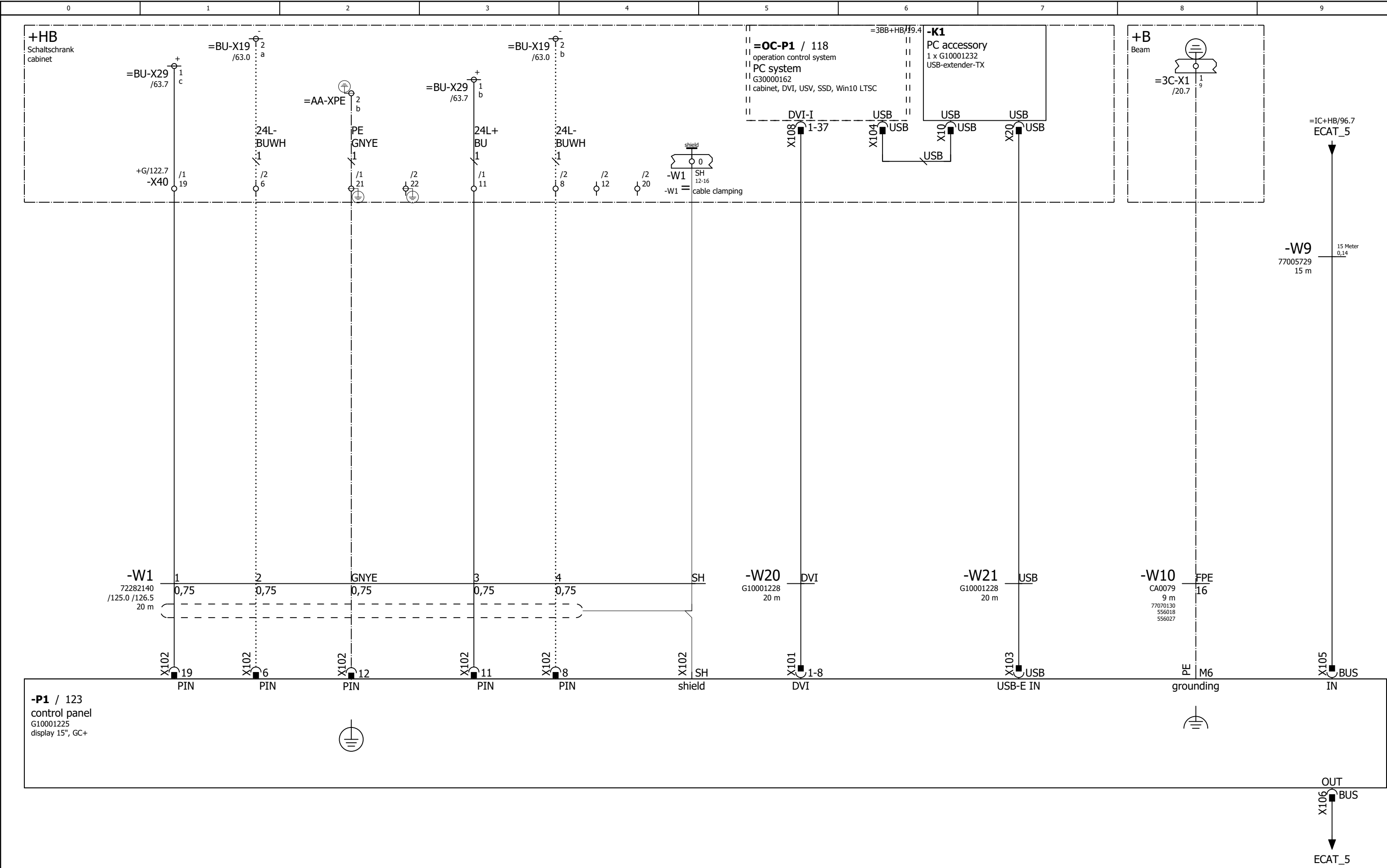


legend operating elements

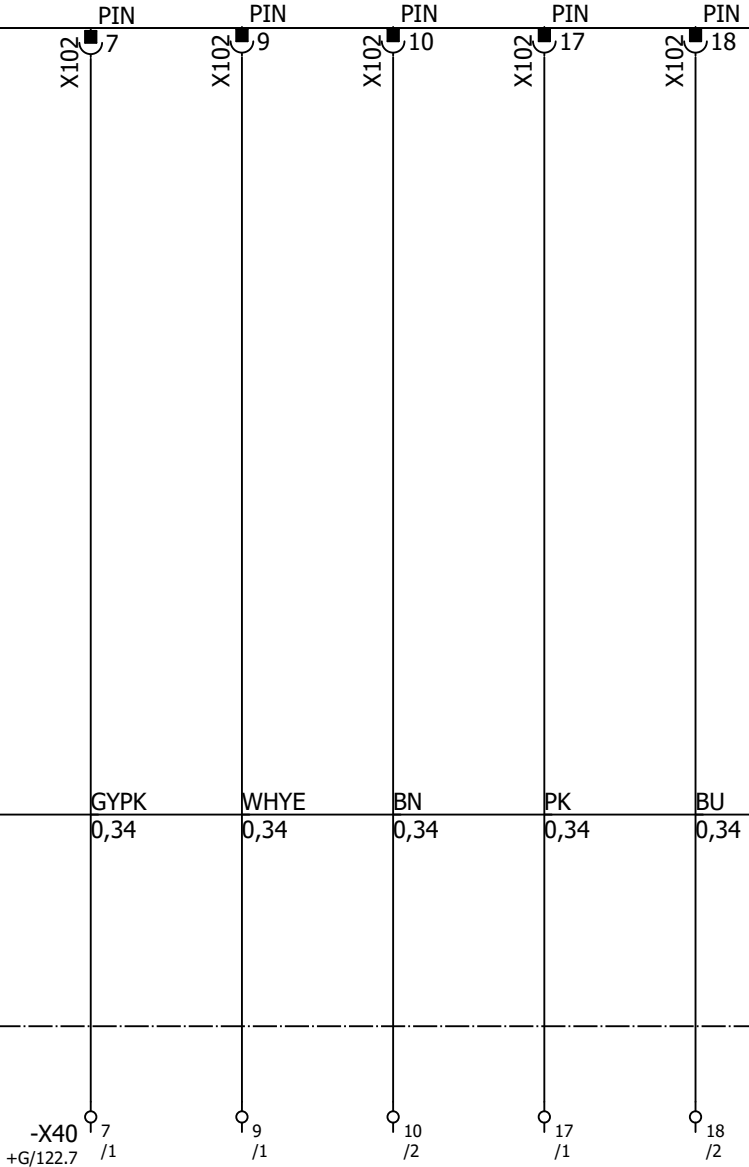
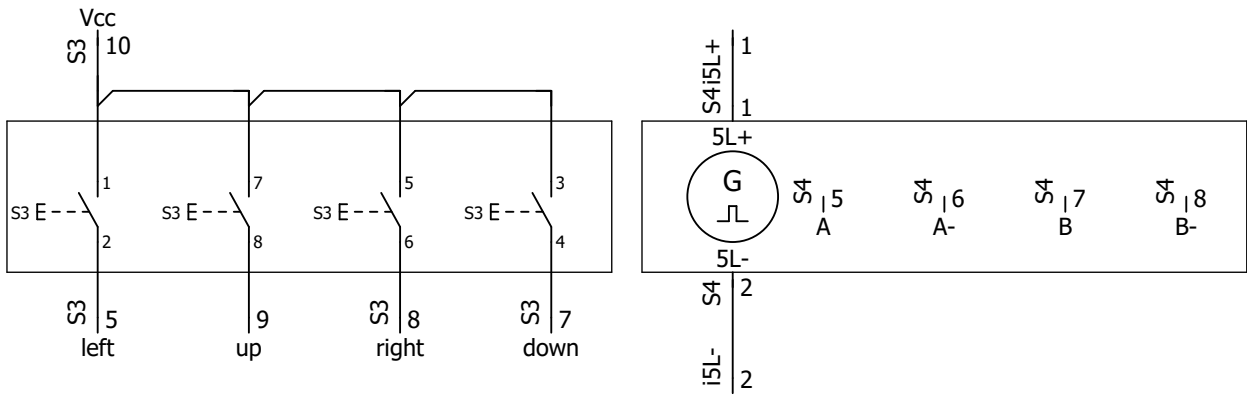
S1	emergency stop
S2	operating voltage
S3	Joystick
S4	hand wheel
S5	start
S6	Stop
S7	operating modes
S11 - S27	lifter

legend symbols

	torch ON
	scanning ON
	torch move
	scanning all
	torch move all
	start
	Stop



-P1 / 123
control panel
G10001225
display 15", GC+



+HB
Schaltschrank
cabinet

Joystick
left

Joystick
high

Joystick
right

Joystick
down

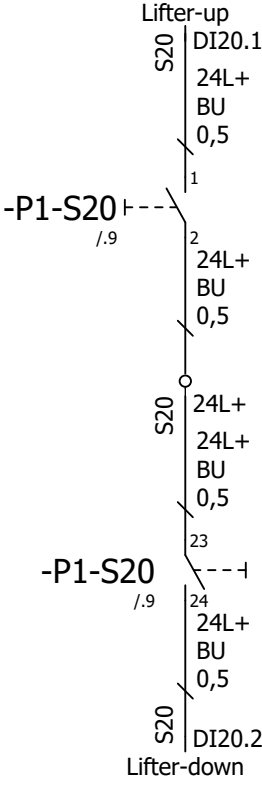
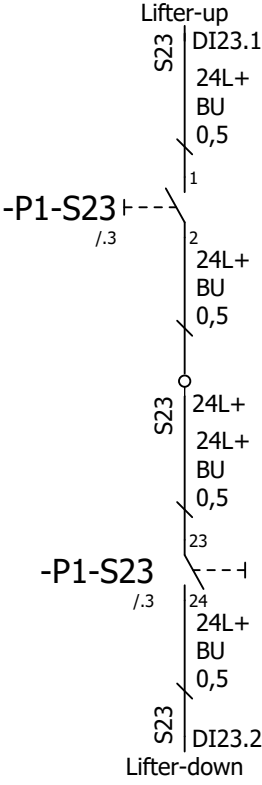
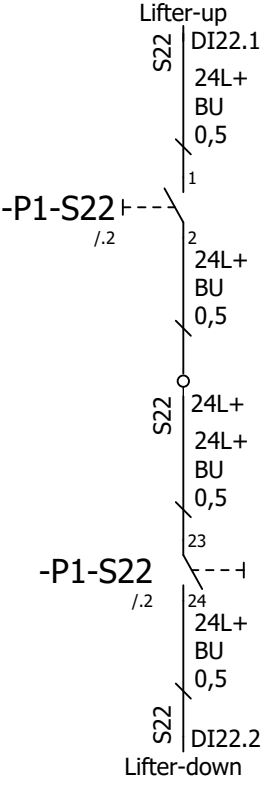
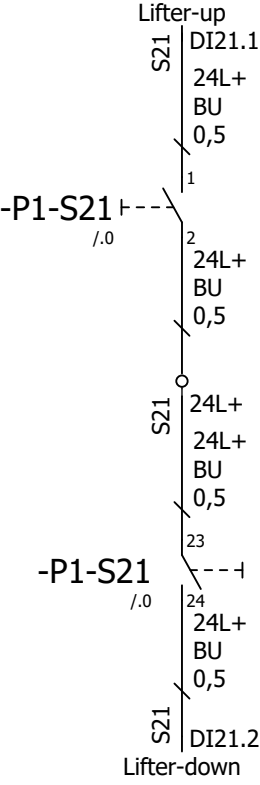
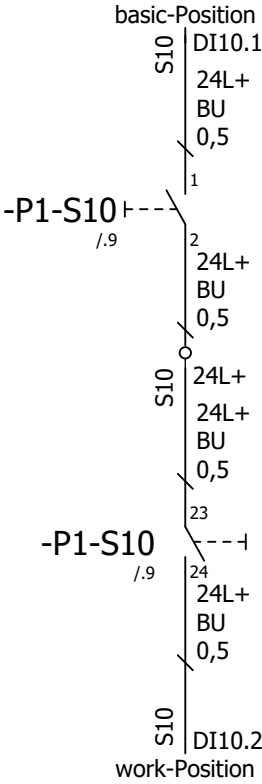
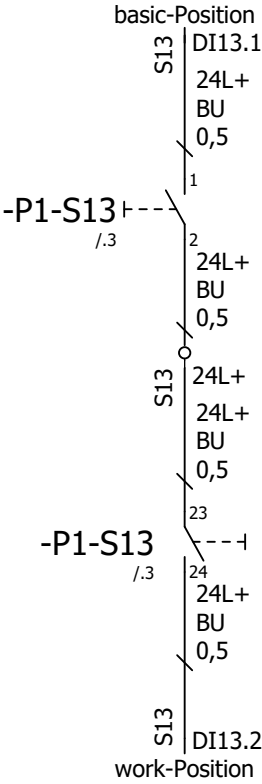
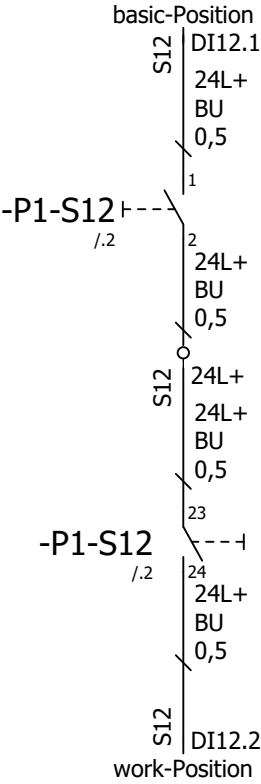
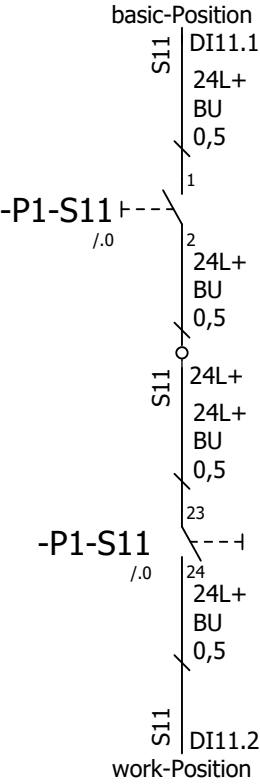
control panel
encoder

control panel
#

-P1 / 123

control panel

G10001225
display 15", GC+



panel
Lifter 1

panel
Lifter 2

panel
Lifter 3

panel
#

panel
#

panel
#

panel
Lifter All

			Toolbox_88	Option: customer: commission:	0103C GC+ +PC {R.34}			Operation interface		module		operation panel CP7902		Page 127	
			Toolbox version		Maghreb Steel			operating elements Lifter							
			OP16 - ABI - 34.80.00 - CC: EU		286002000				project: 286002000						
Modification	Date	Name	Toolbox module designation	Ed. by. Johann.Rais		24.05.2024									

G20000436

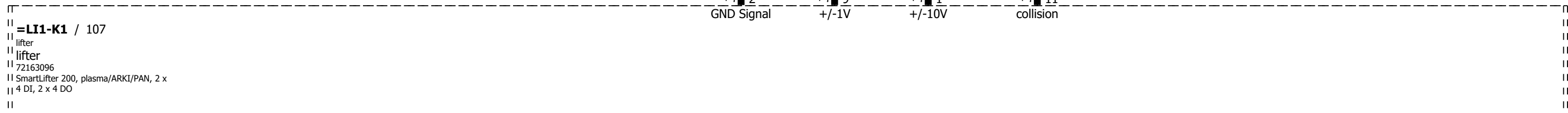
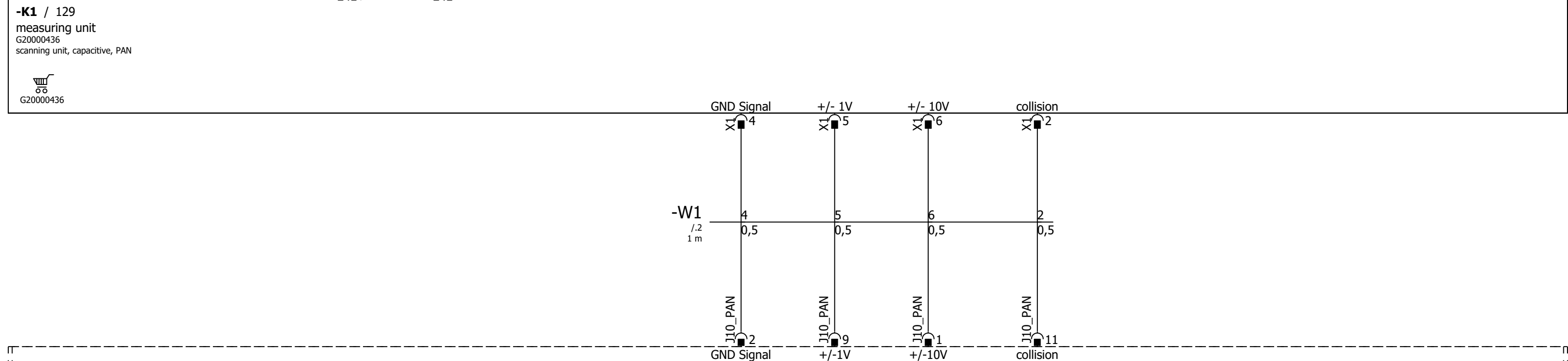
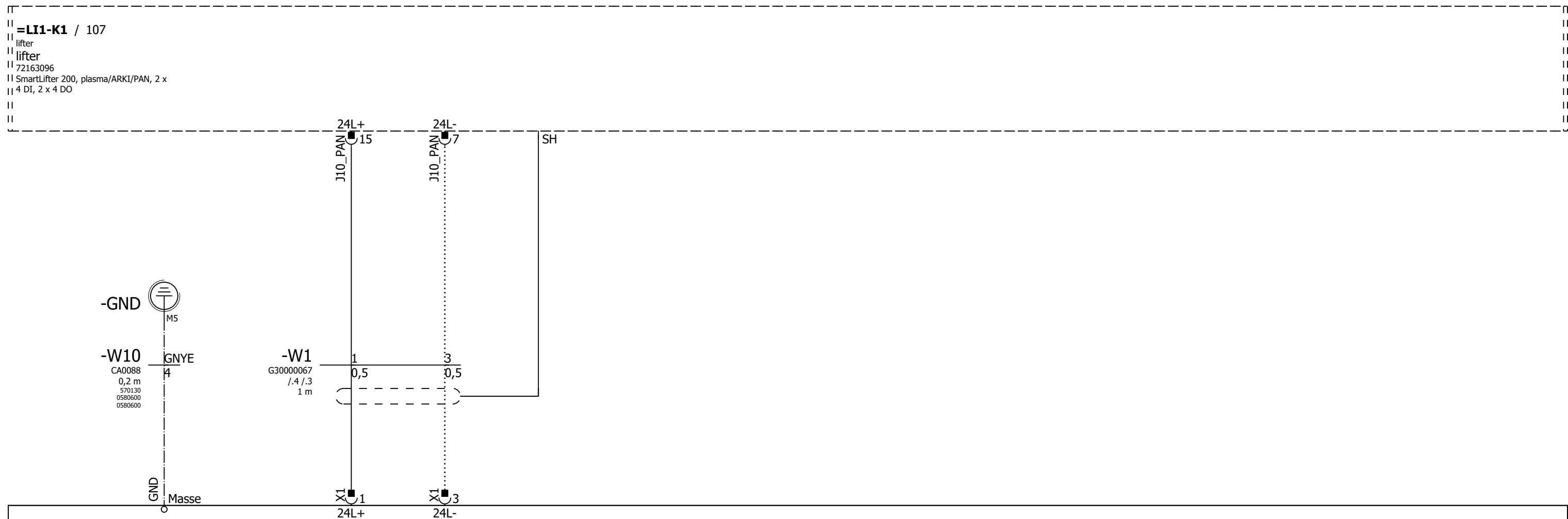
measuring unit

-K1

scanning unit, capacitive, PAN

BNC			
Sensor			
SH	shield	plate scanning	=UM1/143.5
sens	Sensor	=	=UM1/143.4
GND			
reference			
Masse		PAN sensors grounding	/131.1
X1			
D-Sub			
1	24L+	PAN sensors supply	/131.2
2	collision	PAN sensors collision	/131.6
3	24L-	PAN sensors supply	/131.2
4	GND Signal	PAN sensors desired value	/131.4
5	+/- 1V	=	/131.5
6	+/- 10V	=	/131.5

SH1	0	1	2	3	4	5	6	7	8	9
-----	---	---	---	---	---	---	---	---	---	---



PAN sensors grounding			PAN sensors supply		PAN sensors desired value		PAN sensors collision						
			Toolbox_88	Option: customer: commission:	0701A L1-Punch, HF {R.31}		MESSER	PAN		module	sensor height	Page	
			Toolbox version		Maghreb Steel			connection on lifter					129
			SH1 - ABI - 31.68.00 - CC: EU		286002000 project: 286002000								
Modification	Date	Name	Toolbox module designation	Ed. by.	Johann.Rais								



G30000122
measuring unit
-K1 scanning unit, inductive, ARKI

J2

5	MW B2	coil 1	=TO2/140.5
6	Aux	AUX	=TO2/140.6
7	MW C1	coil 2	=TO2/140.7
8	collision	collision	=TO2/140.8

J1

PLC Interface		
8	n.c.	/133.9
PLC Interface		
4	n.c.	/133.9
5	n.c.	/133.8
12	n.c.	/133.8
SH		/133.4

J1

1	setpoint	ARKI sensors desired value	/133.6
2	signal GND	=	/133.7
3	collision	ARKI sensors collision	/133.5
3'	collision		
6	up	ARKI sensors up	/133.3
7	24L-	ARKI sensors 24 VDC	/133.0
9	n.c.		/133.8
10	set	ARKI sensors height set	/133.4
11	n.c.		/133.6
13	automatic on	ARKI sensors automatic ON	/133.1
14	down	ARKI sensors lowering	/133.2
15	24L+	ARKI sensors 24 VDC	/133.0

J2

Torch		
1	MW A1	coil primary =TO2/140.2
2	MW A2	= =TO2/140.2
3	24L+	power supply 24VDC =TO2/140.3
4	MW B1	coil 1 =TO2/140.4



G30000122
measuring unit
-K1 scanning unit, inductive, ARKI

J2

5	MW B2	coil 1	=TO3/142.5
6	Aux	AUX	=TO3/142.6
7	MW C1	coil 2	=TO3/142.7
8	collision	collision	=TO3/142.8

J1

PLC Interface		
8	n.c.	/135.9
PLC Interface		
4	n.c.	/135.9
5	n.c.	/135.8
12	n.c.	/135.8
SH		/135.4

J1

1	setpoint	ARKI sensors desired value	/135.6
2	signal GND	=	/135.7
3	collision	ARKI sensors collision	/135.5
3'	collision		
6	up	ARKI sensors up	/135.3
7	24L-	ARKI sensors 24 VDC	/135.0
9	n.c.		/135.8
10	set	ARKI sensors height set	/135.4
11	n.c.		/135.6
13	automatic on	ARKI sensors automatic ON	/135.1
14	down	ARKI sensors lowering	/135.2
15	24L+	ARKI sensors 24 VDC	/135.0

J2

Torch		
1	MW A1	coil primary =TO3/142.2
2	MW A2	= =TO3/142.2
3	24L+	power supply 24VDC =TO3/142.3
4	MW B1	coil 1 =TO3/142.4

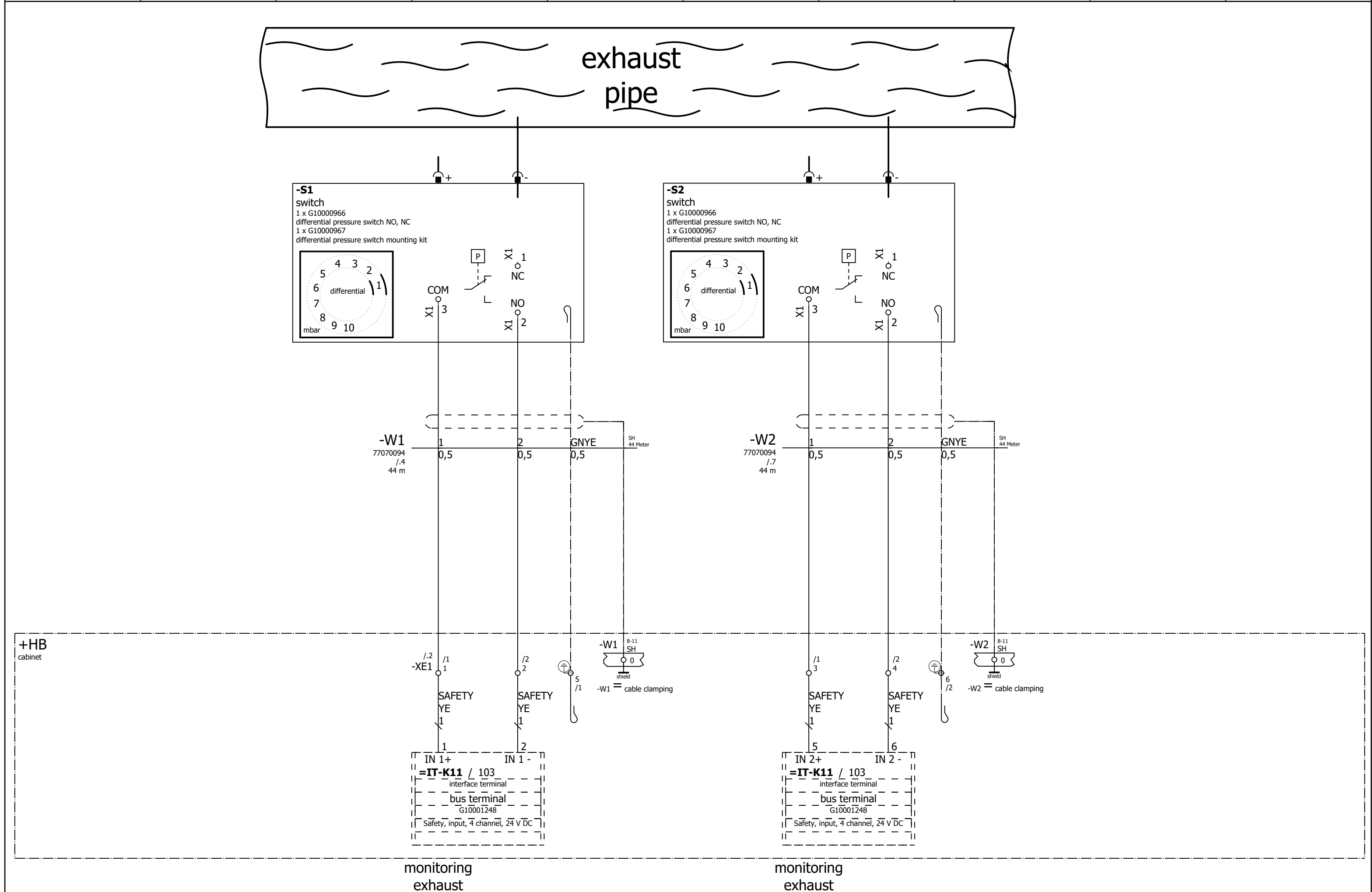
72238202

components machine

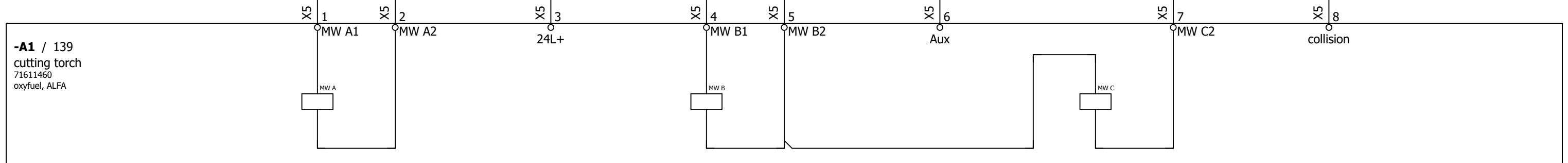
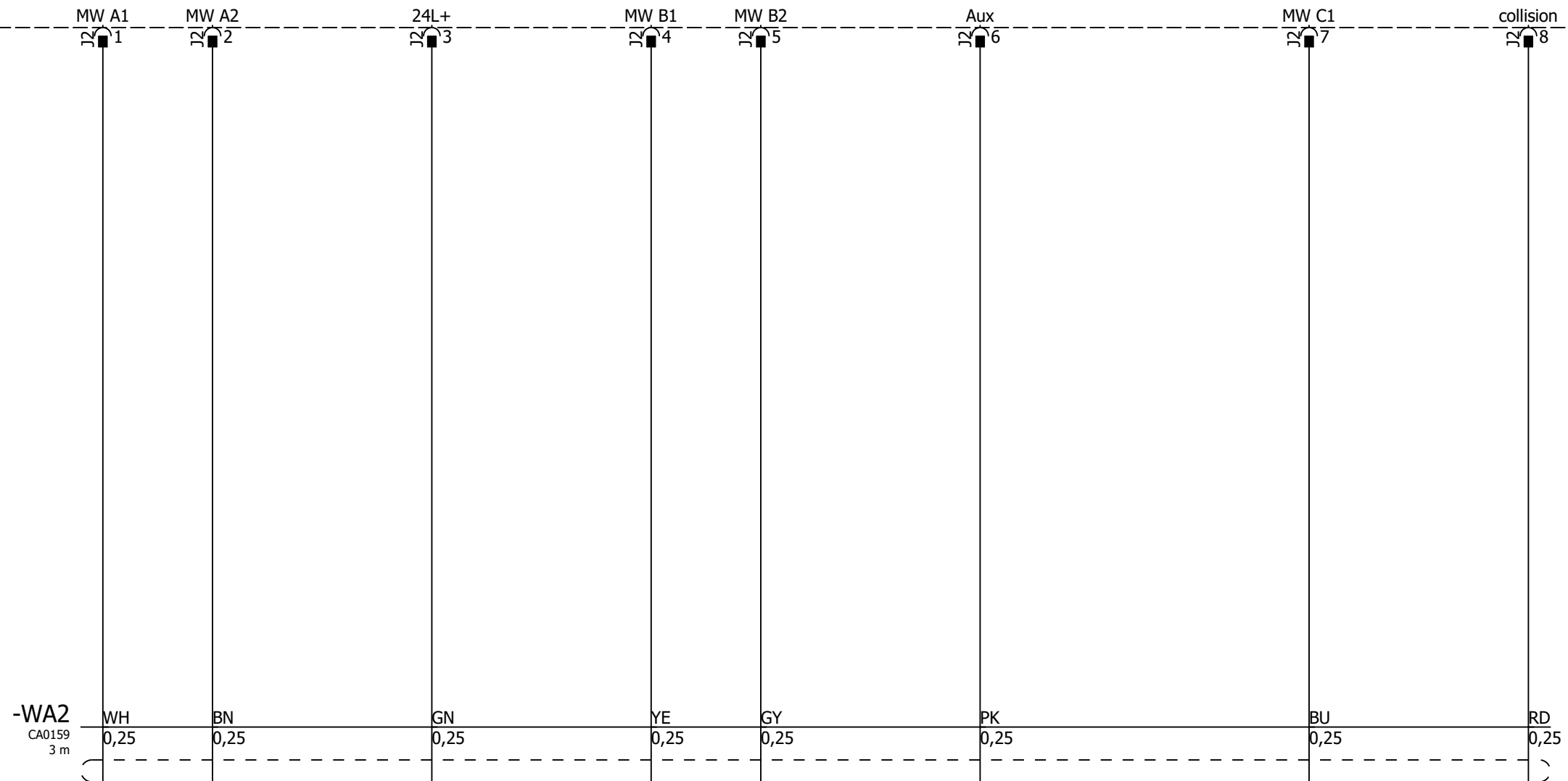
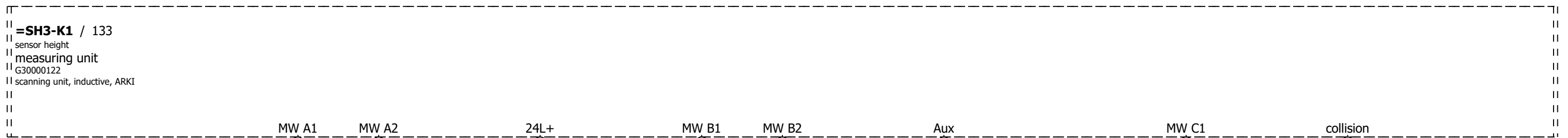
-K1

distributor air, 6 feeder

Air		
IN	Inlet	air supply /137.5
Air1		
OUT	Outlet	air =UM1+L1/143.7
Air2		
OUT	Outlet	
Air3		
OUT	Outlet	
Air4		
OUT	Outlet	
Air5		
OUT	Outlet	
Air6		
OUT	Outlet	



TO2	0	1	2	3	4	5	6	7	8	9
-----	---	---	---	---	---	---	---	---	---	---



coil
primary

power supply
24VDC

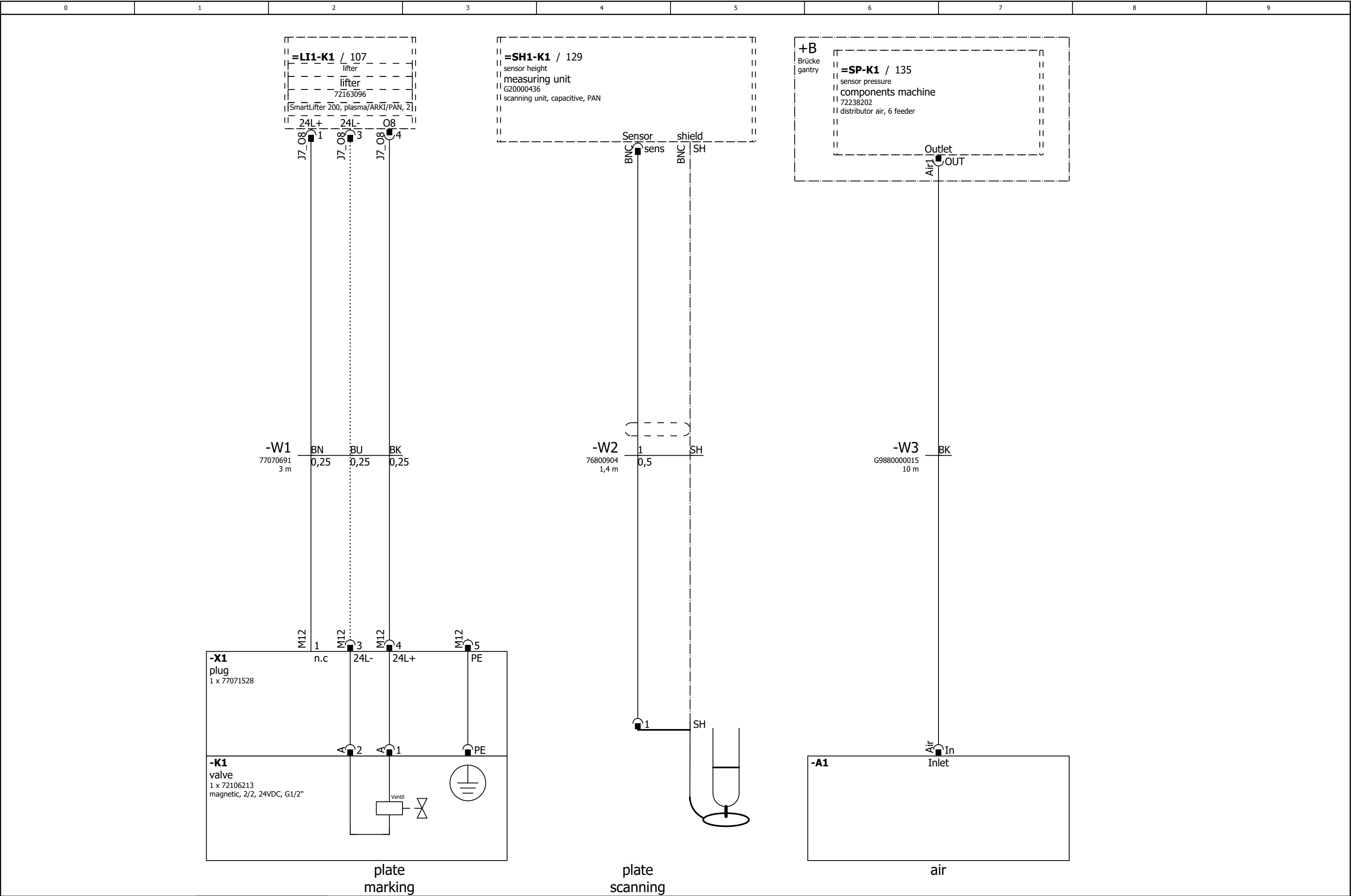
coil 1

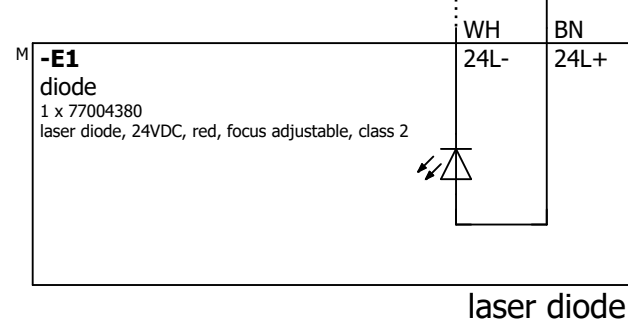
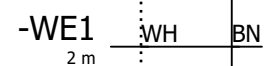
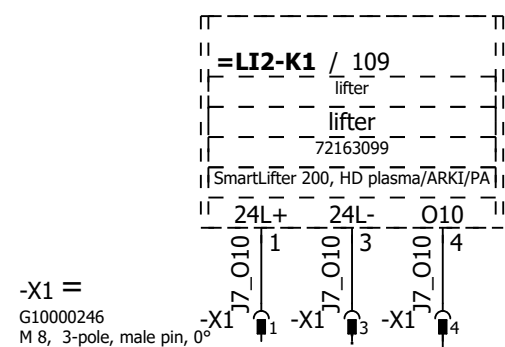
AUX

coil 2

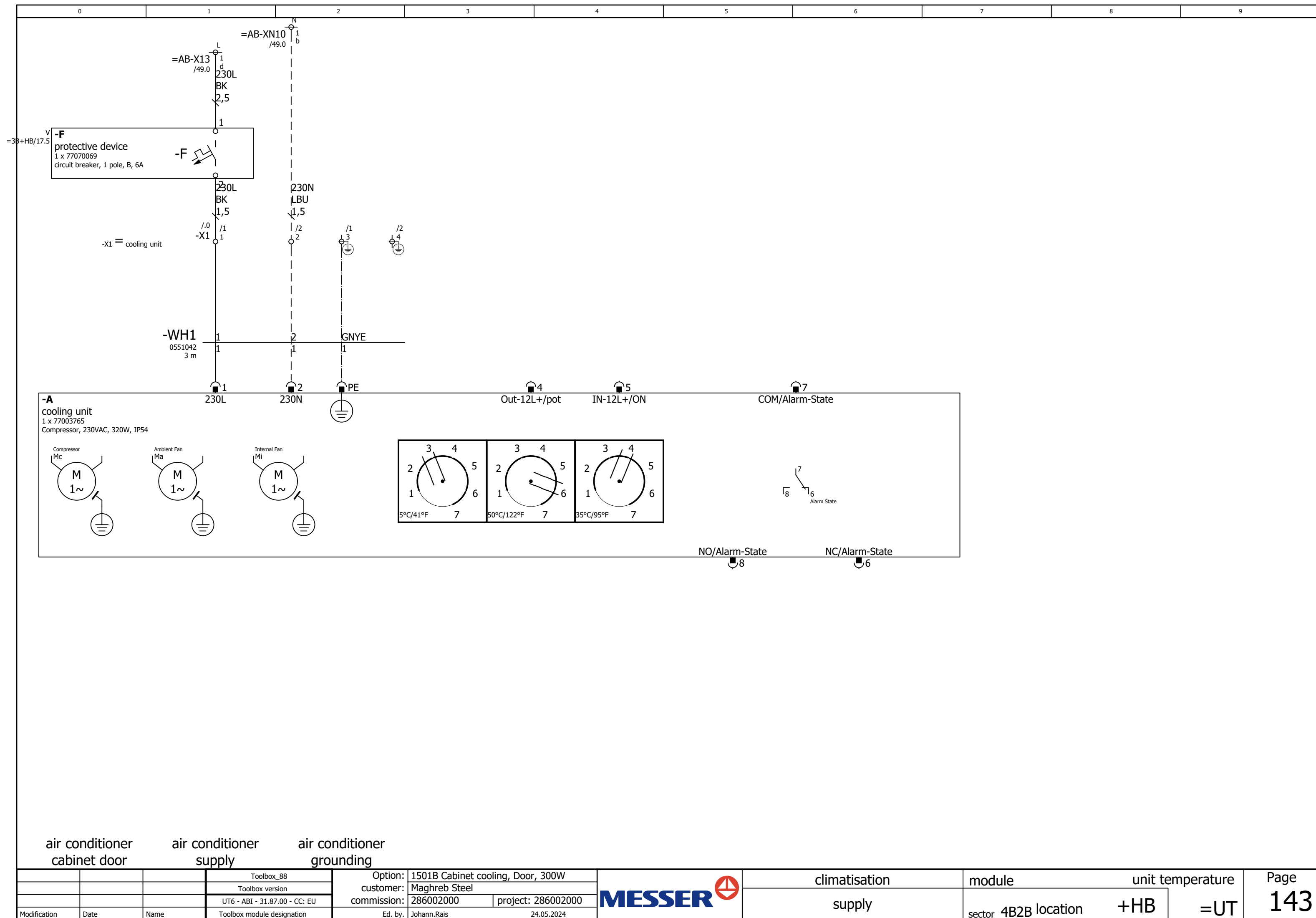
collision

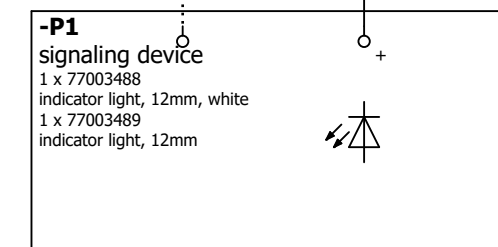
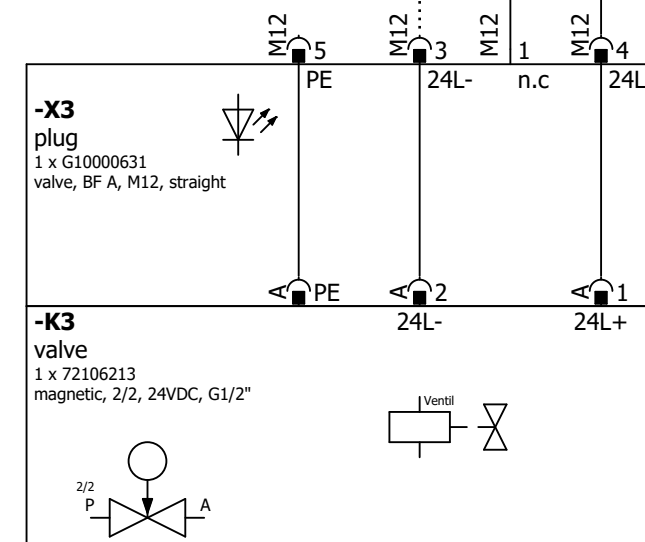
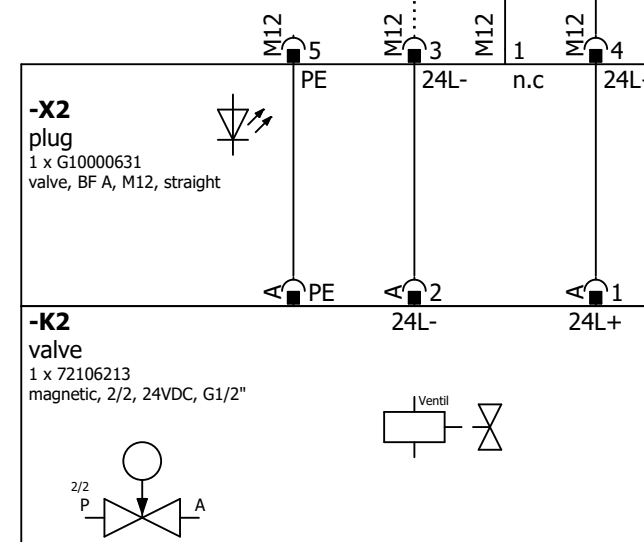
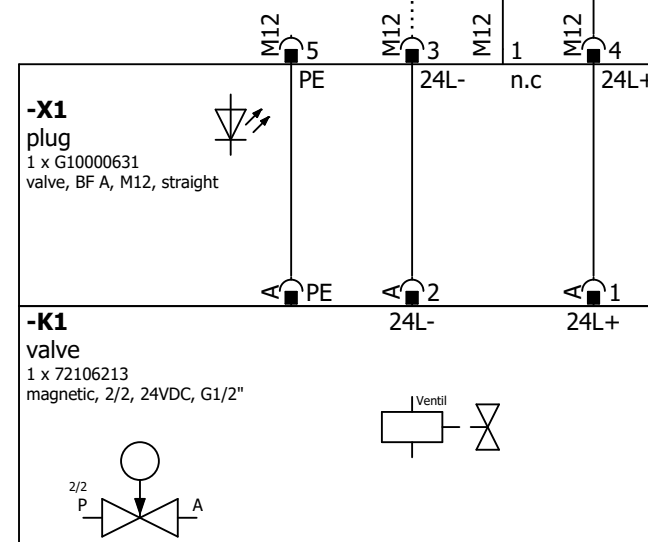
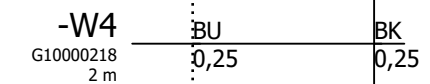
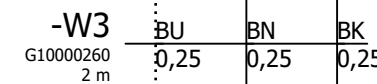
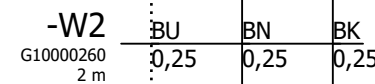
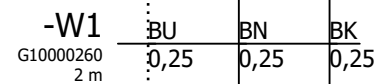
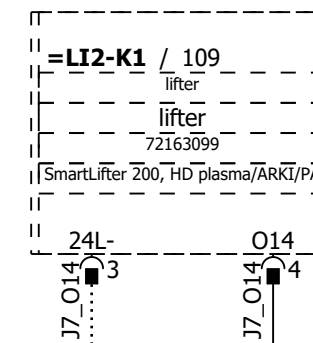
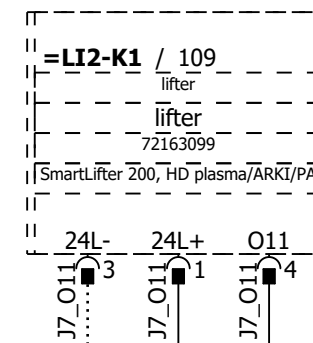
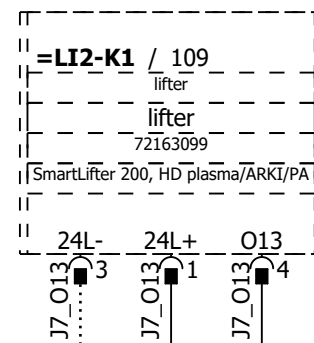
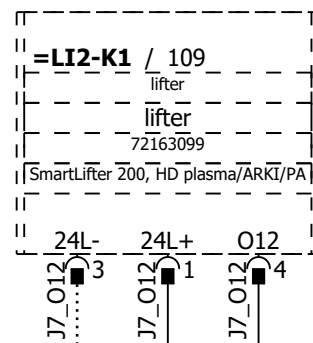
			Toolbox_88	Option:	0801A L3-Torch ALFA {R.31}			type ALFA	module	torch	Page 140
			Toolbox version	customer:	Maghreb Steel			power supply collision	sector CC3L location +L3	=TO3	
			TO2 - ABI - 31.85.00 - CC: EU	commission:	286002000	project: 286002000					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024					





			Toolbox_88	Option:	1001A L2-Laser Pointer {R.31}			laser diode	module	unit pointer	Page 142
			Toolbox version	customer:	Maghreb Steel			connection	sector CB2L location +L2	=UP2	
			UP1 - ABI - 31.32.00 - CC: EU	commission:	286002000	project: 286002000					
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024					





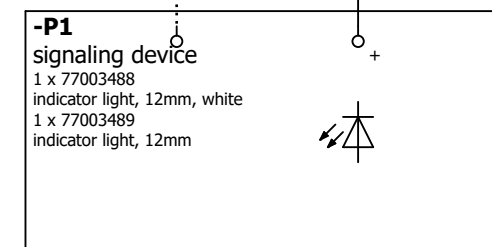
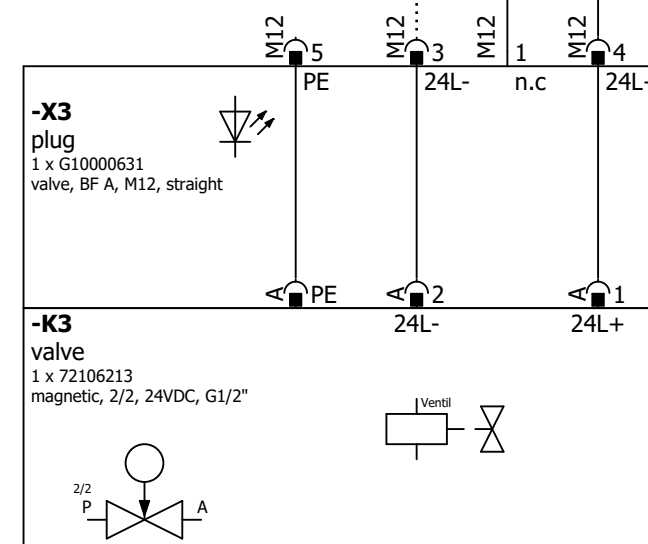
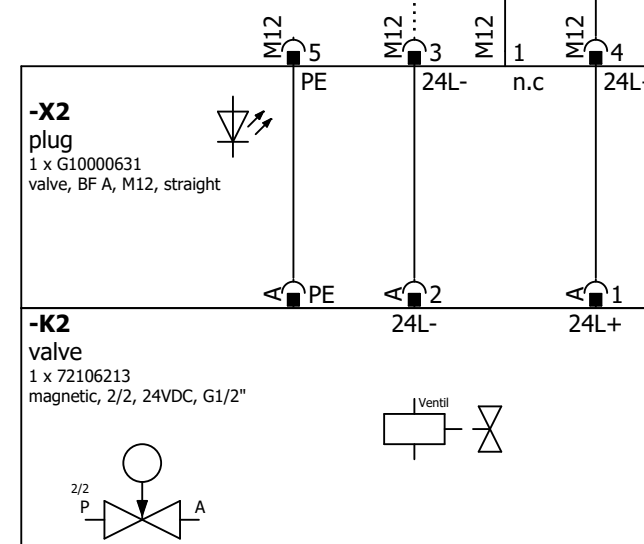
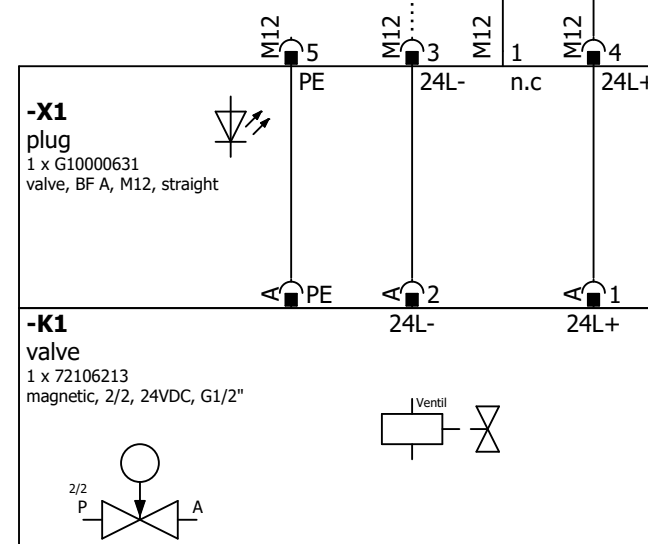
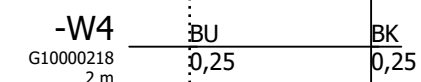
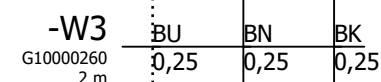
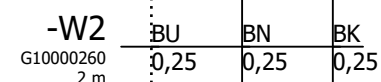
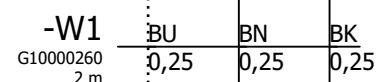
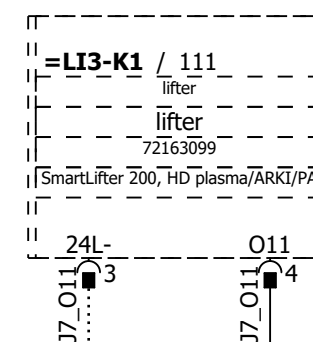
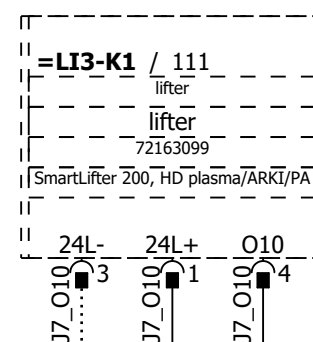
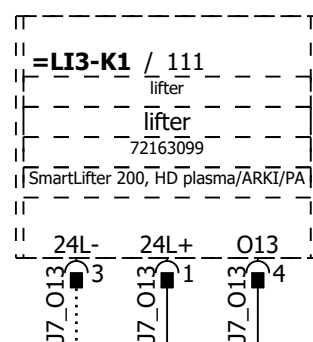
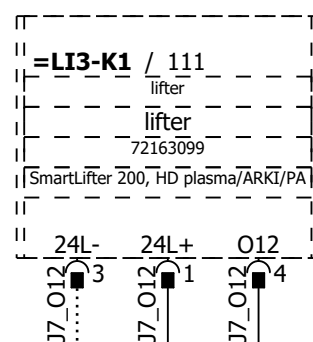
addressing
cutting gas

addressing
fuel gas

addressing
heating gas

addressing
gas flow

			Toolbox_88		Option: 0810A L2-Torch addressing {R.34}			addressing		module		valves addressing		Page 144
			Toolbox version		customer: Maghreb Steel			cutting gas, fuel gas		sector CB1L location		+L2 =VA2		
			VA1 - ABI - 31.81.01 - CC: EU		commission: 286002000 project: 286002000									
Modification	Date	Name	Toolbox module designation		Ed. by: Johann.Rais 24.05.2024									



addressing
cutting gas

addressing
fuel gas

addressing
heating gas

addressing
gas flow

			Toolbox_88	Option:	0810A L3-Torch addressing {R.34}			addressing		module	valves addressing		Page 145
			Toolbox version	customer:	Maghreb Steel			cutting gas, fuel gas	sector	CC1L location	+L3	=VA3	
			VA1 - ABI - 31.81.01 - CC: EU	commission:	286002000	project: 286002000							
Modification	Date	Name	Toolbox module designation	Ed. by:	Johann.Rais	24.05.2024							

CA0078
+HB=3B-W10

MCS* Cable Production

6 m - 16 mm²



77070529 //

M10 - female

+B=3C-X1

Page: 16

OE -

+HB=3B-X0FPE

c

cable // FPE, ring, OE-M10, 16mm²

a

grounding cabinet // cabinet

Cable Plan Formular: =Z6B++111+A/156

CA0078
+HB=3BB-W10

MCS* Cable Production

6 m - 16 mm²



M10 - female

+B=3C-X1

Page: 18

OE -

+HB=3BB-X0FPE

d

cable // FPE, ring, OE-M10, 16mm²

a

grounding cabinet // cabinet

Cable Plan Formular: =Z6B++111+A/157

CA0122
+B=3C-W1

MCS* Cable Production

0,2 m - 25 mm²

M10 - female

+B=3C-X1

Page: 21

M8 - female

+B=3C-GND

a

cable // grounding, PE, ring, M8-M10, 25mm²

grounding bar // grounding bar

Cable Plan Formular: =Z6B++111+A/158

CA0196
+B=3C-W2

MCS* Cable Production

36 m - 35 mm²

M10 - female

+B=3C-X1

Page: 21

M10 - female

+T=3CB-X1

b

cable // grounding, PE, ring, M10-M10, 35mm²

d

grounding bar // grounding bar

Cable Plan Formular: =Z6B++111+A/159

CA0197
+T=3CB-W1

MCS* Cable Production

19 m - 35 mm²



M10 - female

+T=3CB-X1

Page: 23

OE -

+PZ=CUSTOMERGND

b

cable // grounding, PE, ring, OE-M10, 35mm²

grounding bar // grounding bar

Cable Plan Formular: =Z6B++111+A/160

CA0196
+T=3CB-W12

MCS* Cable Production

0,2 m - 35 mm²

M10 - female

+T=3CB-X1

Page: 23

M10 - female

+T=3CB-GND

a

cable // grounding, PE, ring, M10-M10, 35mm²

grounding bar // grounding bar

Cable Plan Formular: =Z6B++111+A/161

G10001038
+B=3D-W1

0,1 m - 16 mm²



M6 - female

+B=3D-GND

Page: 24

M6 - female

+B=3D-GND

ground strap // M6-M6, 0,1 m

grounding machine // grounding machine

G10001039
+E=3D-W1

0,3 m - 16 mm²



M6 - female

+E=3D-GND

Page: 24

M6 - female

+E=3D-GND

ground strap // M6-M6, 0,3 m

grounding machine // grounding machine

G10001039
+G=3D-W1

0,3 m - 16 mm²



M6 - female

+G=3D-GND

Page: 24

M6 - female

+G=3D-GND

ground strap // M6-M6, 0,3 m

grounding machine // grounding machine

G10001038
+E=3D-W2

0,1 m - 16 mm²



M6 - female

+E=3D-GND

Page: 24

M6 - female

+E=3D-GND

ground strap // M6-M6, 0,1 m

grounding machine // grounding machine

G10001038
+G=3D-W2

0,1 m - 16 mm²



M6 - female

+G=3D-GND

Page: 24

M6 - female

+G=3D-GND

ground strap // M6-M6, 0,1 m

grounding machine // grounding machine

G10001108
+PRE=3DB-W0

0,3 m - 70 mm²



M12 - female

+PRE=3DB-GND

Page: 25

M12 - female

+PRE=3DB-GND

cable // single line

grounding machine // grounding machine

G10001108
+PRE=3DB-W3

0,3 m - 70 mm²

M12 - female
+PRE=3DB-GND3.1

Page: 25

M12 - female
+PRE=3DB-GND3.2

grounding machine // grounding machine

G10001108
+PRE=3DB-W6

0,3 m - 70 mm²



M12 - female
+PRE=3DB-GND6.1

Page: 25

M12 - female
+PRE=3DB-GND6.2

cable // single line

grounding machine // grounding machine



G10001108
+PRG=3DB-W6

0,3 m - 70 mm²



M12 - female
+PRG=3DB-GND6.1

Page: 25

M12 - female
+PRG=3DB-GND6.2

cable // single line

grounding machine // grounding machine

G10001108
+PRE=3DB-W7

0,3 m - 70 mm²



M12 - female
+PRE=3DB-GND7.1

Page: 25

M12 - female
+PRE=3DB-GND7.2

cable // single line

grounding machine // grounding machine

G10001108
+PRG=3DB-W7

0,3 m - 70 mm²



M12 - female
+PRG=3DB-GND7.1

Page: 25

M12 - female
+PRG=3DB-GND7.2

cable // single line

grounding machine // grounding machine

G10001108
+PRE=3DB-W8

0,3 m - 70 mm²



M12 - female
+PRE=3DB-GND8.1

Page: 25

M12 - female
+PRE=3DB-GND8.2

cable // single line

grounding machine // grounding machine

G10001108
+PRG=3DB-W8

0,3 m - 70 mm²



M12 - female
+PRG=3DB-GND8.1

Page: 25

M12 - female
+PRG=3DB-GND8.2

cable // single line

grounding machine // grounding machine

CA0212
+T=3DB-W10

MCS* Cable Production

11 m - 25 mm²

M10 - female
+T=3CB-X1

Page: 25

M12 - female
+PRE=3DB-GNDE

cable // grounding, PE, ring, M10-M12, 25mm²

grounding machine // grounding machine

Cable Plan Formular: =Z6B++111+A/162

CA0212
+T=3DB-W20

MCS* Cable Production

11 m - 25 mm²

M10 - female
+T=3CB-X1

Page: 25

M12 - female
+PRG=3DB-GNDG

cable // grounding, PE, ring, M10-M12, 25mm²

grounding machine // grounding machine

Cable Plan Formular: =Z6B++111+A/163

CA0079
+B=3E-W10

MCS* Cable Production

4 m - 16 mm²



M10 - female
+B=3C-X1

Page: 26

M6 - female
+B=3E-GND

cable // FPE, ring, M6-M10, 16mm²

grounding elements // grounding Location

Cable Plan Formular: =Z6B++111+A/164

CA0125
+L1=3E1-W1

MCS* Cable Production

8 m - 16 mm²

M10 - female
+B=3C-X1

Page: 27

M6 - female
+L1=3E1-GND1-1

cable // grounding, PE, ring, M6-M10, 16mm²

grounding elements // grounding Location

Cable Plan Formular: =Z6B++111+A/165

G10001040
+L1=3E1-W2

0,6 m - 16 mm²



M6 - female
+L1=3E1-GND1-2

Page: 27

M6 - female
+L1=3E1-GND2-1

ground strap // M6-M6, 0,6 m

grounding elements // grounding Location

CA0125
+L2=3E2-W1

MCS* Cable Production

8 m - 16 mm²

M10 - female
+B=3C-X1

Page: 28

M6 - female
+L2=3E2-GND1-1

cable // grounding, PE, ring, M6-M10, 16mm²

grounding elements // grounding Location

Cable Plan Formular: =Z6B++111+A/166

G10001040
+L2=3E2-W2

0,6 m - 16 mm²



M6 - female
+L2=3E2-GND1-2

Page: 28

M6 - female
+L2=3E2-GND2-1

ground strap // M6-M6, 0,6 m

grounding elements // grounding Location

CA0125
+L3=3E3-W1

MCS* Cable Production

8 m - 16 mm²

M10 - female
+B=3C-X1
h

Page: 29

M6 - female

cable // grounding, PE, ring, M6-M10, 16mm²

+L3=3E3-GND1-1

grounding elements // grounding Location

Cable Plan Formular: =Z6B++111+A/167

G10001040
+L3=3E3-W2

0,6 m - 16 mm²

M6 - female
+L3=3E3-GND1-2

Page: 29

M6 - female

ground strap // M6-M6, 0,6 m

+L3=3E3-GND2-1

grounding elements // grounding Location

G10000911
+HB=AA-W0

32 m - 4 mm²

OE -

Page: 43

OE -

+HB=AA-X0

cable sold by meter // PVC, 5x 4, PE, CY

+PZ=AA-K1 S

AC power supply cabinet // AC operating voltage

Cable Plan Formular: =Z6B++111+A/168

G10001294
+B=BI-W1

3 m - 2,5 mm²

OE -

Page: 61

7/8 - female

+HB=BI-X1

cable // OE-7/8", socket, 5 x 2,5, 3m

+B=BI-K1 X1

DC power supply interface // EP power - distributor

Cable Plan Formular: =Z6B++111+A/169

77005808
+B=BI-W9

5 m - 0,34 mm²

RJ45 - male

Page: 61

M8 - male

+HB=IC-K1 X8

cable network // EtherCAT, M8-RJ45, 5m

+B=BI-K1 L/A

DC power supply interface // EP power - distributor

G10000253
+HB=CXXa-W9

2 m - 0,16 mm²

RJ45 - male

Page: 66

RJ45 - male

+HB=OC-P1 X110

cable network // EtherCAT, RJ45-RJ45, 2m

+HB=CXXa-K1 X04

controller X,Xa // AX5206 X,Xa

G10000209
+HB=CY2Y3-W9

0,25 m - 0,14 mm²

RJ45 - male

Page: 68

RJ45 - male

+HB=CY2Y3-K1 X04

cable network // EtherCAT, RJ45-RJ45, 0,25m

+HB=CXXa-K1 X05

controller Y2,Y3 // AX5203 Y2,Y3

G10000458
+L1=DL1-W1

15 m - 0,34 mm²

OE -

Page: 71

M12 - female

+HB=DL1-W1

cable // OE-M12, 5-pole, socket, 15m

+L1=DL1-B1

detection limiting // limiting Y+

Cable Plan Formular: =Z6B++111+A/170

G10000458
+L3=DL3-W1

15 m - 0,34 mm²

OE -

Page: 73

M12 - female

+HB=DL3-W1

cable // OE-M12, 5-pole, socket, 15m

+L3=DL3-B1

detection limiting // limit Y-

Cable Plan Formular: =Z6B++111+A/171

G10000516
+E=EA-W1

5 m - 0,34 mm²

M12 - female

Page: 74

OE -

+E=EA-S1 X1

cable // OE-M12, 5-pole, socket, 5m

+HB=IT-K2

emergency stop additions // emergency stop

Cable Plan Formular: =Z6B++111+A/172

G10000957
+G=EAB-W1

10 m - 0,34 mm²

M12 - female

Page: 75

OE -

+G=EAB-S1 X1

cable // OE-M12, 5-pole, socket, 10m

+HB=IT-K2

emergency stop additions // emergency stop

Cable Plan Formular: =Z6B++111+A/173

G10000965
+E=EF-W1

5 m - 0,34 mm²

M12 - male

Page: 76

M12 - female

+B=IBB-K1 01

cable // M12-M12, 5-pole, 5m

+E=EF-S1

emergency stop foot protection // Fusschutzsicherung

G10000965
+E=EF-W2

5 m - 0,34 mm²



M12 - male
+B=IBB-K1 23

Page: 76

cable // M12-M12, 5-pole, 5m

M12 - female
+E=EF-S2

emergency stop foot protection // Fusschutzsicherung

G10000965
+E=EF-W3

5 m - 0,34 mm²



M12 - male
+B=IBB-K1 45

Page: 76

cable // M12-M12, 5-pole, 5m

M12 - female
+E=EF-S3

emergency stop foot protection // Fusschutzsicherung

G10000965
+E=EF-W4

5 m - 0,34 mm²



M12 - male
+B=IBB-K1 67

Page: 76

cable // M12-M12, 5-pole, 5m

M12 - female
+E=EF-S4

emergency stop foot protection // Fusschutzsicherung

G10000965
+G=EFB-W1

5 m - 0,34 mm²



M12 - male
+B=IB-K1 01

Page: 77

cable // M12-M12, 5-pole, 5m

M12 - female
+G=EFB-S1

emergency stop foot protection // Fusschutzsicherung

G10000965
+G=EFB-W2

5 m - 0,34 mm²



M12 - male
+B=IB-K1 23

Page: 77

cable // M12-M12, 5-pole, 5m

M12 - female
+G=EFB-S2

emergency stop foot protection // Fusschutzsicherung

G10000965
+G=EFB-W3

5 m - 0,34 mm²



M12 - female
+G=EFB-S3

Page: 77

cable // M12-M12, 5-pole, 5m

M12 - male
+G=EFB-X1 outG

emergency stop foot protection // Fusschutzsicherung

G10000965
+G=EFB-W4

5 m - 0,34 mm²



M12 - female
+G=EFB-S4

Page: 77

cable // M12-M12, 5-pole, 5m

M12 - male
+G=EFB-X1 outS

emergency stop foot protection // Fusschutzsicherung

G10000965
+G=EFB-W5

5 m - 0,34 mm²



+B=IB-K1 67

Page: 77

cable // M12-M12, 5-pole, 5m

M12 - female
+G=EFB-S5

emergency stop foot protection // Fusschutzsicherung

G10001242
+B=FO-W1

5 m - 1,09 mm²



OE -
+HB=FO-K1

Page: 85

cable // OE-M23, 19-pole, socket (plug female), 5m

OE -
+HB=FO-W1

flow controller oxyfuel // gas supply OmniFlow

G10000787
+B=FO-W2

5 m - 0,34 mm²



OE -
+HB=FO-W2

Page: 87

cable // OE-M12, 5-pole, plug, 5m

OE -
+HB=FO-K4

flow controller oxyfuel // gas supply OmniFlow

Cable Plan Formular: =Z6B++111+A/174

G10001037
+B=FO-W10

0,2 m - 16 mm²



M6 - female
+B=FO-GND

Page: 85

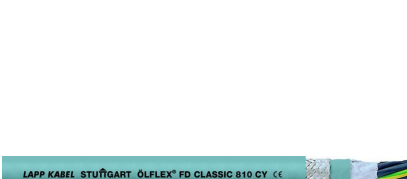
ground strap // M6-M6, 0,2 m

M6 - female
+B=FO-R1 GND

flow controller oxyfuel // gas supply OmniFlow

0551076
+HG=FS-W0

38 m - 1,5 mm²



OE -
+HB=FS-X1

Page: 89

cable sold by meter // PVC, 3x 1,5, PE, CY

OE -
+HG=FS-X1

flow controller safety // Durchfluss Sicherheit

Cable Plan Formular: =Z6B++111+A/175

19500410
+HG=FS-W1

1 m - 0,75 mm²



OE -

+HG=FS-X1

Page: 89

cable sold by meter // PVC, 3x 0,75, PE

OE -

+HG=FS-KH1 X1

flow controller safety // Durchfluss Sicherheit

Cable Plan Formular: =Z6B++111+A/176

19500410
+HG=FS-W2

1 m - 0,75 mm²



OE -

+HG=FS-X1

Page: 89

cable sold by meter // PVC, 3x 0,75, PE

OE -

+HG=FS-KH2 X1

flow controller safety // Durchfluss Sicherheit

Cable Plan Formular: =Z6B++111+A/177

19500410
+HG=FS-W3

1 m - 0,75 mm²



OE -

+HG=FS-X1

Page: 89

cable sold by meter // PVC, 3x 0,75, PE

OE -

+HG=FS-KF1 A1

flow controller safety // Durchfluss Sicherheit

Cable Plan Formular: =Z6B++111+A/178

19500410
+HG=FS-W4

1 m - 0,75 mm²



OE -

+HG=FS-X1

Page: 89

cable sold by meter // PVC, 3x 0,75, PE

OE -

+HG=FS-KF2 A1

flow controller safety // Durchfluss Sicherheit

Cable Plan Formular: =Z6B++111+A/179

19500410
+HG=FS-W5

1 m - 0,75 mm²



OE -

+HG=FS-X1

Page: 89

cable sold by meter // PVC, 3x 0,75, PE

OE -

+HG=FS-KC1 X1

flow controller safety // Durchfluss Sicherheit

Cable Plan Formular: =Z6B++111+A/180

19500410
+HG=FS-W6

1 m - 0,75 mm²



OE -

+HG=FS-X1

Page: 89

cable sold by meter // PVC, 3x 0,75, PE

OE -

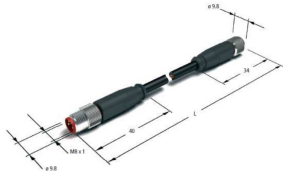
+HG=FS-KC2 X1

flow controller safety // Durchfluss Sicherheit

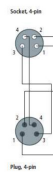
Cable Plan Formular: =Z6B++111+A/181

G10000127
+B=IB-W1

10 m - 0,34 mm²



Power cable | M8 plug, straight, 4-pin – M8 socket, straight, 4-pin



M8 - female

+B=IB-K1 24V IN

Page: 92

cable // power line, M8-M8, 10m

M8 - male

+B=IBB-K1 24V OUT

interface box // Fieldbus module

G10001071
+B=IB-W9

7,5 m - 0,34 mm²



M8 - male

+B=IBB-K1 L/A

Page: 92

cable network // EtherCAT, M8-M8, 7,5m

M8 - male

+B=IB-K1 L/A

interface box // Fieldbus module

G10000124
+B=IBB-W1

0,3 m - 0,34 mm²



Power cable | M8 plug, straight, 4-pin – M8 socket, straight, 4-pin



+B=BI-K1 1

Page: 94

cable // power line, M8-M8, 0,3m

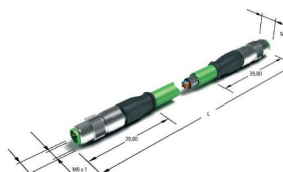
M8 - female

+B=IBB-K1 24V IN

interface box // Fieldbus module

G10001272
+B=IBB-W9

0,3 m - 0,34 mm²



M8 - male

+B=IBB-K1 L/A

Page: 94

cable network // EtherCAT, M8-M8, 0,3m

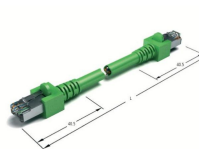
M8 - male

+B=BI-K1 L/A

interface box // Fieldbus module

G10000200
+HB=IC-W9

0,5 m - 0,16 mm²



Industrial Ethernet/EtherCAT patch cable



RJ45 - male

+HB=IT-K1 X2

Page: 96

cable network // EtherCAT, RJ45-RJ45, 0,5m

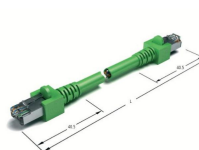
RJ45 - male

+HB=IC-K1 X1

interface control // star hub

G10000253
+HB=IT-W9

2 m - 0,16 mm²



Industrial Ethernet/EtherCAT patch cable



RJ45 - male

+HB=IT-K1 X1

Page: 97

cable network // EtherCAT, RJ45-RJ45, 2m

RJ45 - male

+HB=CY2Y3-K1 X05

interface terminal // bus coupler

CA0000
+L1=LI1-W1

MCS* Cable Production

10 m - 1 mm²

OE -

Page: 107

M22 - female

+HB=LI1-W1

cable // OE-M22, 4-core

+L1=LI1-K1 J5

lifter // SL 200

Cable Plan Formular: =Z6B++111+A/182

CA0168
+L1=LI1-W2

MCS* Cable Production

11 m - 0,34 mm²

M8 - male

Page: 107

M12 - female

+B=BI-K1 2

cable // M8-M12, 5-core, <=15m

+L1=LI1-K1 J3

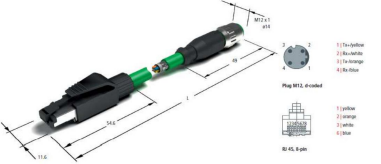
lifter // SL 200

Cable Plan Formular: =Z6B++111+A/183

G10001285
+L1=LI1-W9

MCS* Cable Production

10 m - 0,34 mm²



RJ45 - male

Page: 107

M12 - male

+HB=IC-K1 X2

cable network // EtherCAT, M12-RJ45, 10m

+L1=LI1-K1 J1

lifter // SL 200

EtherCAT cable | M12 plug, straight, d-coded, 4-pin - RJ 45 plug, straight

CA0000
+L2=LI2-W1

MCS* Cable Production

10 m - 1 mm²

OE -

Page: 109

M22 - female

+HB=LI2-W1

cable // OE-M22, 4-core

+L2=LI2-K1 J5

lifter // SL 200 HD

Cable Plan Formular: =Z6B++111+A/184

CA0168
+L2=LI2-W2

MCS* Cable Production

11 m - 0,34 mm²

M8 - male

Page: 109

M12 - female

+B=BI-K1 3

cable // M8-M12, 5-core, <=15m

+L2=LI2-K1 J3

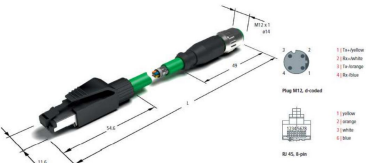
lifter // SL 200 HD

Cable Plan Formular: =Z6B++111+A/185

G10001285
+L2=LI2-W9

MCS* Cable Production

10 m - 0,34 mm²



RJ45 - male

Page: 109

M12 - male

+HB=IC-K1 X3

cable network // EtherCAT, M12-RJ45, 10m

+L2=LI2-K1 J1

lifter // SL 200 HD

EtherCAT cable | M12 plug, straight, d-coded, 4-pin - RJ 45 plug, straight

CA0000
+L3=LI3-W1

MCS* Cable Production

10 m - 1 mm²

OE -

Page: 111

M22 - female

+HB=LI3-W1

cable // OE-M22, 4-core

+L3=LI3-K1 J5

lifter // SL 200 HD

Cable Plan Formular: =Z6B++111+A/186

CA0168
+L3=LI3-W2

MCS* Cable Production

11 m - 0,34 mm²

M8 - male

Page: 111

M12 - female

+B=BI-K1 4

cable // M8-M12, 5-core, <=15m

+L3=LI3-K1 J3

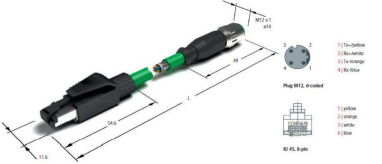
lifter // SL 200 HD

Cable Plan Formular: =Z6B++111+A/187

G10001285
+L3=LI3-W9

MCS* Cable Production

10 m - 0,34 mm²



RJ45 - male

Page: 111

M12 - male

+HB=IC-K1 X4

cable network // EtherCAT, M12-RJ45, 10m

+L3=LI3-K1 J1

lifter // SL 200 HD

EtherCAT cable | M12 plug, straight, d-coded, 4-pin - RJ 45 plug, straight

G10001322
+E=MX-W1

MCS* Cable Production

8 m - 2,57 mm²



M23 - female

Page: 112

X13+X14 - female

+E=MX-M1

cable motor // motor, AMxxxx, OCT, 1,5+0,75+0,32, 8m

+HB=CXXa-K1 X13

motor X // AM8542 OCT X

G10001037
+E=MX-W10

MCS* Cable Production

0,2 m - 16 mm²



M6 - female

Page: 112

M6 - female

+E=MX-GND

ground strap // M6-M6, 0,2 m

+E=MX-GND

motor X // AM8542 OCT X

G10001323
+G=MXa-W1

MCS* Cable Production

10 m - 2,57 mm²



M23 - female

Page: 113

X13+X14 - female

+G=MXa-M1

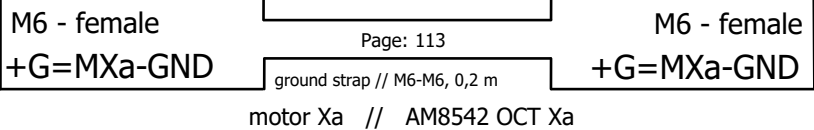
cable motor // motor, AMxxxx, OCT, 1,5+0,75+0,32, 10m

+HB=CXXa-K1 X23

motor Xa // AM8542 OCT Xa

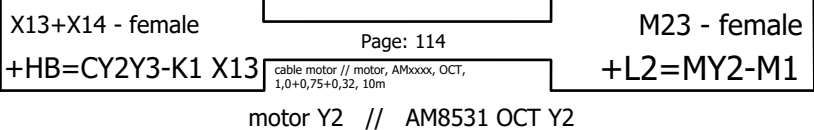
G10001037
+G=MXa-W10

0,2 m - 16 mm²



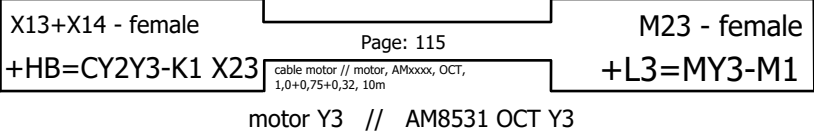
G10001310
+L2=MY2-W1

10 m - 2,07 mm²



G10001310
+L3=MY3-W1

10 m - 2,07 mm²

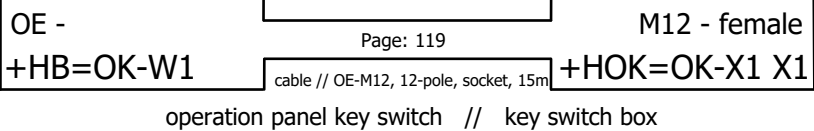


77011241
+HOK=OK-W1

15 m - 0,14 mm²



77003783 //

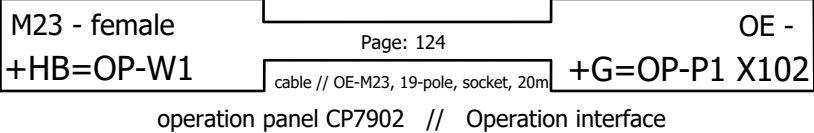


Cable Plan Formular: =Z6B++111+A/189

72282140
+G=OP-W1

MCS* Cable Production

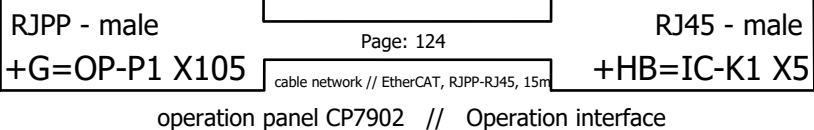
20 m - 1,09 mm²



Cable Plan Formular: =Z6B++111+A/190

77005729
+G=OP-W9

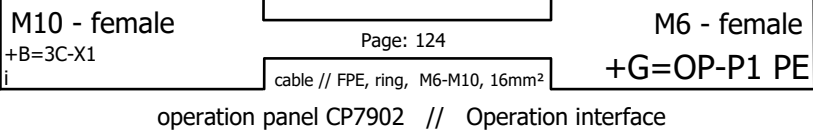
15 m - 0,14 mm²



CA0079
+G=OP-W10

MCS* Cable Production

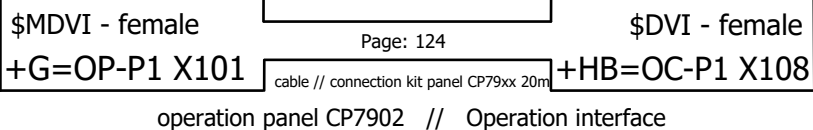
9 m - 16 mm²



Cable Plan Formular: =Z6B++111+A/191

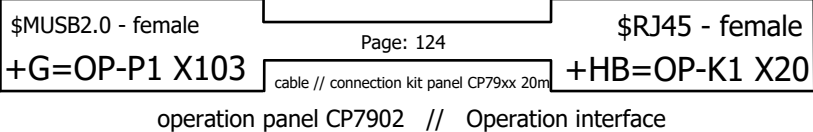
G10001228
+G=OP-W20

20 m - 9999 mm²



G10001228
+G=OP-W21

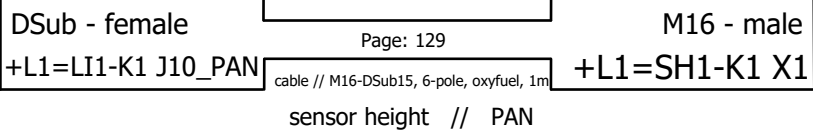
20 m - 9999 mm²



G30000067
+L1=SH1-W1

MCS* Cable Production

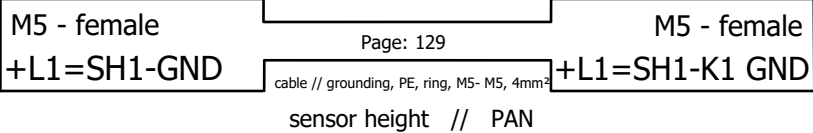
1 m - 9999 mm²



CA0088
+L1=SH1-W10

MCS* Cable Production

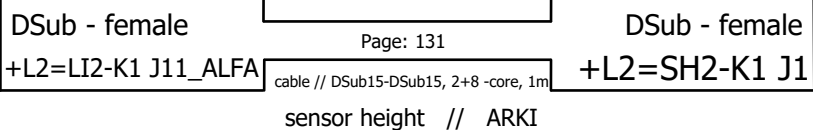
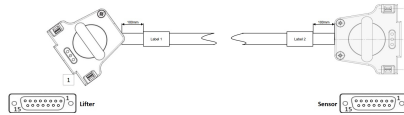
0,2 m - 4 mm²



Cable Plan Formular: =Z6B++111+A/192

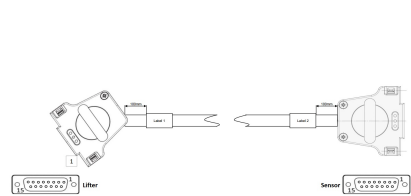
G30000143
+L2=SH2-W1

1 m - 0,75 mm²



G30000143
+L3=SH3-W1

1 m - 0,75 mm²



DSub - female
+L3=LI3-K1 J11_ALFA
Page: 133
cable // DSub15-DSub15, 2+8 -core, 1m
sensor height // ARKI
DSub - female
+L3=SH3-K1 J1

77070094
+P=SPB-W1

44 m - 0,5 mm²



OE -
+HB=SPB-W1
Page: 136
cable sold by meter // PVC, 3x 0,5, PE, CY
sensor pressure // differential pressure monitor
OE -
+P=SPB-S1 X1

Cable Plan Formular: =Z6B++111+A/193

77070094
+P=SPB-W2

44 m - 0,5 mm²



OE -
+HB=SPB-W2
Page: 136
cable sold by meter // PVC, 3x 0,5, PE, CY
sensor pressure // differential pressure monitor
OE -
+P=SPB-S2 X1

Cable Plan Formular: =Z6B++111+A/194

77070691
+L2=TO2-W1

3 m - 0,25 mm²



M8 - male
+L2=LI2-K1 J7_O9
Page: 137
cable // M8-M12, 3-pole, 3m
torch // type ALFA
M12 - female
+L2=TO2-A1 X1

77070679
+L2=TO2-WA1

3 m - 0,25 mm²



+L2=LI2-K1 J7_O8
Page: 137
cable // OE-M8, 3-pole,plug, 3m
torch // type ALFA
+L2=TO2-A1 X2

CA0159
+L2=TO2-WA2

MCS* Cable Production

3 m - 0,25 mm²

+L2=SH2-K1 J2
Page: 138
cable // OE-M12, 8-core, socket, 0°
torch // type ALFA
+L2=TO2-A1 X5

Cable Plan Formular: =Z6B++111+A/195

77070691
+L3=TO3-W1

3 m - 0,25 mm²



M8 - male
+L3=LI3-K1 J7_O9
Page: 139
cable // M8-M12, 3-pole, 3m
torch // type ALFA
M12 - female
+L3=TO3-A1 X1

77070679
+L3=TO3-WA1

3 m - 0,25 mm²



+L3=LI3-K1 J7_O8
Page: 139
cable // OE-M8, 3-pole,plug, 3m
torch // type ALFA
+L3=TO3-A1 X2

CA0159
+L3=TO3-WA2

MCS* Cable Production

3 m - 0,25 mm²

+L3=SH3-K1 J2
Page: 140
cable // OE-M12, 8-core, socket, 0°
torch // type ALFA
+L3=TO3-A1 X5

Cable Plan Formular: =Z6B++111+A/196

77070691
+L1=UM1-W1

3 m - 0,25 mm²



M12 - female
+L1=UM1-X1 M12
Page: 141
cable // M8-M12, 3-pole, 3m
unit marking // HF punch marker
M8 - male
+L1=LI1-K1 J7_O8

76800904
+L1=UM1-W2

MCS* Cable Production

1,4 m - 0,5 mm²

BNCP - male
+L1=SH1-K1 BNC
Page: 141
cable // BNC-BNC, 0°-0°, 1,40m
unit marking // HF punch marker
BNC - male
+L1=UM1-A2

G988000015
+L1=UM1-W3

10 m - 9999 mm²



G1/4L - female
+B=SP-K1 Air1
Page: 141
hose // hose, 8mm
unit marking // HF punch marker
G1/4L - female
+L1=UM1-A1 Air

+L2=UP2-WE1

2 m

+L2=UP2-X1
J7_O10:3

Page: 142

+L2=UP2-E1

potted line end

unit pointer // laser diode

G10000260

+L2=VA2-W1

2 m - 0,25 mm²

M8 - male

Page: 144

M12 - female

+L2=LI2-K1 J7_O12

cable // M8-M12, 3-pole, 2m

+L2=VA2-X1 M12

valves adressung // adressung

G10000260

+L2=VA2-W2

2 m - 0,25 mm²

M8 - male

Page: 144

M12 - female

+L2=LI2-K1 J7_O13

cable // M8-M12, 3-pole, 2m

+L2=VA2-X2 M12

valves adressung // adressung

G10000260

+L2=VA2-W3

2 m - 0,25 mm²

M8 - male

Page: 144

M12 - female

+L2=LI2-K1 J7_O11

cable // M8-M12, 3-pole, 2m

+L2=VA2-X3 M12

valves adressung // adressung

G10000218

+L2=VA2-W4

2 m - 0,25 mm²

OE -

Page: 144

M8 - male

+L2=VA2-P1

cable // OE-M8, 3-pole,plug, 2m

+L2=LI2-K1 J7_O14

valves adressung // adressung

Connection cable | M8 plug, straight, 3-pin – open end

Cable Plan Formular: =Z6B++111+A/197

G10000260

+L3=VA3-W1

2 m - 0,25 mm²

M8 - male

Page: 145

M12 - female

+L3=LI3-K1 J7_O12

cable // M8-M12, 3-pole, 2m

+L3=VA3-X1 M12

valves adressung // adressung

G10000260

+L3=VA3-W2

2 m - 0,25 mm²

M8 - male

Page: 145

M12 - female

+L3=LI3-K1 J7_O13

cable // M8-M12, 3-pole, 2m

+L3=VA3-X2 M12

valves adressung // adressung

G10000260

+L3=VA3-W3

2 m - 0,25 mm²

M8 - male

Page: 145

M12 - female

+L3=LI3-K1 J7_O10

cable // M8-M12, 3-pole, 2m

+L3=VA3-X3 M12

valves adressung // adressung

G10000218

+L3=VA3-W4

2 m - 0,25 mm²

OE -

Page: 145

M8 - male

+L3=VA3-P1

cable // OE-M8, 3-pole,plug, 2m

+L3=LI3-K1 J7_O11

valves adressung // adressung

Connection cable | M8 plug, straight, 3-pin – open end

Cable Plan Formular: =Z6B++111+A/198

			Toolbox_88	Option:		MESSER		module	sector 111 location	+A	=Z6A	Page 155
			Toolbox version	customer:	Maghreb Steel			cable overview				
			automated	commission:	286002000		project: 286002000					
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024						

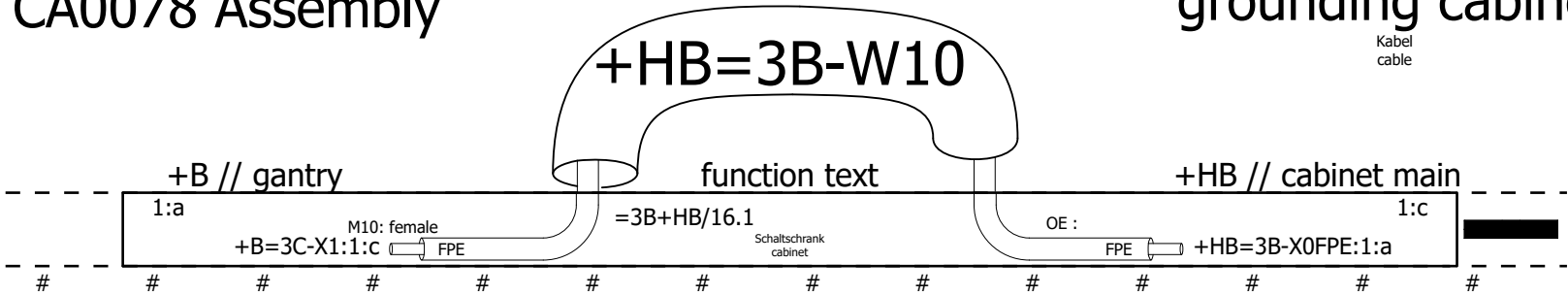
6 Meter

CA0078 Assembly

grounding cabinet

+HB=3B-W10

Kabel
cable

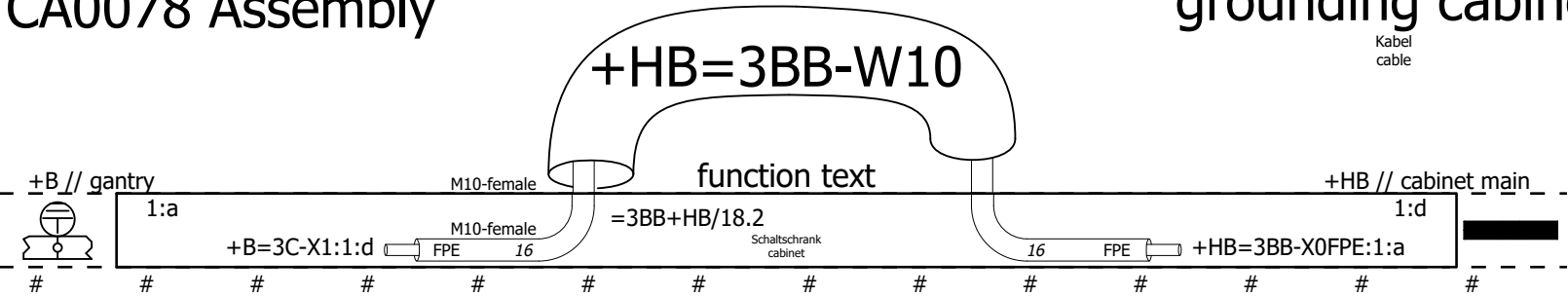


description: FPE, Ring, OE-M10, 16mm²
number of cores: FPE, ring, OE-M10, 16mm²
cross section: 16 mm²
length: 6 m
assembly Article: cable accessory: 77070529

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVCT, 1x16 PVCT, 1x16	77070130	A
1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 16mm ² cable lug, M10, 16mm ²	556027	

CA0078 Assembly

grounding cabinet



description:

FPE, Ring, OE-M10, 16mm²

FPE, ring, OE-M10, 16mm²

number of cores:

cross section

length:

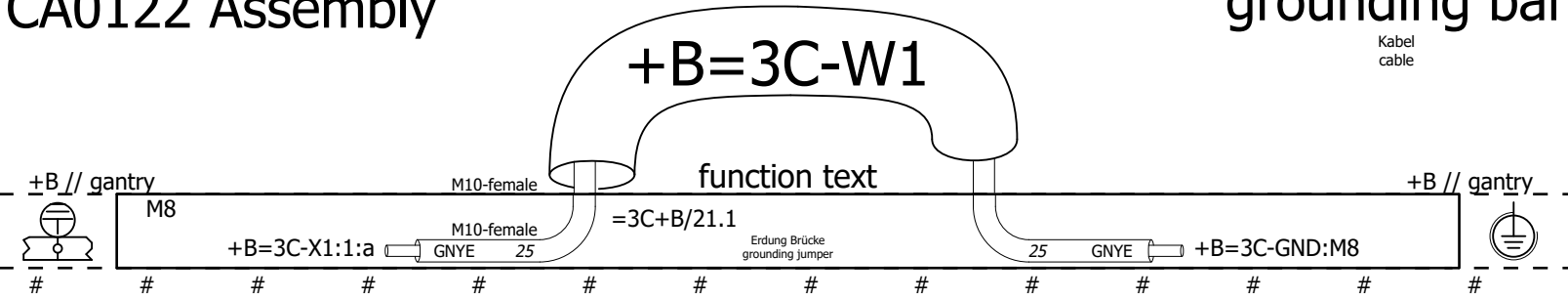
16 mm²

6 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVCT, 1x16 PVCT, 1x16	77070130	A
1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 16mm² cable lug, M10, 16mm²	556027	

CA0122 Assembly

grounding bar

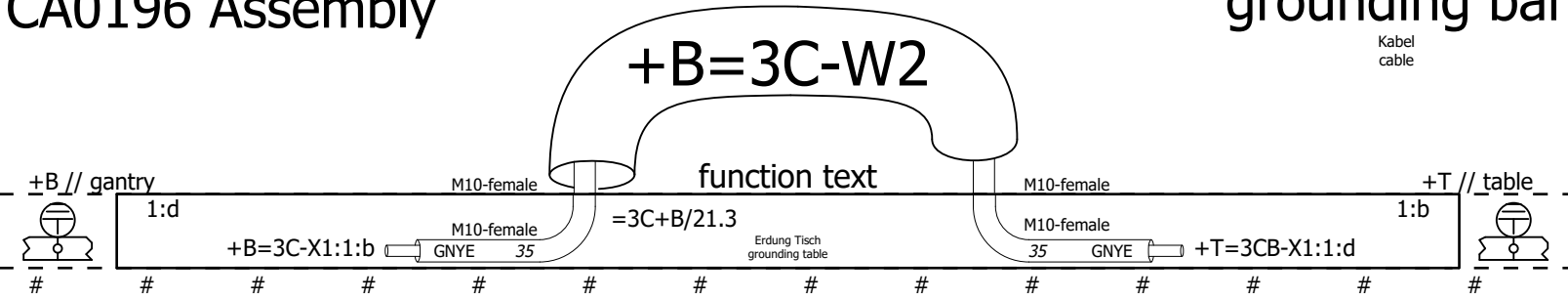


description: Erdung, PE, Ring, M8-M10, 25mm²
number of cores: grounding, PE, ring, M8-M10, 25mm²
cross section: 25 mm²
length: 0,2 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x25, PE	0551079	A B
	PVC, 1x25, PE	Erdung Brücke	
1	Kabel Zubehör cable accessory Rohrkabelschuh, M8, 25mm ²	556077	
Stück	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 25mm ²	556091	
1	cable lug, M10, 25mm ²	=	

CA0196 Assembly

grounding bar

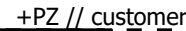


description: Erdung, PE, Ring, M10-M10, 35mm²
number of cores: 1 grounding, PE, ring, M10-M10, 35mm²
cross section 35 mm²
length: 36 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x35, PE	G10001060	
		Erdung Tisch	
Stück 1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 35mm² cable lug, M10, 35mm²	G10000648	A
Stück 1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 35mm² cable lug, M10, 35mm²	G10000648	B

grounding bar

Kabel
cable

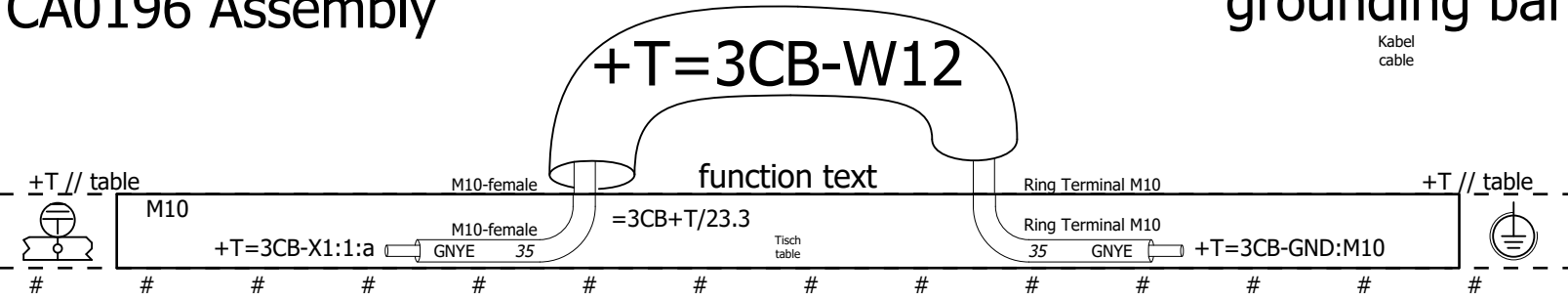
function text

Erdung, PE, Ring, OE-M10, 35mm²
grounding, PE, ring, OE-M10, 35mm²

Quantity	Designation	Part number			
	Kabel Meterware cable sold by meter PVC, 1x35, PE PVC, 1x35, PE	G10001060 Erdung Kunde			
Stück 1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 35mm² cable lug, M10, 35mm²	G10000048 Erdung Kunde =		A	

CA0196 Assembly

grounding bar



description: Erdung, PE, Ring, M10-M10, 35mm²
number of cores: 1 grounding, PE, ring, M10-M10, 35mm²
cross section 35 mm²
length: 0,2 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x35, PE	G10001060	A B
Stück 1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 35mm² cable lug, M10, 35mm²	G10000648 =	
Stück 1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 35mm² cable lug, M10, 35mm²	G10000648 =	

CA0212 Assembly

grounding machine

description:

number of cores:

cross section

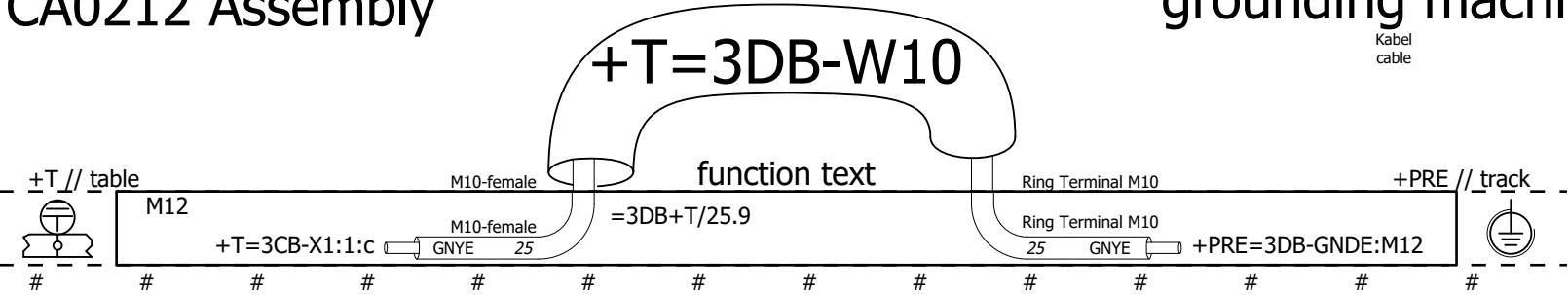
length:

Erdung, PE, Ring, M10-M12, 25mm²
grounding, PE, ring, M10-M12, 25mm²

1

25 mm²

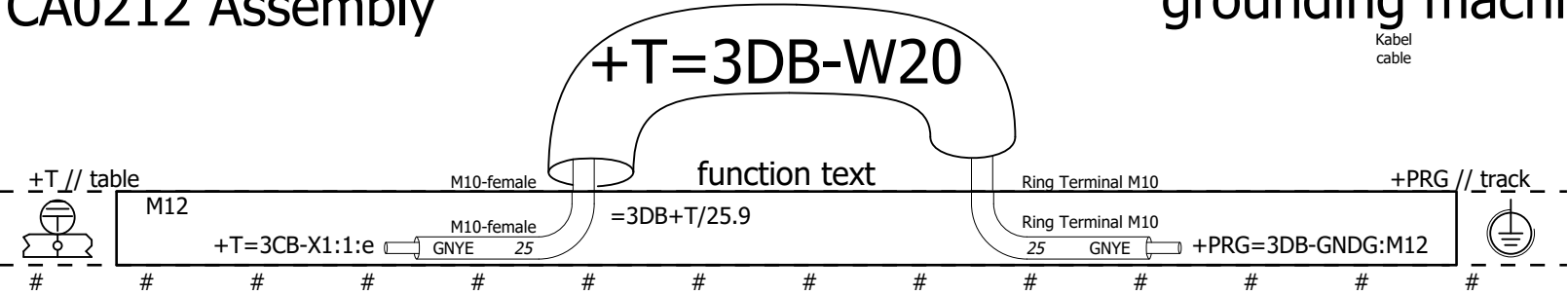
11 m



Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x25, PE	0551079	A B
		Laufbahn 2	
Stück 1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 25mm² cable lug, M10, 25mm²	556091	
Stück 1	Kabel Zubehör cable accessory Rohrkabelschuh, M12, 25mm² cable lug, M12, 25mm²	G10001087	

CA0212 Assembly

grounding machine

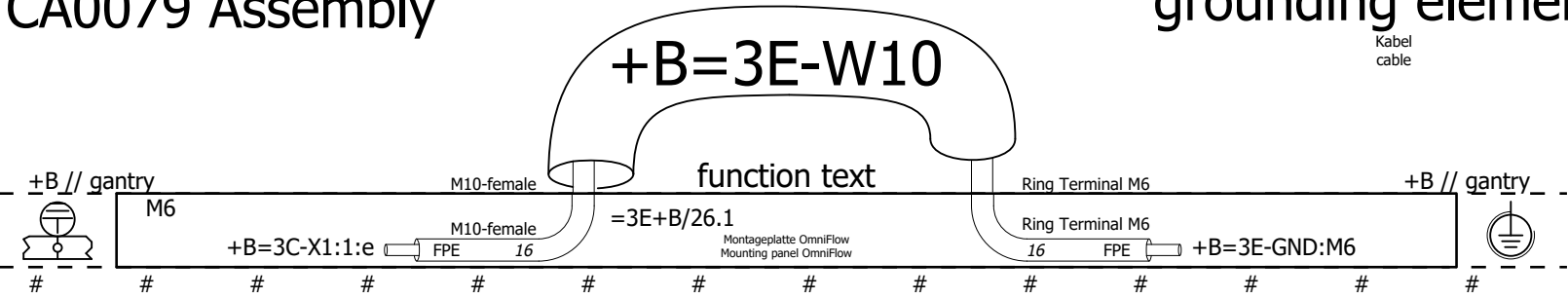


description: Erdung, PE, Ring, M10-M12, 25mm²
number of cores: 1 grounding, PE, ring, M10-M12, 25mm²
cross section 25 mm²
length: 11 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x25, PE	0551079	A B
Stück 1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 25mm² cable lug, M10, 25mm²	556091	
Stück 1	Kabel Zubehör cable accessory Rohrkabelschuh, M12, 25mm² cable lug, M12, 25mm²	G10001087	

CA0079 Assembly

grounding elements

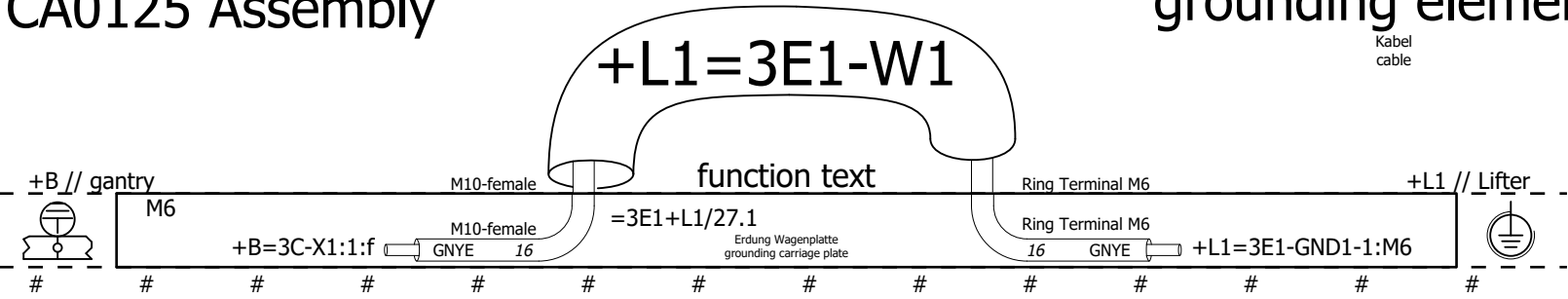


description: FPE, Ring, M6-M10, 16mm²
number of cores: 1 FPE, ring, M6-M10, 16mm²
cross section: 16 mm²
length: 4 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVCT, 1x16 PVCT, 1x16	77070130	A B
1	Kabel Zubehör cable accessory Rohrkabelschuh, M6, 16mm ² cable lug, M6, 16mm ²	Montageplatte OmniFlow 556018 =	
1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 16mm ² cable lug, M10, 16mm ²	556027 =	

CA0125 Assembly

grounding elements

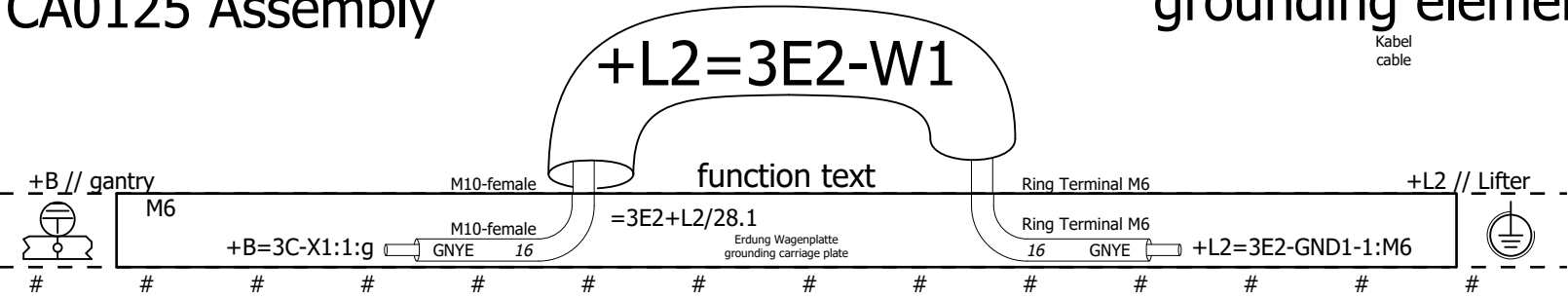


description: Erdung, PE, Ring, M6-M10, 16mm²
number of cores: 1 grounding, PE, ring, M6-M10, 16mm²
cross section: 16 mm²
length: 8 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x16, PE	G10000669	A B
		Erdung Wagenplatte	
1	Kabel Zubehör cable accessory Rohrkabelschuh, M6, 16mm ² cable lug, M6, 16mm ²	556018 =	
1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 16mm ² cable lug, M10, 16mm ²	556027 =	

CA0125 Assembly

grounding elements

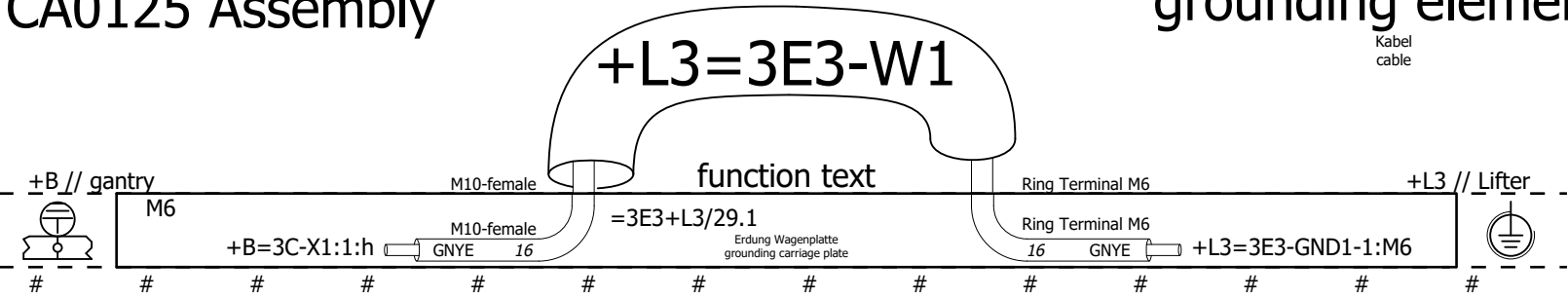


description: Erdung, PE, Ring, M6-M10, 16mm²
number of cores: 1 grounding, PE, ring, M6-M10, 16mm²
cross section: 16 mm²
length: 8 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x16, PE	G10000669	A B
		Erdung Wagenplatte	
1	Kabel Zubehör cable accessory Rohrkabelschuh, M6, 16mm ² cable lug, M6, 16mm ²	556018 =	
1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 16mm ² cable lug, M10, 16mm ²	556027 =	

CA0125 Assembly

grounding elements



description: Erdung, PE, Ring, M6-M10, 16mm²
number of cores: 1 grounding, PE, ring, M6-M10, 16mm²
cross section: 16 mm²
length: 8 m

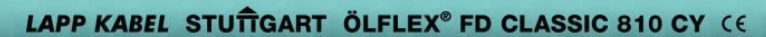
Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x16, PE	G10000669	A B
		Erdung Wagenplatte	
1	Kabel Zubehör cable accessory Rohrkabelschuh, M6, 16mm ² cable lug, M6, 16mm ²	556018 =	
1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 16mm ² cable lug, M10, 16mm ²	556027 =	

AC power supply cabinet

$$+HB=AA-W0$$

1xXXX
//

PVC, 5x 4, PE, CY
PVC, 5x 4, PE, CY
5
4 mm²
32 m



MESSER 

DC power supply interface

The diagram illustrates a 24 VDC distribution system. A central terminal block is connected to two side cabinets. The central block has a main terminal block at the top and a distribution block below it. The distribution block is divided into two sections: '1:c' and '2:b'. Each section has five terminals labeled BK, BU, GY, BN, and WH. The left cabinet has a main terminal block with five terminals labeled BK, BU, GY, BN, and WH. The right cabinet has a main terminal block with five terminals labeled BK, BU, GY, BN, and WH. The diagram shows the wiring connections between the central block and the side cabinets, including the distribution of power to the side cabinets.

Central Terminal Block:

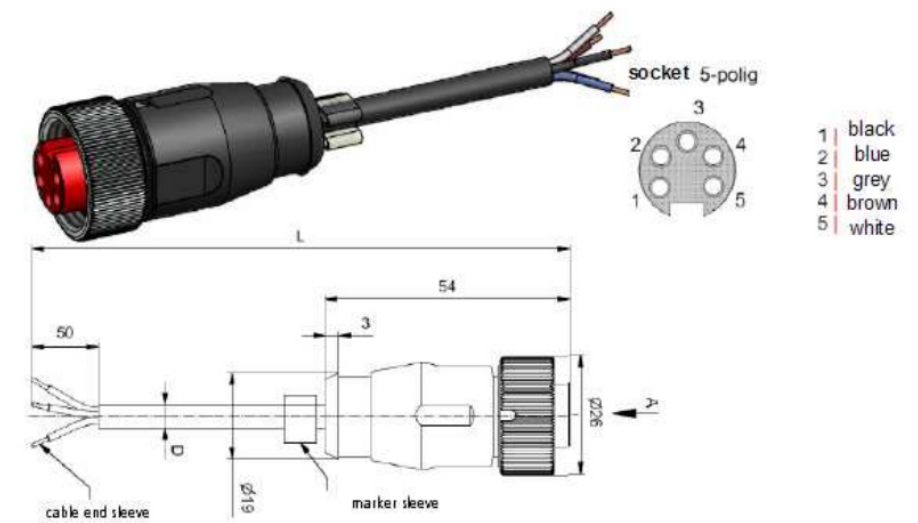
- Top Section:**
 - Terminal: **+HB** // cabinet main
 - Terminal: **+B** // gantry
- Bottom Section:**
 - Terminal: **1:c**
 - Terminal: **2:b**

Wiring Connections:

- Left Cabinet:**
 - +HB=BI-X1:1** (BK, 2,5) to **1:c**
 - +HB=BI-X1:3** (BU, 2,5) to **1:c**
 - +HB=BI-X1:5** (GY, 2,5) to **1:c**
 - +HB=BI-X1:2** (BN, 2,5) to **2:b**
 - +HB=BI-X1:4** (WH, 2,5) to **2:b**
- Right Cabinet:**
 - +B=BI-K1:X1:1** (BK, 2,5) from **1:c**
 - +B=BI-K1:X1:2** (BU, 2,5) from **1:c**
 - +B=BI-K1:X1:3** (GY, 2,5) from **1:c**
 - +B=BI-K1:X1:4** (BN, 2,5) from **2:b**
 - +B=BI-K1:X1:5** (WH, 2,5) from **2:b**

Labels and Notes:

- 1:c**: Distribution 24 VDC extern
- 2:b**: Distribution 24 VDC external
- +HB**: cabinet main
- +B**: gantry
- 1xG10000103**: distributor power// EtherCAT-box, power distribution, M8, 4 x 4A



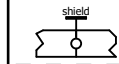
			Toolbox_88	Option: customer: commission:					module	Page 169
			Toolbox version		Maghreb Steel					
			automated		286002000	project: 286002000				
Modification	Date	Name	evaluation has been generated	Ed. by.	Johann.Rais	24.05.2024		cable connection plan	sector 111 location +A =Z6B	

detection limiting

Kabel
cable
1x77002698

1x77002698

```
+L1 // Lifter
```



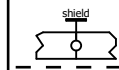
			Toolbox_88	Option: customer: commission:					module			Page 170
			Toolbox version		Maghreb Steel							
			automated		286002000	project: 286002000						
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024		cable connection plan		sector 111	location +A	=Z6B

detection limiting

Kabel
cable
1x77002698

1x77002698

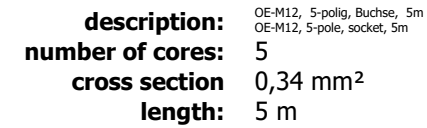
```
+L3 // Lifter
```



			Toolbox_88	Option: customer: commission:					module			Page 171
			Toolbox version		Maghreb Steel							
			automated		286002000	project: 286002000						
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024		cable connection plan		sector 111	location +A	=Z6B

emergency stop additions

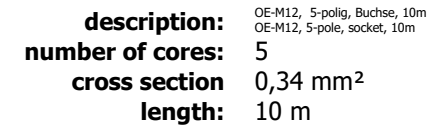
1xG10001248
channel, 24 V DC



			Toolbox_88	Option: customer: commission:					module			Page 172	
			Toolbox version		Maghreb Steel								
			automated		286002000	project: 286002000							
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024		cable connection plan		sector 111	location +A	=Z6B	

emergency stop additions

1xG10001248
channel, 24 V DC

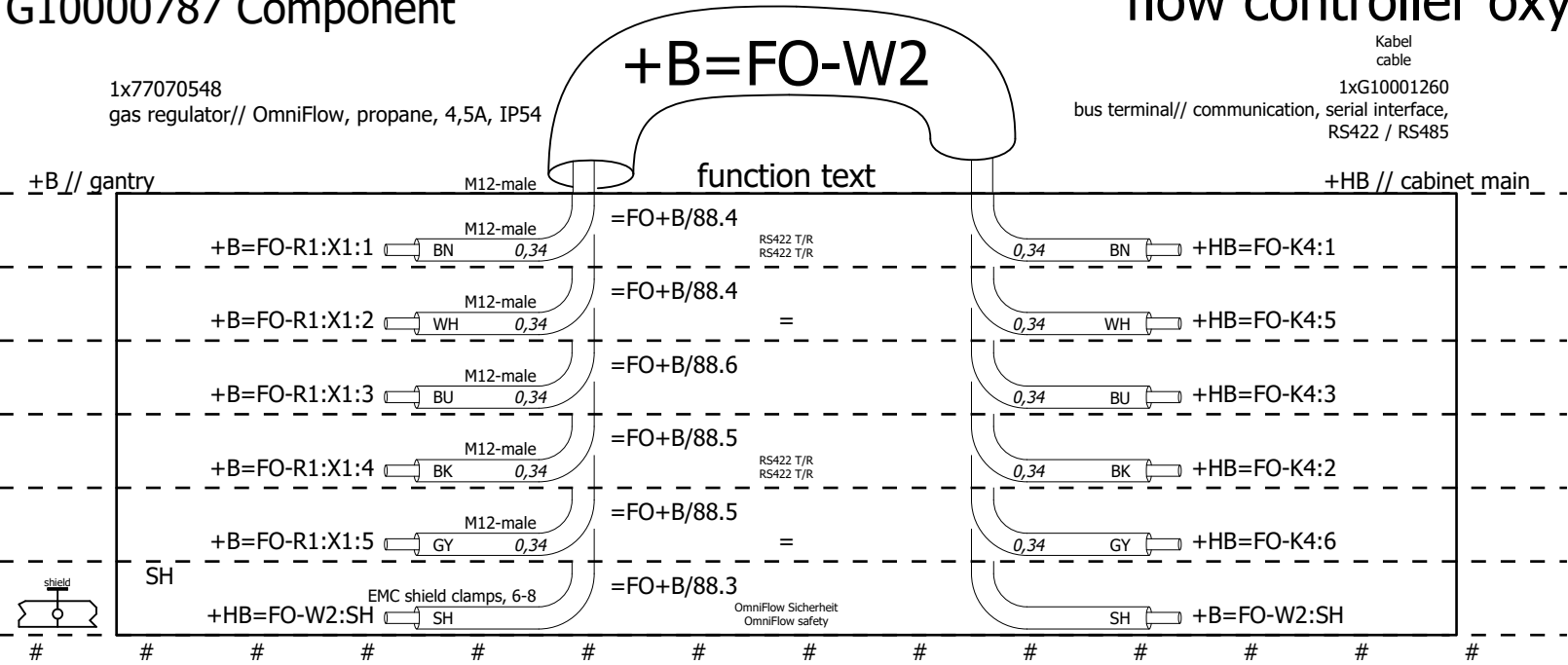


			Toolbox_88	Option: customer: commission:					module			Page 173
			Toolbox version		Maghreb Steel							
			automated		286002000	project: 286002000						
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024		cable connection plan	sector 111	location	+A	=Z6B

G10000787 Component

flow controller oxyfuel

description: OE-M12, 5-polig, Stecker, 5m
number of cores: 5
cross section: 0,34 mm²
length: 5 m



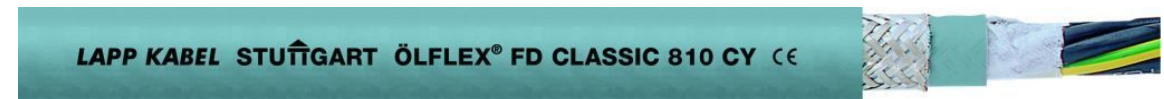
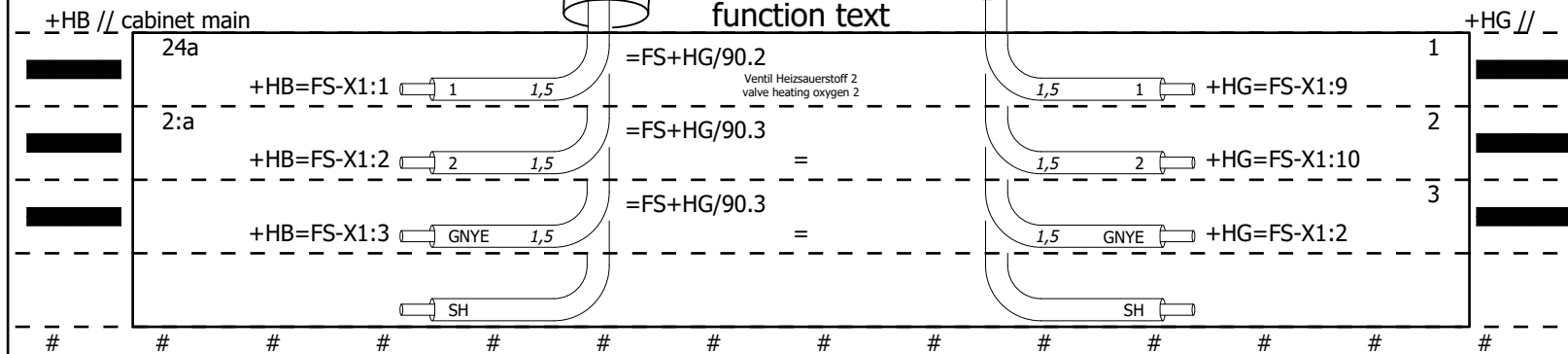
flow controller safety

description:	PVC, 3x 1,5, PE, CY PVC, 3x 1,5, PE, CY
number of cores:	3
cross section	1,5 mm ²
length:	38 m



+HG=FS-W0

function text

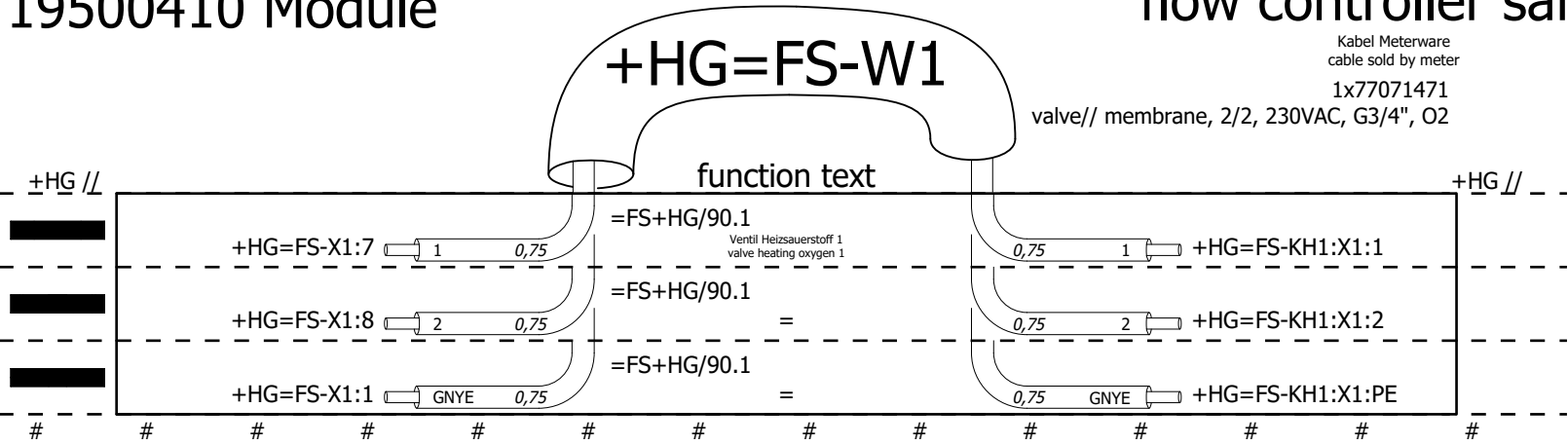


			Toolbox_88	Option:					module		Page 175
			Toolbox version	customer:	Maghreb Steel						
			automated	commission:	286002000	project: 286002000					
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024			cable connection plan	sector 111 location +A =Z6B	

19500410 Module

flow controller safety

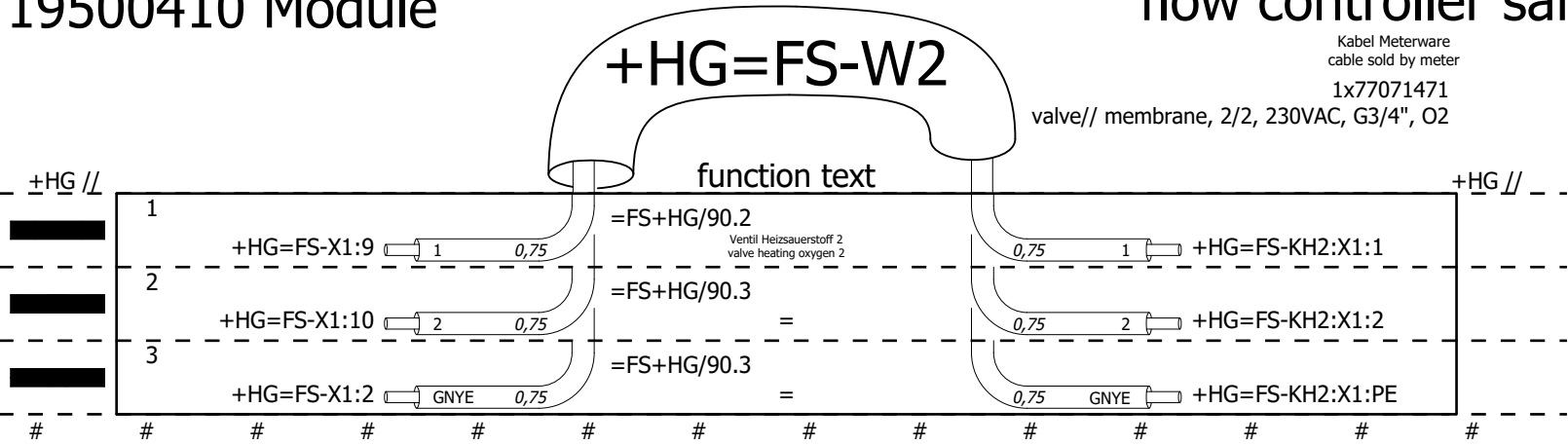
description: PVC, 3x 0,75, PE
number of cores: 3
cross section: 0,75 mm²
length: 1 m



19500410 Module

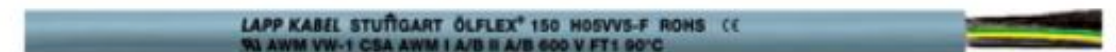
flow controller safety

description: PVC, 3x 0,75, PE
number of cores: 3
cross section 0,75 mm²
length: 1 m



flow controller safety

+HG //										function text										+HG //									
+HG=FS-X1:11										=FS+HG/90.4										+HG=FS-KF1:A1:1									
+HG=FS-X1:12										=FS+HG/90.4										+HG=FS-KF1:A1:2									
+HG=FS-X1:3										=FS+HG/90.4										+HG=FS-KF1:A1:PE									



flow controller safety

Kabel Meterware
cable sold by meter
1x77071623
G3/4", fuel gas

valve// magnetic, 2/2, 230VAC, G3/4", fuel gas

function text									
+HG=FS-X1:13	1	0,75	=FS+HG/90.5	0,75	1	+HG=FS-KF2:A1:1			
+HG=FS-X1:14	2	0,75	=FS+HG/90.6	=	0,75	2	+HG=FS-KF2:A1:2		
+HG=FS-X1:4	GNYE	0,75	=FS+HG/90.6	=	0,75	GNYE	+HG=FS-KF2:A1:PE		

description:	PVC, 3x 0,75, PE
number of cores:	3
cross section	0,75 mm ²
length:	1 m

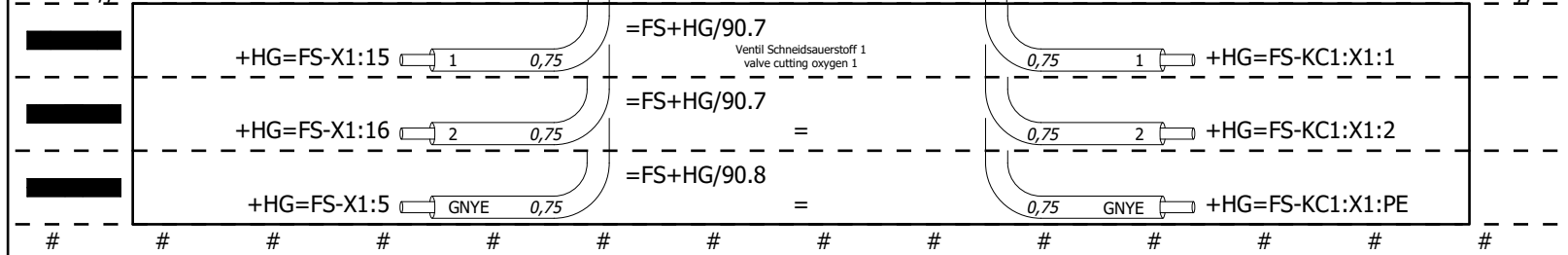


			Toolbox_88	Option: customer: commission:					module		Page 179
			Toolbox version		Maghreb Steel						
			automated		286002000	project: 286002000					
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024		cable connection plan	sector 111 location +A	=Z6B	

flow controller safety

Kabel Meterware
cable sold by meter

valve// membrane, 2/2, 230VAC, G3/4", O2

$$+ \underline{\text{HG}} //$$


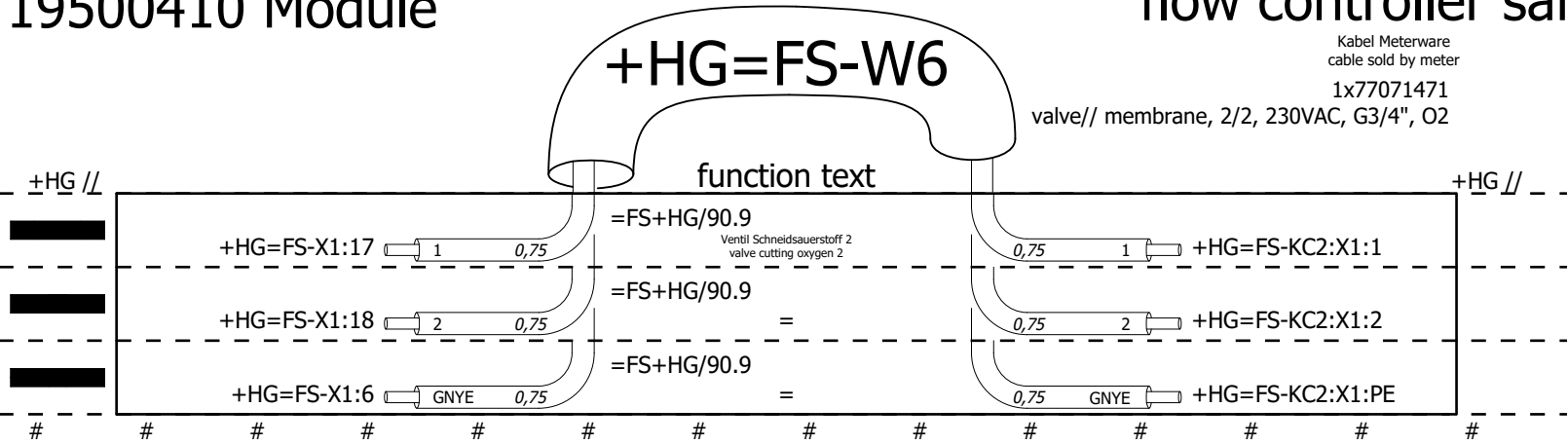
PVC, 3x 0,75, PE
PVC, 3x 0,75, PE



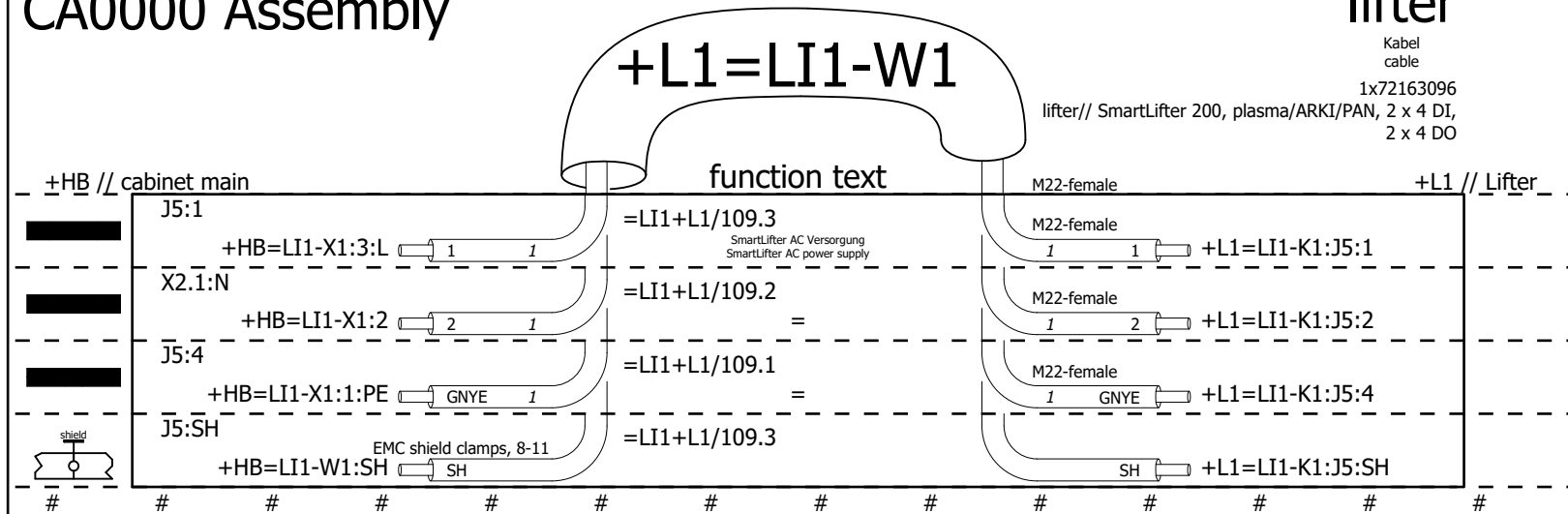
19500410 Module

flow controller safety

description: PVC, 3x 0,75, PE
number of cores: 3
cross section: 0,75 mm²
length: 1 m



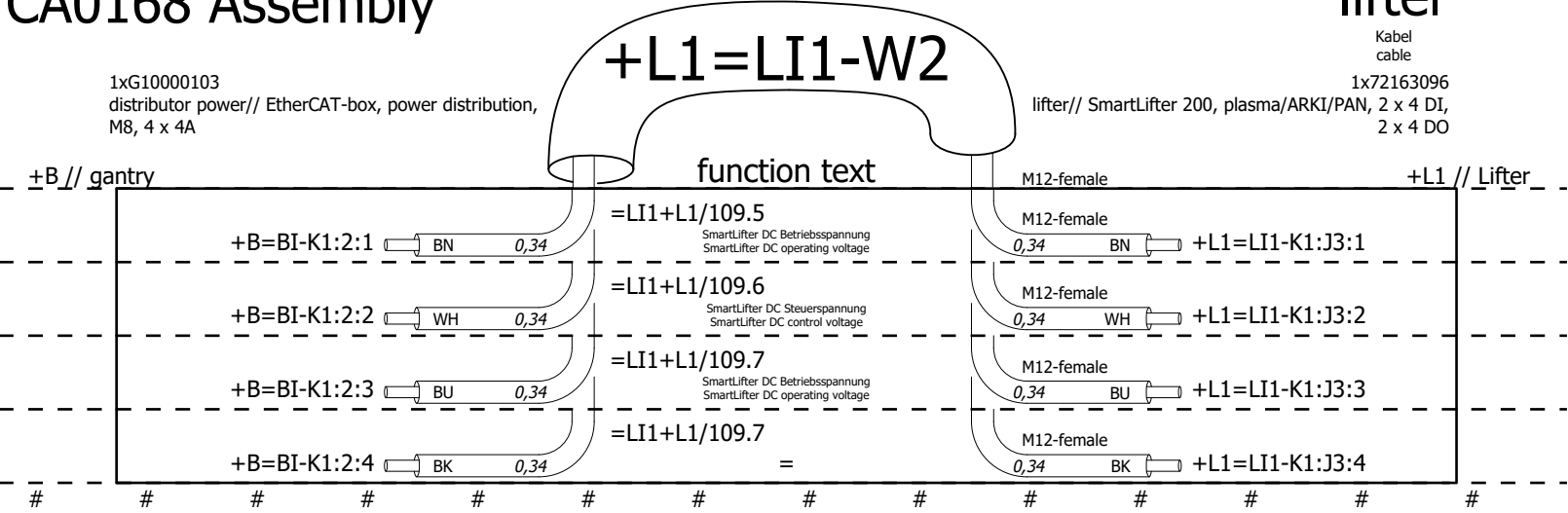
lifter



Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 3x 1, PE, CY PVC, 3x 1, PE, CY	0551042 SmartLifter AC Versorgung	A
Stück 1	Stecker plug M22, 4-polig, Buchse, 0° M22, 4-pole, socket, 0°	SmartLifter AC power supply 77070191 =	

CA0168 Assembly

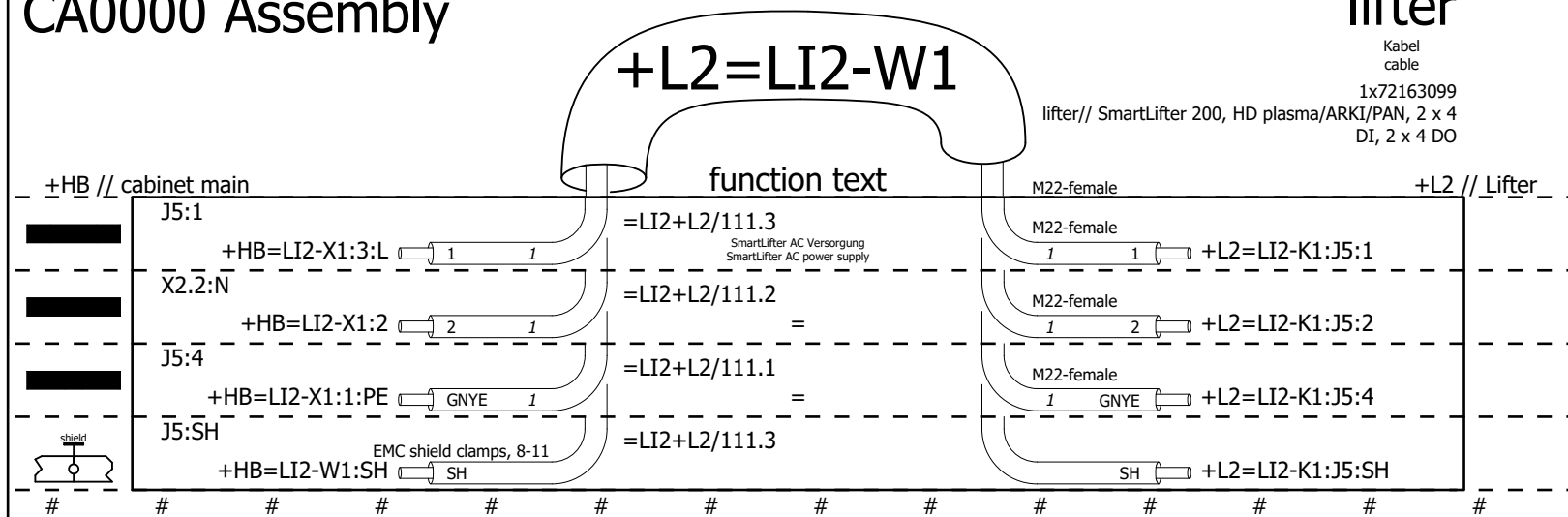
lifter



description: M8-M12, 5-adrig, <=15m
number of cores: M8-M12, 5-core, <=15m
cross section 0,34 mm²
length: 11 m

Quantity	Designation	Part number	
Stück 1	Kabel cable Powerleitung, M8-M8, 15m power line, M8-M8, 15m	77005851	B
Stück 1	Stecker plug M12, 4-polig, Buchse, 0° M12, 4-pole, socket, 0°	SmartLifter DC Betriebsspannung 77005053 =	

lifter

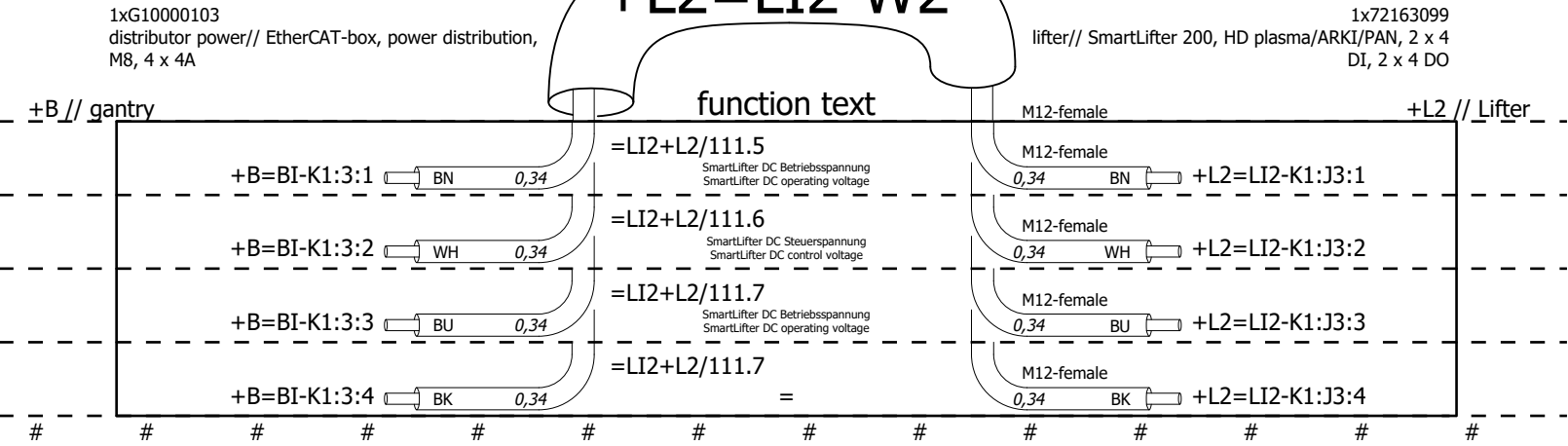


description:	OE-M22, 4-adrig
number of cores:	OE-M22, 4-core
cross section	1 mm ²
length:	10 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 3x 1, PE, CY PVC, 3x 1, PE, CY	0551042 SmartLifter AC Versorgung SmartLifter AC power supply	
Stück 1	Stecker plug M22, 4-polig, Buchse, 0° M22, 4-pole, socket, 0°	77070191 =	A

CA0168 Assembly

lifter



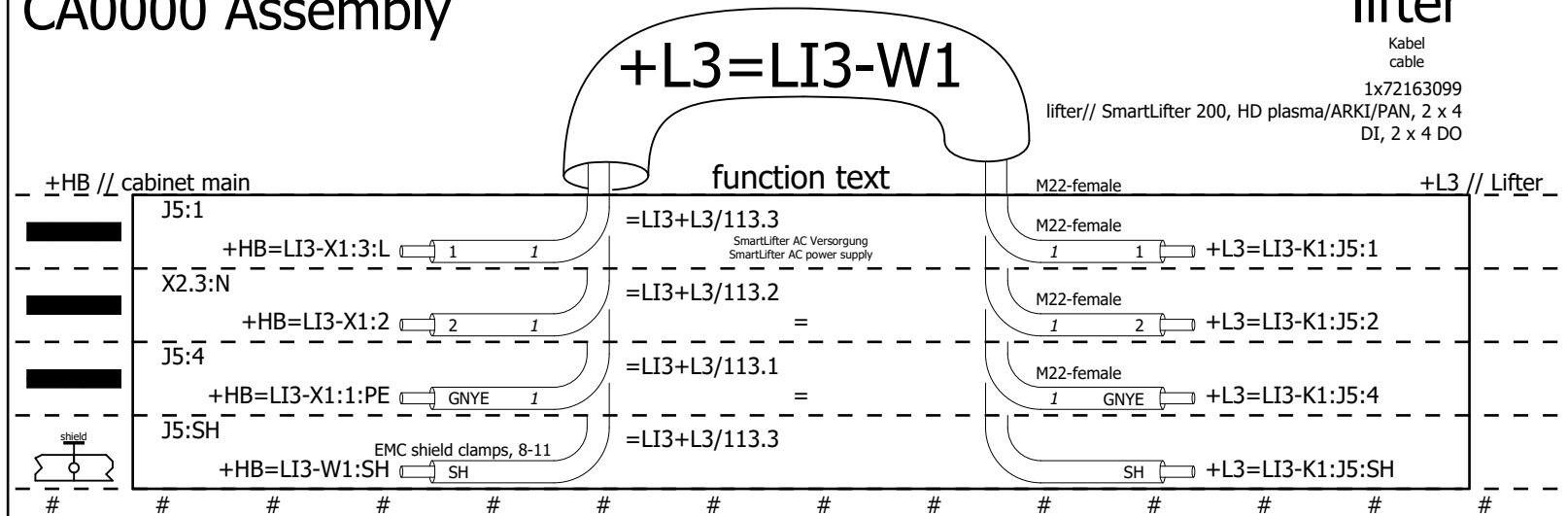
CA0000 Assembly

lifter

Kabel
cable

1x72163099
RKI/PAN, 2 x 4
DI, 2 x 4 DO

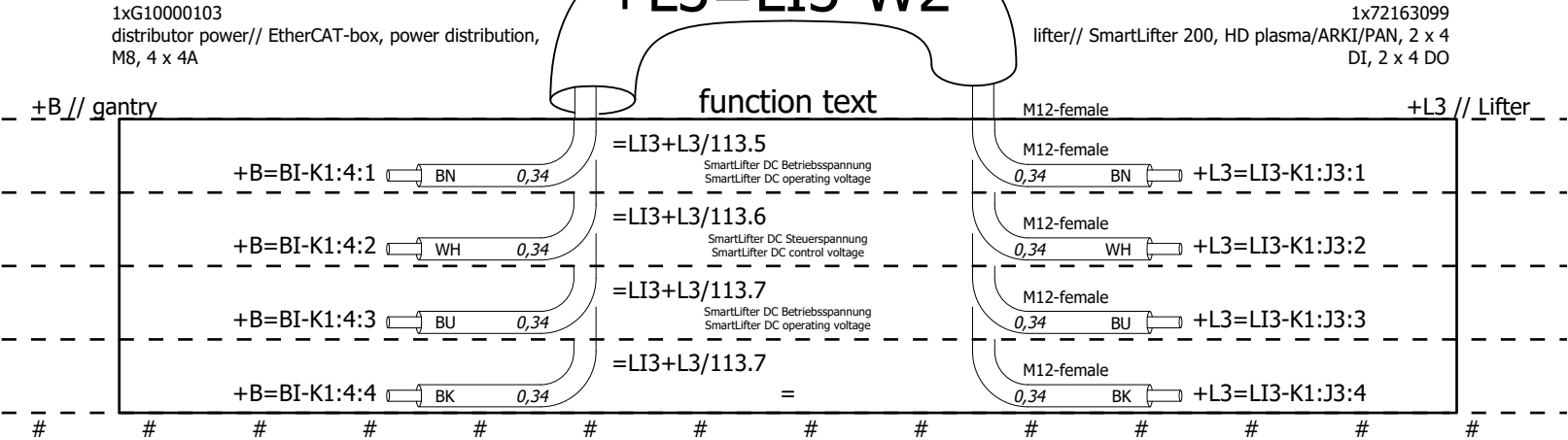
description:	OE-M22, 4-adrig
number of cores:	OE-M22, 4-core
cross section	1 mm ²
length:	10 m



Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 3x 1, PE, CY PVC, 3x 1, PE, CY	0551042 SmartLifter AC Versorgung	A
Stück 1	Stecker plug M22, 4-polig, Buchse, 0° M22, 4-pole, socket, 0°	SmartLifter AC power supply 77070191 =	

			Toolbox_88	Option:				module	Page 186
			Toolbox version	customer:	Maghreb Steel				
			automated	commission:	286002000		project: 286002000		
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024		cable connection plan sector 111 location +A =Z6B	

CA0168 Assembly



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lifter

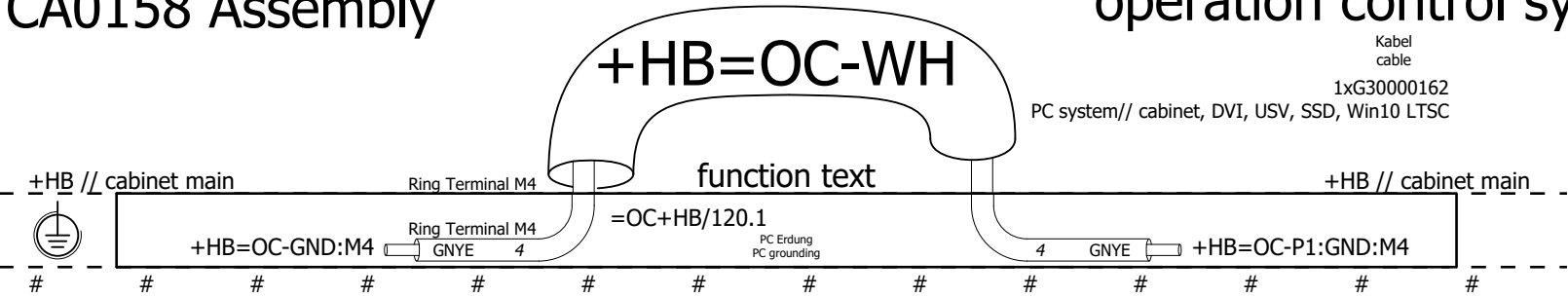
Kabel
cable

description: M8-M12, 5-adrig, <=15m
number of cores: M8-M12, 5-core, <=15m
cross section 0,34 mm²
length: 11 m

Quantity	Designation	Part number	
Stück 1	Kabel cable Powerleitung, M8-M8, 15m power line, M8-M8, 15m	77005851	B
Stück 1	Stecker plug M12, 4-polig, Buchse, 0° M12, 4-pole, socket, 0°	SmartLifter DC Betriebsspannung 77005053 =	

CA0158 Assembly

operation control system



description: Erdung, PE, Ring, M4- M4, 4mm²
grounding, PE, ring, M4- M4, 4mm²
number of cores:
cross section 4 mm²
length: -1 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x 4 PVC, 1x 4	570130 PC Erdung	A B
1	Kabel Zubehör cable accessory Ringkabelschuh, M4, 6mm² ring cable lug, M4, 6mm²	0556124 =	
1	Kabel Zubehör cable accessory Ringkabelschuh, M4, 6mm² ring cable lug, M4, 6mm²	0556124 =	

operation panel key switch

Add-on: cable accessory: 77003783



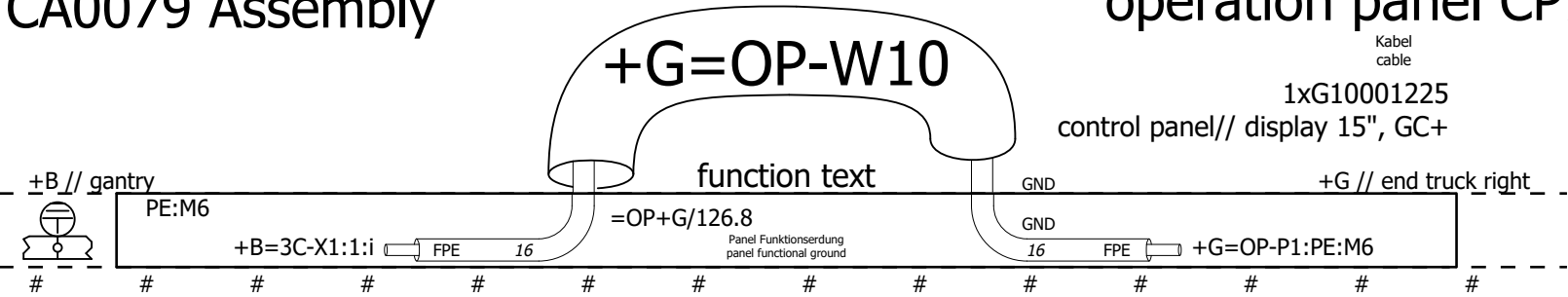
operation panel CP7902

OE-M23, 19-polig, Buchse, 20m
OE-M23, 19-pole, socket, 20m



CA0079 Assembly

operation panel CP7902

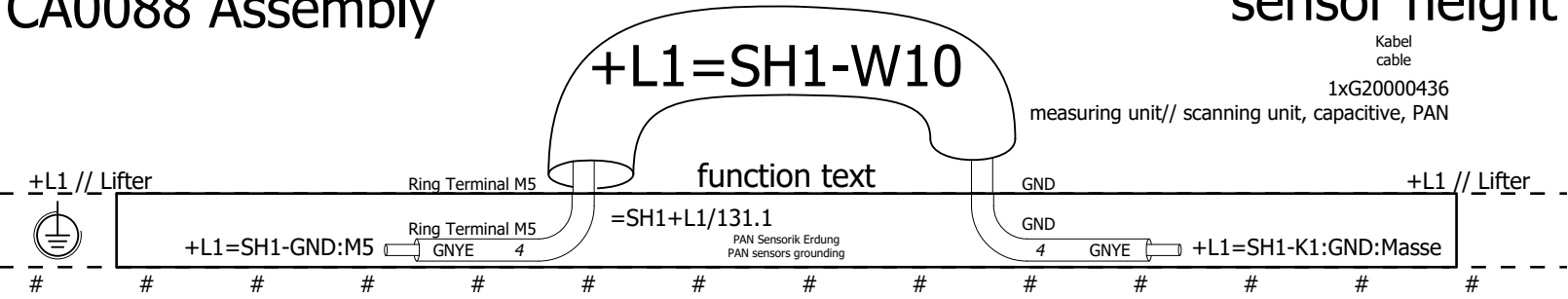


description: FPE, Ring, M6-M10, 16mm²
number of cores: 1 FPE, ring, M6-M10, 16mm²
cross section: 16 mm²
length: 9 m

Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVCT, 1x16 PVCT, 1x16	77070130	A B
1	Kabel Zubehör cable accessory Rohrkabelschuh, M6, 16mm² cable lug, M6, 16mm²	Panel Funktionserdung 556018 =	
1	Kabel Zubehör cable accessory Rohrkabelschuh, M10, 16mm² cable lug, M10, 16mm²	556027 =	

CA0088 Assembly

sensor height



description: Erdung, PE, Ring, M5- M5, 4mm²
number of cores: grounding, PE, ring, M5- M5, 4mm²
cross section 4 mm²
length: 0,2 m

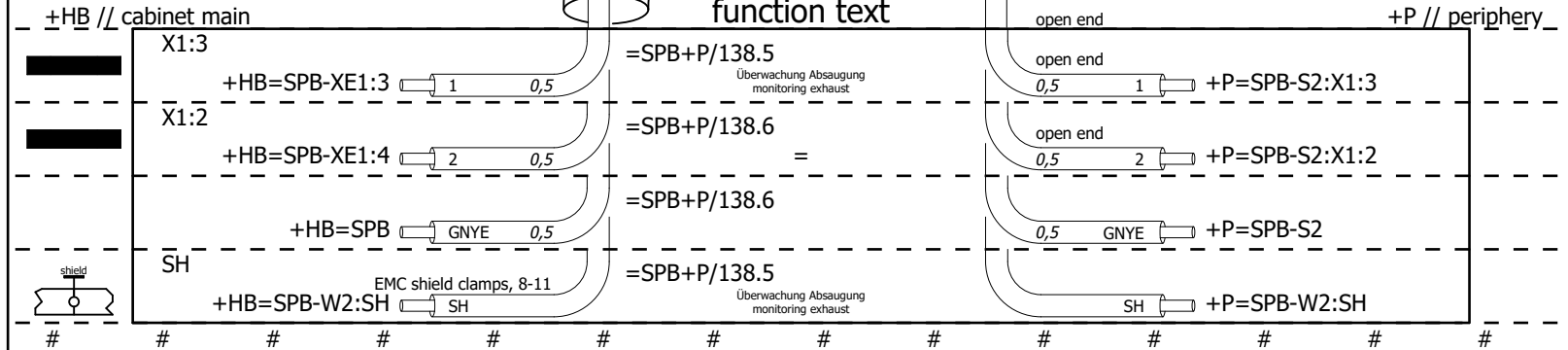
Quantity	Designation	Part number	
	Kabel Meterware cable sold by meter PVC, 1x 4	570130	B
	PAN Sensorik Erdung PAN sensors grounding	0580600	
1	Kabel Zubehör cable accessory Ringkabelschuh, M5, 6mm² ring cable lug, M5, 6mm²	=	
1	Kabel Zubehör cable accessory Ringkabelschuh, M5, 6mm² ring cable lug, M5, 6mm²	0580600	=

sensor pressure

Kabel Meterware
cable sold by meter
1xG10000966

open end

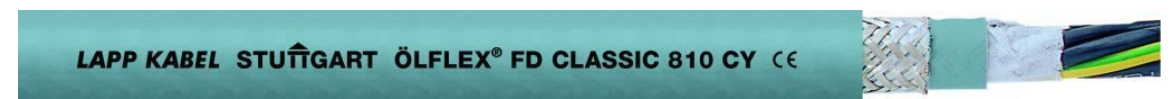
+P // periphery



PVC, 3x 0,5, PE, CY
PVC, 3x 0,5, PE, CY

0,5 mm²

44 m



			Toolbox_88	Option:				module	Page 194
			Toolbox version	customer:	Maghreb Steel				
			automated	commission:	286002000		project: 286002000		
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024		cable connection plan sector 111 location +A =Z6B	

CA0159 Assembly

1xG30000122
measuring unit// scanning unit, inductive, ARKI

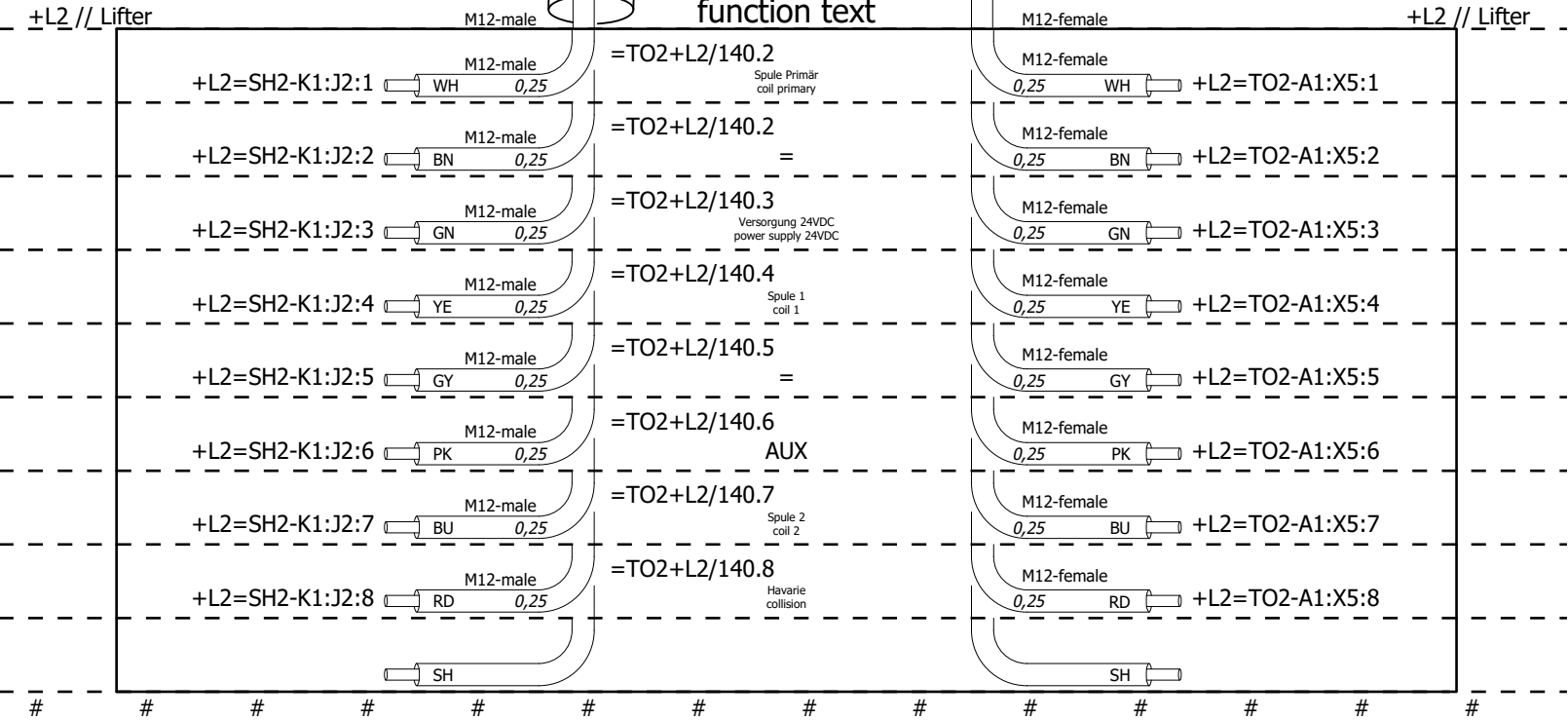
+L2=TO2-WA2

torch

1x71611460

cutting torch// oxyfuel, ALFA

description: OE-M12, 8-adrig, Buchse, 0°
number of cores: OE-M12, 8-core, socket, 0°
cross section: 0,25 mm²
length: 3 m



Quantity	Designation	Part number	
Stück	Kabel cable	77005405	
1	OE-M12, 8-polig, Buchse, 0°, 20m OE-M12, 8-pole, socket, 0°, 20m	Spule Primär coil primary	

CA0159 Assembly

1xG30000122
measuring unit// scanning unit, inductive, ARKI

+L3=TO3-WA2

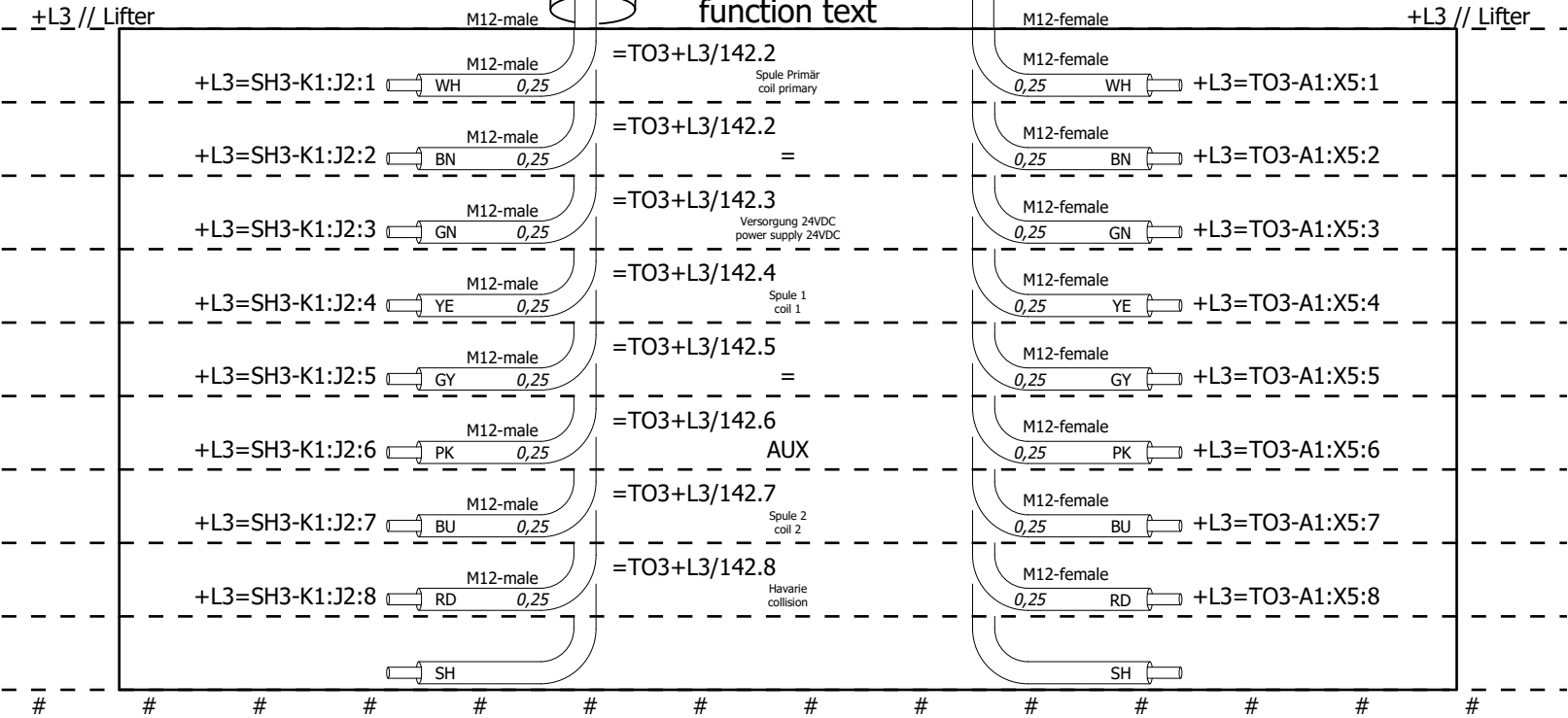
torch

Kabel
cable

1x71611460

cutting torch// oxyfuel, ALFA

description: OE-M12, 8-adrig, Buchse, 0°
number of cores: OE-M12, 8-core, socket, 0°
cross section: 0,25 mm²
length: 3 m

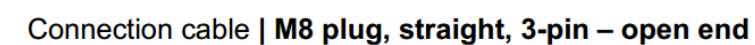
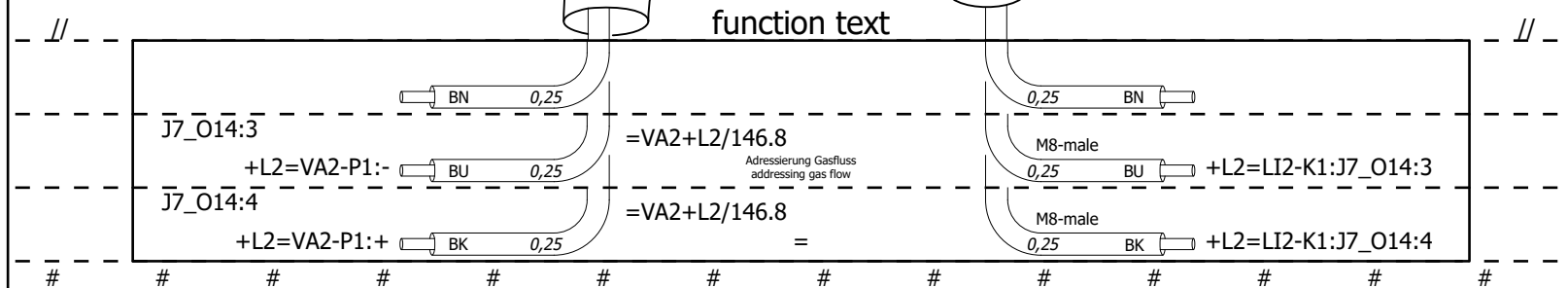


Quantity	Designation	Part number	
Stück	Kabel cable	77005405	
1	OE-M12, 8-polig, Buchse, 0°, 20m OE-M12, 8-pole, socket, 0°, 20m	Spule Primär coil primary	

valves addressing

$$+L2=VA2-W4$$

Kabel
cable

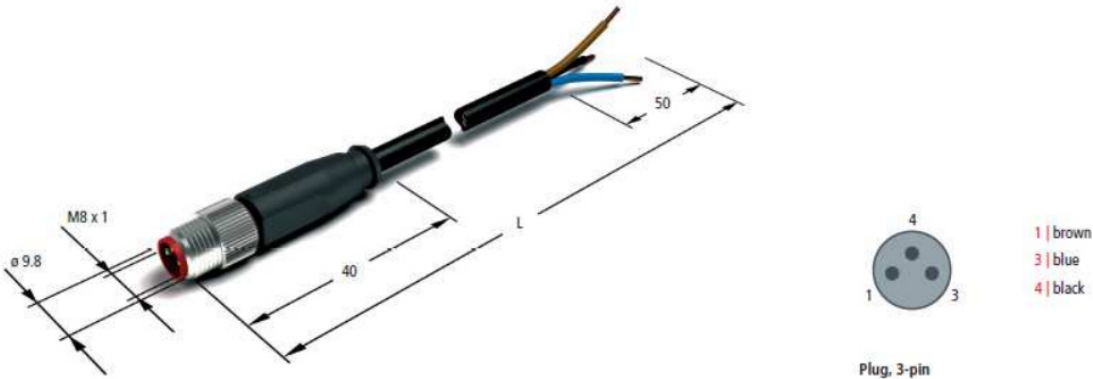
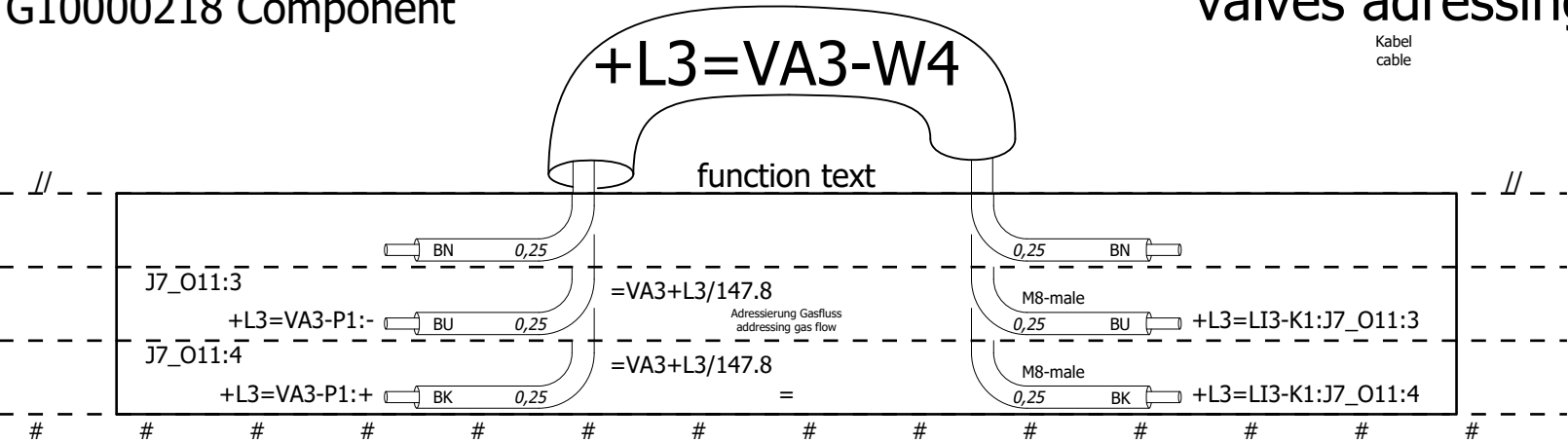


			Toolbox_88	Option:				module	Page 197
			Toolbox version	customer:	Maghreb Steel				
			automated	commission:	286002000		project: 286002000		
Modification	Date	Name	evaluation has been generated	Ed. by:	Johann.Rais	24.05.2024	cable connection plan	sector 111 location +A =Z6B	

G10000218 Component

valves adresssing

description: OE-M8, 3-polig, Stecker, 2m
number of cores: 3
cross section: 0,25 mm²
length: 2 m



Connection cable | M8 plug, straight, 3-pin – open end